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BEHAVIORAL SCIENCE FOR DEVELOPMENT

Insights and Strategies for **Global Impact**



EDITED BY

LUIS ARTAVIA-MORA & ZARAK KHAN

BEHAVIORAL SCIENCE FOR DEVELOPMENT

Insights and Strategies for **Global Impact**

EDITED BY
LUIS ARTAVIA-MORA & ZARAK KHAN



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First and foremost, we say thank you to each of the authors that made this book possible. Your expertise, insights and constructive feedback have been invaluable to produce this work. We are humble to have the opportunity to share our names and experience together in this volume, and we look forward to collaborating on more projects that develop and expand the impact of our field around the world.

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that ignited this entire project and, from that moment, he pursued it with unflagging enthusiasm and passion. Luis's dedication not only ensured the accomplishment of this endeavor, but also transformed the process into an engaging, fun-filled journey.

We are providing the digital version of this book for free thanks to Bescy Publishing, a part of the non-profit Bescy. Bescy has grown to over 18,000 followers since it was founded in 2012 (as the Action Design Network) and promotes the work of many leading behavioral science researchers, academics and practitioners. In line with that mission, the intention of this book is to share the lessons learned, challenges, and best practices of behavioral science to practitioners and the general public.

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Part I:

INTRODUCTION

1.

Let's Begin

Luis Artavia-Mora, Independent consultant and Bescy

Zarak Khan, Bescy

Social and behavioral science is generating groundbreaking solutions that are improving international development initiatives around the world. By understanding a wider and richer framework that includes the social, psychological, and cultural factors that influence human behavior across contexts, behavioral change practitioners are leading the design and implementation of more cost-effective development projects that are based on evidence.

This book does two things: first, it draws on the latest research and case studies to provide practical insights and guidance for practitioners, policymakers, and researchers from around the world. Second, it also offers an appraisal of the challenges, lessons learned and ways in which social and behavioral change initiatives are improving the lives of people around the world.

When speaking with both practitioners and enthusiasts, it is common to hear a desire for a practical manual documenting the evidence at the intersection of behavioral science and

international development. Some of the questions this book aims to answer include:

- ▶ What behavioral tools and approaches are often used in the field of international development?
- ▶ What agents and organizations lead these efforts?
- ▶ What areas of impact are they working on?
- ▶ What challenges have they faced?
- ▶ What lessons have they learned?

By bridging practical and academic knowledge from practitioners, policymakers, along with researchers working in international organizations, the private sector and academia, readers will find answers to these questions in this book.

We hope this work fosters interconnectivity and collaboration in the practical application of behavioral science in international development by unifying resources and experiences from professionals working in every corner of the world. We do not aim to provide a definitive guide or standard; instead, we offer a resource to help advance the field as we move forward. Each project is different in terms of resources, capabilities, needs and related factors. Context matters!

The information in this book will help practitioners, aspiring practitioners and the general public interested in our discipline create a more inclusive and prosperous future for everyone. Hence, we reiterate our deepest gratitude to every author that contributed to this book and for their leadership in the field. Your work is inspiring a new generation of behavioral scientists to improve lives.

This book does not provide an in-depth review of the theoretical underpinning of behavioral science in international development. A reader with that goal could orient themselves by reading the 2015 World Bank Report titled “Mind, Society, and Behavior,” and the books *Thinking, Fast and Slow* (2011),

and *Nudge* (2009). These texts provide insight into the way modern humans behave and the implications for policymaking in international development.

“Mind, Society, and Behavior” is possibly the ideal material to start with. Setting a pivotal breakpoint at the strategic core of development policy in 2015, the report explains that understanding the processes and societal influences that shape how people think can enhance the design and implementation of development work.

The report’s pivotal feature is a shift to a more intrinsically human-centered vision that recognizes how resolving systemic biases and removing barriers result in policies that have the potential to achieve better outcomes.¹ “Mind, Society, and Behavior” categorizes these new social and behavioral insights into three systems of human decision-making:

- ▶ First, people make most judgments and choices using automatic processing, which depends on simpler rules of thumb that follow unconscious biases, fast associations, interpretive frames, and default assumptions (narrow frame). A second, more effortful system, then considers the broader range of relevant factors (wide frame) based on reasoning and a more reflective, deliberate thought process.
- ▶ Second, humans shape their choices according to what others do and say. We follow complex social group dynamics involving norms, networks, common identities, rules of cooperation and competition.

¹ For instance, it aims to identify and resolve biases in beliefs (overconfidence, availability, optimism, etc.), information processing (cognitive overload, hassle factors, framing, etc.), and preferences (present bias, social norms, status quo, etc.), and to overcome systemic barriers in physical, social and psychological factors.

- ▶ Third, a set of factors like experience and context help human beings create “mental models” that help them navigate the world we live in.

These new insights give behavioral specialists an opportunity to create better policies in practically all areas of international development including poverty alleviation, health, education, the management and collection of financial assets, environmental and climate goals, the inclusion of migrants and vulnerable groups in our societies, and many more.

This practitioners’ guide is meant to be a timely complement to the theoretical work in this space. Behavioral science units have spread across international organizations and regions, creating an opportunity to generate new solutions that better address the pressing needs of our times.

Despite the positive efforts to raise standards of living, severe challenges threaten the future of our world. Increased political polarization and military confrontations, the first war in Europe since the Second World War, repeated economic crises and social disruptions across regions, the rise of global pandemics, and the hazardous impacts of climate change have all occurred since the turn of the Millennium.

The entire planet faces the consequences of these events, but sadly the most vulnerable groups and communities suffer their worst effects. As people look for ways to improve the world, behavioral science has become another vital tool to address these challenges.

This book is about action, not theory. The ideas, processes, resources, and lessons come directly from hands-on experiences in the field. We encourage readers to look for opportunities to implement and test the advice and strategies included in this book to find potential solutions in your own work.

It is with this aspiration in mind that the digital version of this book is available for free in collaboration with the non-profit

Bescy and Bescy Publishing. We hope to make this book widely accessible for practitioners and the general public across hemispheres.

The book is presented in five parts:

- ▶ The introduction outlines the motivation for the book, offers the building blocks that are important to grasp its content, and contextualizes the relevance of conducting non-WEIRD research.
- ▶ The second section showcases a host of behavioral resources and tools that we often use in the field.
- ▶ The third section discusses the current state of development professionals and the roles various types of organizations play.
- ▶ The fourth section highlights case studies in different areas of impact; and,
- ▶ The last section briefly reflects on the future of the field based on the main conclusions of the book.

Before moving to the next chapter, we want to note that we are eager to learn and collaborate. Please do not hesitate to contact us if you would like to work together, discuss ideas, or to ask for general advice. We are happy to help, and you can find more information on how to reach us at the end of this book.

In the next chapter we discuss insights about the “WEIRDness” of our field and what we can do about it.

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2.

We're WEIRD

Steve Wendel, Busara

In Behavioral Science for International Development, we often talk about the challenges of relying on research that comes from **WEIRD** populations: **W**estern, **E**ducated, **I**ndustrialized, **R**ich and **D**emocratic. This raises two important questions:

- ▶ Can we apply that research to other people around the world?
- ▶ What's required to do non-WEIRD research?

In this chapter, we will look at what WEIRD Behavioral Science means for international development and why it matters, along with why this very book and the research we use is WEIRD.

Research by and About a Narrow Slice of People

While international development professionals can use the term WEIRD loosely, we'll use 'WEIRD' descriptively²: most research in behavioral science comes from a narrow slice of people. Specifically, the research our community usually cites and applies comes predominantly from the US, UK, The Netherlands, Germany, and a few other countries.³ It's conducted by researchers at leading universities within these countries and often focuses on well-educated, well-off people within them. Those facts are indisputable, and that's the definition of WEIRDness we'll focus on here.

We can break behavioral research into four parts, each of

² Joe Henrich, who originally coined the term, has used the term both descriptively and as part of a broader model of the development of countries. Henrich argued that WEIRD people have a fundamentally *different culture* than everyone else in the world, and that culture means that WEIRD people both think and act differently: in diverse areas from how people think (for example, Asians think “holistically” versus WEIRDs thinking “reductively”) and relate to one another (non-WEIRDs focus on “family”; WEIRDs focus on “voluntary social groups”). He also argued that the Catholic church drove the cultural divergence by stopping people in Europe from marrying close relatives (thus implying that non-WEIRD groups are in-bred). Like a host of other researchers, I personally disagree with Henrich's argument about the process and culture of WEIRDness. I find it reductionist and even patronizing about different groups of people around the world. But that's not the point here. Instead, we focus on the more common *descriptive* use of the term, as “a narrow subsegment of the world population.”

³ WEIRD is a catchy acronym, but it is itself incomplete: the original usage doesn't address the concentration of research by race for example. We've defined it here to better capture that dimension of the narrowness of BeSci research. See <http://getsyeducated.blogspot.com/2021/06/weird-times-three-reasons-to-stop-using.html> for a critique of the use of the term altogether.

which draws from a narrow (WEIRD) subsegment of the population⁴:

- ▶ **Agenda Setters:** The people that prioritize research and their cultural milieu.
- ▶ **Researchers:** The people who plan and execute the research.
- ▶ **Location:** The place the research is conducted.
- ▶ **Participants:** The people who are researched in that location.

The field is making progress in broadening “the participants” and “the location” of behavioral research but has made little progress in other areas. As researchers from Busara recently discussed in an article in “Nature Human Behavior,”: 17% of the world’s population lives in Africa, but less than 1% of authors and of research participants in top psychology journals are from Africa (Mughogho, et al., 2023). Psychological research is becoming less centralized in the US, but largely through the inclusion of WEIRD Europe (Thalmayer et al., 2021).

This problem isn’t unique to behavioral science. It’s a problem in social science research and the development community more broadly. One study found that development journals are dominated by Global North researchers, and that this has *not significantly improved over the last three decades*.

We Can See There’s a Problem

WEIRDness directly affects our work because it shapes whether we can effectively drive beneficial behavior change in international development. Blindly taking behavioral insights and techniques developed from well-off college students in the US

⁴ Konecni (2010; doi:10.1017/S0140525X10000142) and others have similarly looked at the dimensions of WEIRDness beyond study samples.

can mean that we do ineffective, or simply foolish, things. Most of us in the development community have stories of behavioral science applied blindly and badly.

For example, at Busara our first experiment in the field sought to measure the effect of stress on economic decisions-making. We needed a method of stressing out our participants, so we borrowed a method called the Trier Social Stress Test, a well-known and often used technique originally developed in Germany that uses public speaking to induce stress.

We asked our participants to give a mock interview in front of a panel of experts dressed in white lab coats. In Western settings, this method produces a reliable and large impact on measured stress: speeches are stressful, and white coats conjure an image of an authoritative scientist. There was just one problem: it did not stress our participants at all.

After speaking with our participants and our Kenyan colleagues, we realized speeches are not a major stressor for many Kenyans, since public speaking is more common and expected than in the West. In addition, white lab coats are associated in Kenya with a very different group of people: butchers. This prompted one of our participants to ask, “Why are we being interviewed for a job by butchers?”

This humorous and embarrassing first study helped Busara recognize the importance of local knowledge and context before applying accepted psychological tools like the Trier Social Stress Test. But sometimes, blindly adopting insights from Western studies can have far more serious consequences.

Multiplier and Backfire Effects

In a current project we’re undertaking in Brazil, we are working with an organization that supports victims of domestic violence. There are countless studies in the US and elsewhere of

the power of reminders – emails or text messages that remind people to do something they said they wanted to do – to close the intention-action gap. As we looked at ways to encourage women who reported domestic violence but didn't follow up to receive support, we naturally thought about contacting them via WhatsApp with a reminder.

Thankfully, a Brazilian gender-based violence specialist on the team quickly pointed out how harmful that could be. Many women in dangerous domestic situations in Brazil won't have a *private* WhatsApp number or place to view it without being seen. If their partner sees that communication, it can trigger severe violence. Blindly taking lessons from existing research on using reminders would have meant risking women's lives (Amarante et al., 2021).

By not understanding the local context, behavioral science for international development can become less effective. Academic and Western development research provides ideas about what could work. It does not pinpoint what is likely to actually work in a given context. For that, we need qualitative behavioral science that is embedded in the local context to guide intervention design. We need unbiased, local impact assessments. The first study is an example of that.

Similarly, when we don't understand the local context, we can ignore the broader effects of our work. We usually have a particular outcome we want to achieve and focus our analysis on that chosen outcome. If we don't welcome the participation and insight of local participants, our programs can miss potential multiplier (spillover) and backfire effects.

The project in Brazil shows how that can work. Reminders might increase survey responses (the target outcome) but trigger a disastrous broader impact – increased violence.

Finally, by not including local participants and their aims and desires in a human-centered way, international development can

quickly fall into coercion or condescension. The most ‘effective’ program in the world will falter if people see it as an imposition or insult to their dignity.⁵

Details Matter... but We can't say How Much

If there's one overriding lesson from applied behavioral science, it's that the details matter, including:

- ▶ How we frame decisions.
- ▶ The environment in which an action is taken.
- ▶ The person's existing habits.

In international development, we label those details “context.” I don't know of any established framework in behavioral science for “which details matter,” but across the literature we can identify that culture, personal experience, and the local decision-making environment all affect choices and behavior.⁶

Culture may create early-socialization trait differences in cognitive mechanisms or activate group-specific cognitive mechanisms or responses like social norms. Systematically different personal experiences (with localized tendencies) may activate different cognitive mechanisms or responses. The local decision-making environment may create persistent artifacts that activate different cognitive mechanisms or responses among people in that context.

Each of these tell us that context overall *should* affect impact. Unfortunately, our limited and WEIRD research means that our field is not in a position to make an *empirically rigorous* statement about the impact of WEIRDness. We have some good

⁵ See, for example, <https://tomweinresearch.me/the-dignity-project/>.

⁶ With some authors exploring genetics as well, but we'll stay away from that here.

initial research on how particular mechanisms – like temporal discounting – may vary within and across countries.⁷ This research helps us to imagine how specific interventions *might* have different impacts in different contexts.

However, the number of cross-context impact assessments, linked to cross-context differences in cognitive and behavioral mechanisms, is truly scarce. Our lack of empirical data is compounded by the broader challenges of replicating research in psychology and social science overall.

The best way to quantify the effect of WEIRDness is to change it. We must create behavioral research that isn't WEIRD and compare the types of research and their outcomes to our existing body of knowledge.

Expanding Where We Can

Even without quantified impact, we must address the dominance of WEIRD behavioral science for simple reasons of equity. If you work in the field, you probably believe that behavioral science is an effective tool for doing good in the world. Our own (presumed) effectiveness has an unintended consequence on equity, however.

The research that powers our work is created by WEIRD people who usually already have many resources to support their research, including a high-quality education, well-funded research institutions, and grant funding options. Research is usually directed towards the interests – however benign and well-intentioned – of Western governments and philanthropies.

Let's return to the four aspects we talked about in the beginning: agenda setters, researchers, locations, and research

⁷ See <https://www.nature.com/articles/s41562-022-01392-w> on temporal discounting; see <https://psyarxiv.com/hmnrx/> for a broader study of replication across contexts.

participants. The latter ones are the easiest to address. We start by questioning our own role. Often, we discuss our work as “applying” behavioral science to international development and being “practitioners” who are distinct from academics.

We are forced to be researchers, largely because we can’t trust existing research. Nor can we apply it blindly. Specifically, we are researchers with a unique set of **locations** and **research participants**. If we were to share our results regularly and rigorously, we could help address WEIRDness.

To make this a reality, we must insist on qualitative or quantitative (or both) evaluations. We also can’t stop at target outcomes like uptake, usage, and program impact. We need to be watching and searching for our broader impact on the community and potential unintended consequences. This doesn’t mean that we insist on being the ones who conduct end-to-end research; rather, we must be clear with ourselves and our clients that *someone* must do an assessment.

Then, we should publish our lessons and tools through platforms like the Behavioral Evidence Hub and the Science of Behavior Change.⁸ Anything else is a mirage: our field simply isn’t advanced enough to apply behavioral science in international contexts without a thoughtful evaluation of impact; those evaluations then help us de-WEIRD behavioral science itself.

We can also change how we do design work in the field. Participatory design is a great start. It brings **research participants** into the design process and enables them, partially, to serve as **researchers** themselves. It is incomplete, though. Participation is still constrained and guided by an external agenda often set by WEIRD researchers.

Where possible, we should start with local priorities instead of embarking on a behavior change program that we, or our

⁸ See <https://www.bhub.org/> and <https://measures.scienceofbehaviorchange.org/>

funder, have decided is important. We rarely know what behavioral questions local participants would prioritize and pursue themselves if they had the funding and tools to do so.

Funding is often where we are the most constrained. In behavioral science and international development, the primary **agenda setters** are the funders: philanthropies, governments, and multilateral institutions. Applied behavioral scientists are usually brought in to solve a behavioral problem that a funder has already identified. As a result, our work reinforces the agenda-setting power of funders, primarily from the US and Europe. These agendas are not malicious or badly designed. They are simply from a relatively *narrow* segment of the world's population.⁹

Applied behavioral scientists, as recipients of external funding, should advocate for a broader perspective, especially from the Global South. We should applaud the growth of behavioral science groups *within* funding institutions like the World Bank, UN and IDB, who can help support a broader, inclusive agenda. We should also support funder efforts to *localize* international development: despite all its imperfections, localization is a step in the right direction.¹⁰ We can also learn from Global South researchers by participating in events like the Center for Effective Global Action's Africa Evidence Summit.¹¹

⁹ A somewhat different dynamic occurs in academic behavioral science: there, the researchers can often set the agenda and approach funders. However, the researchers who can do this most effectively are themselves in well-placed, WEIRD contexts.

¹⁰ More information at: <https://www.devex.com/news/opinion-localization-isn-t-working-here-s-why-104178>; and <https://www.brookings.edu/blog/future-development/2022/12/01/rethinking-the-constraints-to-localization-of-foreign-aid/>.

¹¹ For more information see 2023 Africa Evidence Summit, <https://cega.berkeley.edu/event/2023-africa-evidence-summit/>; and, this blog: <https://www.devex.com/news/opinion-localization-isn-t-working-here-s-why-104178>.

Finally, we can look at our hiring and talent development practices and work to broaden the **researchers** and future **agenda setters** of our field. This is an area that Busara (where this author works) has struggled with, as have most international groups in behavioral science. We are based in Kenya but have a broad mix of people from around the world – including a leadership team that is disproportionately from well-off and well-educated populations in the US and Europe (this author included).

As someone who has hired many behavioral scientists over the years, I can say that WEIRDness is almost always an implicit or explicit criteria: either through having a particular educational background (until recently, there simply weren't any non-WEIRD behavioral science programs) or job experience (again, until recently, there weren't jobs at non-WEIRD behavioral science organizations).

At Busara, we've tried to address this by focusing on using skills-based job interviews and building our networks to attract a broader range of behavioral scientists. We've also sought to hire people without a behavioral science background and then train and promote within, to develop skilled non-WEIRD researchers that reflect the places in which we work. These efforts are imperfect and incomplete, but useful. For more on this, see the chapter on Building a Global Team.

This Book is WEIRD¹²

The limitations, challenges and opportunities that we've described about applied Behavioral Science apply to this book as well. The academic research we cite comes primarily from WEIRD contexts, and most of the authors are based in

¹² This section was jointly written by this author (Steve Wendel) and the editors of the book (Zarak Khan and Luis Artavia-Mora).

or trained in WEIRD countries. This is the case even though many are originally from the Global South and have decades of experience living or working there.

In part, this reflects the historical roots of our field: most research and innovation in behavioral science began in the US, UK, Israel and a few other WEIRD contexts. Most people – even those who grew up in the Global South – had little choice but to go to WEIRD contexts to hear about and be trained in the field. Similarly, most of us had the financial resources to embark on that course of study.

One result of this is that most of the current leaders in applied behavioral science around the world are WEIRD, since we effectively had a “head start” in the field. That statement isn’t a judgment. It’s simply an observation. It’s a statement that applies to this author as well. As a result, the people with the most experience are predominately, but not exclusively, WEIRD. Similarly, the authors of this book are predominantly, but not exclusively, WEIRD.

The editors and this author recognize the limitations of our field and, correspondingly, for the book. The editors decided, intentionally and consciously, to focus on those experts and groups with the most experience and impact around the world to share their lessons learned, challenges, and best practices. By sharing their insights, we hope the book helps strengthen and expand our field, encouraging new and better initiatives in new settings.

We also chose not to encourage the book’s WEIRD authors to pull in other people from non-WEIRD contexts as potential co-authors if they did not already have a true partnership and joint contribution. We didn’t want to make the book *appear* to be less WEIRD than it is. This book reflects the strengths and limitations of our field overall.

Conclusion & Key Points

Behavioral Science in international development seeks to improve the lives of people all around the world, and it has seen great successes. The field is also developing and seeks to become more representative of the Global South and address WEIRDness in our research participants, locations, researchers and agenda setting. Early efforts have improved the quality of research by studying the impact of behavioral science efforts on populations in the Global South.

This journey is far from over, however. The field is still lopsided, especially regarding setting the agenda for behavioral science. WEIRDness creates practical challenges that limit the effectiveness of international development, including less effective interventions, unseen unintended impacts, and inequity in the use of this powerful tool.

Once we recognize that the WEIRDness of our field requires us to be both practitioners and researchers, we can play a central role in fixing that problem. Our projects expand the base of research participants and locations, if we share our work. Our projects can support and demonstrate the promise of localization among agenda setters. And as applied researchers, our staff and our perspectives can help broaden the pool of researchers and agenda setters over time.

Yes, our research and our field are both WEIRD – but it needn't stay that way.

Join us for Part II on behavioral tools and frameworks, beginning with learning more about localizing your research.

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Part II:

BEHAVIORAL TOOLS AND FRAMEWORKS

In the landscape of international development, having the right tools at one's disposal can significantly amplify the impact of interventions. With this recognition, we've curated this section to spotlight essential tools, providing practitioners with a tangible roadmap for their application. But beyond the tools themselves, understanding the ethos and approach to the work is paramount.

- ▶ How do we navigate the intricacies of human behavior?
- ▶ How do we ensure that our strategies resonate within specific cultural and socio-economic contexts to create more cost-effective solutions?
- ▶ How do we overcome prototyping and scaling-up challenges? And, above all,
- ▶ How do we ensure that our work uplifts and preserves the dignity of the individuals we serve?

To address these pivotal questions, we also delve into frameworks that serve as compasses for the practitioner's journey. These frameworks offer a structured approach to applying behavioral science insights, ensuring that interventions are not just scientifically robust but also deeply empathetic.

3.

Localizing Research

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Imagine a plan to increase voter participation that has worked repeatedly in the United States and is now being tried in a country with a recent history of political instability and contested elections. Democracy International ran just such a pilot project in Tunisia, with very encouraging results.¹³

However, a similar experiment attempted by the United Nations in another country in the region had to be withdrawn because of political opposition. As behavioral scientists, we know context drives people's preferences and decisions. Countless studies show preferences are not fixed but respond to features of context. And yet, we rarely pay as much attention to context when designing or measuring interventions.

¹³ See the study information at http://democracyinternational.com/media/DI_Tunisia_Study_One-Pager_-_English_v2.pdf

What needs to be localized?

Localization can occur at many levels, including intervention design, stimulus design, and measurement. Many years ago, one of us visited a project for a banking product in rural Andhra Pradesh in South India. Pregnant women (and their families) would make regular monthly deposits that translated into a loan for delivery. The insight behind this intervention was clear: many families did not plan for childbirth related expenses despite having roughly nine months to do so.

What was the fatal flaw in an otherwise well-designed program?

Traditionally, the first child is born at the mother's natal home, leaving the in-laws less invested in childbirth expenses. Further, the natal home could be in a different district or state in which the bank may not have a local branch.

The stimulus presented to people is how behavioral theory becomes practice. Subtle design choices matter. We were reminded of this when we had to re-do a lot of visuals in a goal-setting tool. The visuals had been tested in Tanzania and the team was seeking to implement this in Kenya. In Tanzania, the gender-neutral icons used to show people choosing between various options were interpreted as intended. However, in neighboring Kenya, women interpreted the visuals as applying only to men who were seen as the traditional business gender.

Localized Measurement

Localizing research also means localizing the measurement of responses. One of us was piloting basic behavioral experiments around mental accounting and asked respondents in Kenya what they thought was the localized version of a classic question: Let's say you buy a cinema ticket and when you arrive at the cinema,

you find it has fallen out of your wallet. You have 20 KES in your wallet, which is the price of a ticket.

Do you buy a new ticket?

The research team had focused all their attention on what the “right” price for a cinema ticket was. That is not what people were focusing on at all. They had a more basic question: what is a wallet? Several respondents answered the question in the negative, simply because they didn’t understand the idea of money in a wallet.

Sometimes questions are simply unanswerable to the population. When a group of researchers from the World Bank asked people from Jakarta, Lima, and Nairobi about life-expectancy, expecting to compare responses with the World Bank staff’s expectations, they found many non-responses and hard to interpret data.

The reason?

Many respondents thought answering such a question was bad luck under some circumstances.¹⁴

Other questions require more piloting for precise interpretation. Measuring social norms requires assessing the reference group of the people responding. For a project about women’s workforce participation in Jordan, researchers wanted to know people’s best guesses of the percentage of Jordanian women who worked in their community.

Because communities were so varied (rural to urban, migrants to natives), they had to try several versions of the reference group (“people around here”, “people where you live”, “people like you”) before settling on an acceptable one.

¹⁴ To review the full report, see <https://www.worldbank.org/en/publication/wdr2015>.

Perspective Taking vs. Perspective Getting

How can behavioral scientists get better at developing interventions, designing stimuli and creating surveys that adequately capture local context? We are often told to “put ourselves in others’ shoes” in order to practice “perspective-taking.”¹⁵

On the surface, this seems like good advice. Perspective taking is hard, however, if you have never had the experience of trying to survive extreme poverty, or living in a conflict-ridden area, or suffering from constant ill-health. In fact, perspective taking in such cases may even decrease accuracy.

A recent survey of World Bank professionals showed that even seasoned professionals harbored inaccurate beliefs about the poor, despite being deeply immersed in the contexts of poverty alleviation. They imagined poor individuals differ from themselves in self-perception, though responses by bottom, middle and top thirds of income distribution are similar.

Even development professionals often assume poor individuals are less autonomous, less responsible, less hopeful, and less knowledgeable than they are in fact.

A better way to get accurate information about others is perspective getting, which occurs when we ask people directly. Perspective getting is equally hard. Decades of research in behavioral science show us the pitfalls. People exhibit social desirability bias, particularly when power relationships between the researcher and the respondent exist. We often cannot access genuine reasons and substitute the “easier to answer” question instead.

Because of this, perspective-getting will always remain a work-in-progress.

¹⁵ For more information about this topic see <https://behavioralscientist.org/be-mindwise-perspective-taking-vs-perspective-getting/>.

You May Not Have to Look Far

As a first step, getting feedback from the local population is key. Sometimes, we can find this local population in our own offices.

At the Center for Social and Behavior Change in India, we thought we had a great messaging intervention. We called in our administrative manager to react to it. She was from the local community, was of a similar socio-economic class to our audience, and had family and friends who were community health workers (unlike us behavioral scientists). She immediately pointed out several flaws that were embarrassingly obvious to an insider.

Expanding our idea of who can provide feedback can be an enriching and humbling experience. Lived experience often trumps expertise.

Formal prototyping and early feedback sessions are also critical. Building such sessions into a project may seem like a waste of time, but these sessions often pay rich dividends. Two of us were doing a cash transfer project in Madagascar and took early-stage prototypes of a planning exercise to farmers for feedback. The stimuli involved pictures depicting what local farmers might consider doing with the money they were to receive, which was appropriate for a largely non-literate population.

We got a lot of blank stares from people. Finally, someone came forward and said “these people just don’t look like us. They have too many teeth.” In a place where dental care is often inaccessible, perfect teeth stood out to people and derailed comprehension.

People in the community can also be sources of insights on framing messaging vividly. When doing initial fieldwork for a project on birth spacing in rural Uttar Pradesh in India, researchers were struggling to articulate the ideal spacing between births. A young mother stepped in and provided memorable wisdom:

“Only have the second child when the first one can wipe his or her butt after pooping.”.

While prototyping and getting feedback is an excellent step, ultimately building local talent is the only sustainable way to get better at perspective getting. It is also important, however, that “local” is not just defined as a person from the same location, physically or geographically.

Researchers need local talent who can really understand and appreciate the insights, beliefs, and norms of participants while serving as a bridge between communities and international stakeholders.

Attitude Matters More than Formal Training

In our work, we have found that hiring for empathy, curiosity, and eagerness to learn is critical. A Kenyan manager we hired was from a relatively elite local background and dismissive of people we worked with. However, we fondly remember our local manager in Madagascar. Though they came from a similarly elite background, they brought experience with the context, a willingness to engage with the local communities on their own terms and an adeptness in liaising with policy makers.

Personality and attitude matter. We can teach behavioral science knowledge.

Building local talent is not restricted to one’s own teams. Training local partners is an important part of building local talent, particularly in the Global South, where training in behavioral science is scarce.

In co-creation sessions, we may have to insist that local partners with knowledge of context are present in the room with the behavioral designers. In a good co-design process, local partners will be exposed to the theory and practice of behavioral design,

even getting to the point when they can act as surrogates in the field. That's when we know we have truly built capacity.

We have learned that training needs to be hands-on and in the service of a project, rather than a stand-alone theory training. The training helps people develop their own ideas systematically for a project. After training, they can get immediate feedback by deploying the intervention.

Conclusion & Key Points

Localizing research is about making mistakes over and over. No intervention, measure, stimulus, or scientist is perfect from the start. Feedback, however painful (or funny in hindsight), is a key part of the process. We should all be sharing more stories of failure and mistakes so that we can keep reflecting on this.

We still grapple with many issues in localizing research. For example, the question of informed consent is hard to operationalize. For a project assessing how people in Sierra Leone managed their legal disputes, we interviewed women and girls in private to protect their confidentiality about sensitive concerns, such as divorce. But brothers and fathers kept wandering over to listen. The women's answers weren't their own. We kept having to pause, move, or change our questions.

Another tough challenge involves avoiding social desirability bias, Hawthorne effects or demand effects. In our research, we have found that people think they are doing us a favor by telling us what we would like to hear and that pleasing us will deliver benefits to them or their community. There are various approaches – incentivizing answers when you can, participant-observation, vignettes – but we have yet to find one that is consistently satisfactory. As researchers, we need to better understand power and its effects.

The best test of an intervention or measure is when people

think it has been locally developed. Many years ago, one of us interviewed for a job with Colgate. She had grown up in India using Colgate toothpaste and brushes and had watched countless Colgate ads in Hindi. She was shocked to find out that Colgate was not an Indian company, but an American multinational with headquarters in New York!

We need our own Colgate moments as behavioral scientists. We need the people we serve to feel like the work we are doing originated in their own community, rather than from some far-off institution headquartered in a place the local population has likely never seen.

In the next chapter we present a review of the essential behavioral frameworks used in behavioral science projects.

4.

A Review of Behavioral Frameworks

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In just a decade, applied behavioral science has moved from having very few conceptual tools to an overwhelming array of frameworks and approaches.

Behavioral science frameworks help guide and standardize approaches to problem-solving. They clarify - both within the organization and to partners - the steps required for a project, as well as the potential inputs and outputs, stakeholder responsibilities, and timeframes. Many organizations also construct frameworks to consolidate and categorize the knowledge and expertise they typically bring to bear. The best of them are easy to understand and can help someone grapple with a problem and generate potential solutions.

In this chapter, we walk the reader through the major frameworks and tools that are available for an applied behavioral scientist in international development. There are distinct ‘types’

of tools designed for different purposes, and within each type of tool there are remarkable similarities. When we organize these tools by purpose, the overwhelming and messy literature becomes more manageable, understandable and implementable.

The Process of Applying Behavioral Science

Around the world, consensus has emerged on the process of applying behavioral science in the field. While each group has its own terminology and emphasis, the core process is remarkably similar. Specifically, behavioral problem-solving looks like this five-step process:

- ▶ **Step 1:** Align on behaviors of interest.
- ▶ **Step 2:** Understand and diagnose their causes.
- ▶ **Step 3:** Design interventions to increase or decrease the targeted behavior.
- ▶ **Step 4:** Assess, impact and iterate.
- ▶ **Step 5:** Share and scale the insights.

Change a few of the words, and this is simply a general framework for problem solving. What makes this process *applied behavioral science*, instead of generic problem-solving or design thinking, is three-fold:

- ▶ It builds upon a specific body of empirical literature in behavioral science that encompasses the fields of behavioral economics, cognitive psychology, experimental economics, social psychology, anthropology, sociology and others. This literature provides empirical evidence on decision-making and behavioral mechanisms, and a framework for thinking about behavior as contextual, malleable, and social. That is essential to Step 2, “*Understand and Diagnose Causes.*”

- ▶ It focuses on *changing behavior*, instead of merely *understanding* behavior or *changing attitudes, beliefs, or knowledge*. The discipline only focuses on the extent that we can empirically drive behavior change. That is a core part of Step 3, the “*Design*” stage and enriches Step 1, the “*Align*” stage.
- ▶ Behavioral science is committed to empirical revalidation and testing. Behavior change is hard, many well-meaning efforts have no effect or can backfire; hence, we need to “*Assess and Iterate*” and be thoughtful in our efforts to “*Share and Scale*.”

Other differences from a standard problem-solving approach occur in the specific techniques used at each step. For example, behavioral scientists place additional emphasis on defining and understanding key behaviors. We usually include some sort of discovery work to situate key behaviors in a broader system, rather than treating them as independent. In the assessment phase, behavioral practitioners might use prototypes, randomized controlled trials (RCTs), and other techniques and methods to better understand the effects of a new idea.

The First Tool in the Toolkit: Process Frameworks

Given this understanding of what behavioral scientists do, the first resource that a behavioral scientist has in their toolkit is a process framework for behavioral problem-solving. Almost every other tool is designed to help *within* one of the stages of the process.

The following are example process frameworks for behavioral problem-solving:

- ▶ **AUDAS** from Busara. Designed for international development and global impact, the acronym stands for **A**lign, **U**nderstand, **D**esign, **A**ssess, and **S**hare.
- ▶ **DDDT** from Ideas42. Designed for consulting, the acronym stands for **D**efine, **D**iagnose, **D**esign, **T**est.¹⁶
- ▶ **3B** from Irrational Lab. Focused on design and created for product development, the acronym stands for **B**ehavior, **B**arriers, and **B**enefits.¹⁷
- ▶ **IDP** from Matt Wallaert. Designed for product development, this tool focuses on behavioral strategy, insights, design, and impact evaluation. The acronym stands for **I**ntervention **D**esign **P**rocess (Wallaert, 2019).¹⁸
- ▶ **BASIC** from OECD. This tool is focused on policy, and the acronym stands for **B**ehavior, **A**nalysis, **S**trategy, **I**ntervention, and **C**hange (Hansen, 2018).
- ▶ **TESTS** from The Behavioral Insights Team. This tool is a general framework with a particular focus on policy. The acronym stands for **T**arget, **E**xplore, **S**olution, **T**rial, **S**cale.¹⁹
- ▶ **DECIDE**, from Designing for Behavior Change 2nd Edition. Designed for applied research and product development, the acronym stands for **D**efine the Problem, **E**xplore the Context, **C**raft the Intervention, **I**mplement the Solution, **D**etermine the Impact, **E**valuate Next Steps (Wendel, 2020).

¹⁶ More information available at: https://www.ideas42.org/wp-content/uploads/2019/10/I42-1152_ChangingBehaviorPaper_3-FINAL.pdf

¹⁷ More information available at: <https://irrationalabs.com/content/uploads/2022/07/3B-Framework-2022.pdf>

¹⁸ Available also at: <https://mattwallaert.com/startattheend/>

¹⁹ More information available at: <https://www.bi.team/wp-content/uploads/2022/11/BIT-Handbook-How-to-run-simple-BI-projects.pdf>

The words differ and the acronyms are cute, but in the end they all provide a blueprint for behavioral problem solving. That might lead you to wonder why we need a framework at all. Most mature organizations will have process frameworks for how they solve a problem that are specific to their field and the nuances of the problems they want to solve. Since our community tends to solve similar problems, our frameworks look alike.

Frameworks provide a useful starting point for planning a project and understanding the steps involved. Even with them, organizations still rely on the expertise of practitioners to guide the process and adapt as roadblocks arise. As the authors of such frameworks ourselves, we can honestly say that it doesn't matter which framework you use, as long as you stick with it, are clear on what it does, and adapt it to your needs.

Tools to Target or Align the Behavioral Process

Now that we have a sense of the process frameworks, we'll review the other tools in an applied behavioral science toolkit. At each step in the process, the practitioner community has developed a set of tools that help us execute on these stages effectively. Different groups of practitioners have different tools that define their 'special contribution' within this problem-solving framework, but the framework and many aspects of the tools are quite similar. We'll start with the first stage, which at Busara we call "Align."

In this stage, the behavioral community draws heavily on long-standing practice in the design field of a design brief. We tend to call it a 'behavioral brief', but the idea is the same: write down what it is that we're trying to accomplish (or that a client is asking us to accomplish) and any key constraints or considerations. This is done before any actual behavioral design work

is conducted and is an essential tool to help stakeholders get on the same page.

Here are some specific tools that practitioners use:

- ▶ Busara’s **Applied Behavioral Science Brief** includes the scope of the project, the target audience and behavior change, and key details of who does what and how on the project. For example, a recent project at Busara sought to encourage Bajan parents (the agent of change) to use local produce (the specific behavior change) in the meals they prepare for their kids (the target population) to order to reduce non-communicable diseases like diabetes and benefit the local economy (real world outcome). In our briefs we then add notes on planned measurement approaches (e.g., an incentive aligned RCT) and the constraints (e.g. digital-only contact via WhatsApp).
- ▶ Matt Wallaert’s **Behavioral Strategy** process, in particular his “behavioral statement”: *When [population] wants to [motivation], and they [limitations], they will [behavior] (as measured by [data]).*
- ▶ Designing for Behavior Change’s **Actor-Action-Outcome**. This tool works by identifying the Actor (who is going to do something differently to drive that outcome), the action (what is the ‘big thing’ they do differently?), and the outcome (the observable real-world impact that the process should deliver).
- ▶ Rajesh Nerlikar and Prodigy’s **A Behavioral Hypothesis**. This tool helps identify how helping an actor start or stop taking a specific action will create specific outcomes.
- ▶ The BIT’s **Target Statements**, which help drive a specific policy objective.

When there are multiple possible target behaviors (or policies), groups like the OECD and BIT have prioritization tools to help identify which to pursue. For the OECD, their “priority filter questionnaire”, asks about each behavior in terms of its Importance, Ethics, Impact, Feasibility, Data Access, and Frequency. The BIT accomplishes the same goal through a pre-mortem on impact and feasibility.

Tools to Understand and Diagnose Causes

Once we understand what we want to accomplish, we start by trying to understand why the desired behavior isn’t happening already. What is stopping the person from taking a positive action of interest, or why are they currently taking a negative action? Practitioners (and developers of frameworks) sometimes jump straight to the next stage: what can we as applied behavioral scientists *do* to change that behavior – i.e., how do we design interventions? However, they are conceptually distinct, and we have distinct tools designed for each stage.

Tools to understand and diagnose causes can be divided into ‘macro models’ and ‘micro-obstacle tools. **Macro-models** tend to look at behavior in a single, consistent way, assuming that all behavior follows a similar trajectory. Most of these macro-models are actually long-standing models of human behavior from across the social sciences, and not unique to behavioral science at all. For example:

- ▶ **Self-Determination Theory** (Deci and Ryan, 2012)
- ▶ **Health Belief Model (HBM)** from US Public Health Service.
- ▶ **Theory of Planned Behavior (TPB)** (Ajzen, 1991).
- ▶ **Transtheoretical Model, aka Stages of Change** (Prochaska, 1997).

Micro-obstacle tools take a different approach by helping us find the micro-causes of a behavior via a four-step process:

- ▶ **Step 1:** Gather data about what is currently occurring.
- ▶ **Step 2:** Map out the specific sequence of actions.
- ▶ **Step 3:** Find which step(s) are the problem.
- ▶ **Step 4:** Identify the behavioral cause of that problem.

These tools then build on a key insight in behavioral science: small obstacles in a moment can get in the way of action, even if the macro-environment is conducive. Thus, they aim to understand the specific moment and situation in which things go awry.²⁰

The first step, mapping the sequence, usually goes under the title of “behavioral maps.” Here are a few examples you can use:

- ▶ **Behavioral Mapping Case Study & Cheat Sheet** from the Center for Advanced Hindsight at Duke University.²¹
- ▶ **Behavioral Profile** from the Manoff Group.²²
- ▶ **Target Behavioral Funnel** from Ingenious Behavior.²³
- ▶ **Key Behavior Workshops** are sessions with stakeholders to understand the decision-making environment and identify candidate behaviors for interventions.
- ▶ The term “**journey map**” is also used to describe this process (e.g. BIT TESTS framework), though that can be a bit confusing. The term is already used in the

²⁰ See the worksheet at <https://www.bi.team/wp-content/uploads/2022/11/BIT-TESTS-worksheets.pdf>

²¹ Access the case study at advanced-hindsight.com/blog/introducing-the-behavioral-mapping-case-study-cheat-sheet/

²² Access here: thinkbigonline.org/behavior_profile_p

²³ See the Behavioral Profile explanation at www.behavioraldesignmodels.com/about

design community for a more general diagram that shows the sequence of stages a person goes through along a process of interest.

The step to identify the problematic moment can be either qualitative or quantitative by examining the data that shows where in the funnel people drop off, or simply asking them which micro-behavior is a problem. Once one has prioritized a key moment for intervention, then there are **obstacle diagnosis** tools that help the practitioner determine *why* that moment is a problem.

Here are a few additional tools you can use:

- ▶ **COM-B: Capability + Opportunity + Motivation -> Behavior.** This is the most widely used framework for identifying obstacles (Michie, Van Stralen, & West, 2011).²⁴
- ▶ **CREATE: Cue, Reaction, Evaluation, Ability, Timing, Experience.**²⁵
- ▶ **Barriers and Levers:** Busara's Framework for identifying specific types of psychological and environmental factors that block individuals, and the resources (levers) that can encourage it.
- ▶ The OECD's **ABCD Framework: Attention, Belief Formation, Choice and Determination.**²⁶
- ▶ **Pressure Mapping**, which is part of the Intervention Design Process from Matt Wallaert.²⁷

Applying these frameworks to a target behavior allows us to

²⁴ Explore the Barrier Identification Tool at <https://www.bitbarriertool.com>

²⁵ Free workbook available at <http://behavioraltechnology.co/workbook>

²⁶ Framework document available at <https://www.oecd.org/gov/regulatory-policy/BASIC-Toolkit-Draft-for-Consultation.pdf>

²⁷ Available at <https://mattwallaert.com/startattheend/>

identify gaps that might prevent someone from taking action. For example, the COM-B model could reveal that while a person has the Capability (they know what to do) and Opportunity (they have time to do it), they lack sufficient Motivation (the desire to take action) and the desired behavior does not occur.

These same tools that help the practitioner determine *why* a particular moment is a problem can also be used to audit a potential user experience and identify potential behavioral obstacles. This process, not surprisingly, is known as a “behavioral audit.”

Two frameworks in particular – the BIT TESTS framework and Busara’s AUDAS – add an additional “preparation step” to the Understand phase before developing a behavioral map. These two frameworks explicitly call upon the practitioner to gather applicable data and conduct field research as needed with the population of interest.

Both the BIT and Busara work in the development sector, where the context of applied behavior varies widely from project to project (and location to location); hence the explicit step of understanding that context. The specific tools used in this phase are the standard tools of field research: surveys, in depth interviews, analysis of existing administrative data, and more. However, it’s important to remember that all frameworks incorporate this process.

Tools to Design Interventions

With an understanding of *why* there is a behavioral problem, next up we have tools to select interventions. Obstacle diagnosis tools often come with an intervention-selection tool as part of the package, for example:

- **The Behavior Change Wheel**, linked to COM-B (Michie, Van Stralen, & West, 2011).

- ▶ **Busara's mapping of interventions** to specific structural and psychological Barriers and Levers.
- ▶ The table of **CREATE Interventions**, linked to CREATE (Wendel, 2020).
- ▶ The menu of strategies within the OECD's BASIC framework (mapped to the same ABCD obstacle model used in the prior step)²⁸

What makes those tools special is that they are directly tied to specific behavioral barriers. A variety of other tools are available that more generally seek to support a new behavior but aren't directly tied to a specific cause. These more general, less targeted, tools include:

- ▶ **EAST:** Easy, Attractive, Social, Timely. From the UK BIT (BIT, 2014).
- ▶ **MAP:** Motivation – Ability – Prompt, the Fogg Behavior Model from BJ Fogg (Fogg, 2019).
- ▶ **CAR:** Cue – Action – Reward (a model of habit formation)
- ▶ **Hook Model:** Trigger Action, provide a (variable) reward, then increase investment (Eyal, 2014).²⁹

No matter what framework one uses, often there will be too many intervention ideas than can be implemented in practice. The Designing for Behavior Change, the BIT's TESTS framework, and Busara's AUDAS each provide tools to prioritize the intervention ideas. These prioritization tools are also generally quite similar: they ask a series of questions to assess which interventions are likely to be most cost-effective and ethical.

²⁸ Available at: <https://www.oecd.org/gov/regulatory-policy/BASIC-Toolkit-Draft-for-Consultation.pdf>

²⁹ Further explanation available at: <https://www.nirandfar.com/how-to-manufacture-desire/>

Tools to Assess Impact

Practically speaking, *behavioral science* practitioners assess impact just like any other practitioner would, with techniques ranging from randomized control trials (experimental) to difference in difference models (statistical) to baseline-endline comparisons (algebraic) to embedded observation and process tracing (qualitative).

A wealth of literature exists on these research methods, from books on the practicalities of field experiments (e.g., Gerber and Green) and A/B testing, to more theoretically-based analyses of causality (e.g. Pearl).

Similarly, a wealth of tools help practitioners use the following:

- ▶ **Tools to design** and implement experiments. These are medium-specific, and good professional packages can be found for email experiments, website testing, app testing, and non-digital field work.
- ▶ **Tools to analyze** the results of experiments or other causal data, often general-purpose statistical tools like R, Python, Stata, and such, and their various packages. For example, at Busara, there is an internal code repository to help with the impact assessment process - with samples for R and Stata.

These techniques are rarely specialized for “behavioral interventions” versus other types of impact assessment. The field is well developed and has existed for much longer than applied behavioral science. Behavioral science practitioners rightfully draw upon these existing lessons and tools.

Tools to Iterate, Share and Scale Up

Guidance on how, practically, to share and scale up behavioral science research is largely underdeveloped: there are few ready to use tools for this process. Sharing and Scaling is often just a command in current frameworks, as in: “you should really do this”. The BIT provides some practical guidance with its SCALE acronym: Sponsorship, Cost/benefit, Accountability, Logistics, Evidence. Overall however, this is an area that would benefit from new thinking and codification.

A recent addition to the scholarship in this space, John List’s *The Voltage Effect* can offer advice and serve as a starting point for practitioners. Some considerations when scaling include:

► False Positives

A false positive is an initial signal that an idea is successful, but that turns out to be incorrect when scaled up. For example, a small-scale trial of a new program might show promising results, but when the program is implemented at a larger scale, the results might not hold up. To avoid false positives, replicate successful trials multiple times to ensure that the results are robust and not driven by chance.

► Representativeness of the Population

The success of an idea often depends on the specific population it is designed for. For example, a program that works well for wealthy families might not work as well for lower-income households. To avoid scaling an idea that won’t work for everyone, ensure that test groups at smaller scales reflect the larger population you’re aiming to reach, and adjust your expectations or approach if necessary.

► **Representativeness of the Situation**

The success of an idea can depend on specific circumstances that might not be easily replicable at larger scales. For example, a program that relies heavily on a few talented individuals might not be scalable if those individuals are difficult to find or retain. To avoid this, identify the core drivers of success and make sure they can be replicated in different situations and with different people.

► **Unintended consequences and misaligned incentives**

An idea may be piloted successfully but fail to scale due to misaligned incentives and unintended consequences. Consider, for example, an organization that has a customer experience team, a collections team, and a leader tasked with increasing revenue. The customer experience team may recognize that the collections experience is quite negative for customers. In working to improve it, a behavioral scientist develops a successful intervention to reduce the number of customers going into debt collection. It may be a success for two parties, but the leader in charge of revenue may shelve the idea because it impacts the money collected via fees. To mitigate this issue, consider the unintended consequences of scaling and work to ensure alignment across the system as part of a pilot.

Integrative Toolkits

Finally, we can return to our initial discussion. What does an applied behavioral scientist do? Here, we've structured the discussion using Busara's AUDAS framework, and the individual

tools that fit into each stage of the process of behavioral problem solving. Some authors have developed integrated toolkits that try to provide an end-to-end package of the tools needed at each step of the way. Here are a few of those:

- ▶ **The Behavioral Insights Team’s TESTS handbook** and its worksheets.³⁰
- ▶ *Designing for Behavior Change*, and the workbook that goes with it, was an effort to create such an integrative toolkit.³¹
- ▶ Matt Wallaert’s book, *Start at the End*, similarly provides an integrative toolkit, unified by his Intervention Design Process.
- ▶ **Behav’s Behavior Kit**.³²
- ▶ **The Brains, Behavior and Design Toolkit**.³³

Conclusion & Key Points

Throughout this chapter, we’ve discussed applied behavioral science as problem-solving. That is the dominant paradigm, and there are a wealth of tools and frameworks available for behavioral problem solving.

However, some alternatives are starting to appear. The most exciting work is likely on systemic or structural behavioral science. We can apply behavioral insights to better understand a dynamic system of interacting individuals, organizations and resources, and the opportunities for change.

³⁰ Further explanation available at https://www.bi.team/publications/testsguide/#sq_h939705cy0; and workbook available at <https://www.bi.team/wp-content/uploads/2022/11/BIT-TESTS-worksheets.pdf>

³¹ Available at: <https://www.behavioraltechnology.co/workbook/>

³² Available at <https://behaviorkit.com>

³³ Available at <http://www.brainsbehavioranddesign.com/kit.html>

The furthest along of these comes from Ruth Schmidt at the Institute of Design, and her work on “choice infrastructure” and “behavioral brittleness.” At this point, the non-problem-solving approaches to applied behavioral science are still in their infancy. We hope that soon enough our community will develop and share tools and resources like the well-established problem-solving resources described here.

In the next chapter we review the importance of qualitative data in behavioral science.

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5.

Importance of Qualitative Insights

Sania Ashraf, Rare & The University of Pennsylvania

Designing successful high impact interventions that achieve desired outcomes requires understanding the goals of the people involved. What are their motives? How do they feel about the topic of interest? What words are they using to talk about it? How can we include their perspectives to inform the next steps?

This is where qualitative research becomes useful. Qualitative research tools are used to gain insight into behaviors, values, motivations, attitudes and contextual factors like culture and lifestyle. It helps behavioral scientists better understand the subjective experience of the target population, generate depth, and provide nuance in your understanding of the topic. I will draw from my work and experience in global research studies to highlight the use of qualitative research methods.

Which qualitative research tools are relevant to you?

There are many research tools that will help you achieve your goals. What you pick depends on the goals, the nature of your questions, and the time you have available. You can zoom out and include dissimilar study subjects to get multi-faceted insights on the same topic. You can ask similar people questions to get their personal life stories and perspectives. You can conduct qualitative research individually, with a group of people, or even just observe subjects in their natural setting to learn about a topic.

Based on your area of research, *both* qualitative and quantitative methods may be relevant. We often conduct qualitative research at the beginning of a study to help scope the relevant dimensions of a study and to understand best practices to engage the actors. Qualitative research also adds depth to our understanding of the literature as it applies to the study population. These insights can inform how to ask specific questions and sections included in a large-scale survey. You might review your qualitative findings to see if they align with your survey or quantitative research.

It is crucial you spend time setting up your research objectives. They should include your target behavior (if applicable), the target audience and context. These will influence the logistics of conducting research.

In-depth Interviews

You can think of interviews as structured or semi-structured conversations that are conducted in an open-ended way to discover details about a topic from relevant participants. The interviewer can probe ‘in-depth’ by asking the respondent questions like:

- ▶ Do you have an example of this?
- ▶ Could you clarify what they mean when you say that?
- ▶ What challenges do you face when doing that?
- ▶ What advice would you have for this?

This technique is best used to ask about personal thoughts, experiences, and beliefs. Depending on the topic, you might need a private place to have a confidential conversation. You should develop an interview or a moderator guide which details the flow of the conversation, uses open-ended questions, prompts for deeper questions that delve into specific areas of interest. As an interviewer, you will have the flexibility to adjust the interview style in response to the needs of the participant.

As an interviewer, you will need to be open-minded, patient, and a good listener. Consider taking a few minutes in the beginning to use ice-breaker questions and build rapport with the participant so that they feel comfortable sharing details with you. Be aware of your facial and body language to avoid leading the participant in any particular direction. Always end with an invitation to share anything that was not covered in the conversation and leave your contact information for them to reach you later if needed.

It depends on the topic, but it is common to reach ‘saturation’, or a level where you do not uncover new information from additional interviews. If you are hearing similar things in your interviews, consider broadening your sample to include those who might have divergent views to understand the factors underlying the difference.

An Example of an Interview

Our team wanted to understand the barriers faced by married women when making household financial decisions in rural communities in Bihar, India. Acknowledging the sensitive

nature and underlying gender norms guiding the behavior, we conducted one-on-one interviews with the women spanning a wide age range (18-45) in several villages.

We got consent from the participants and relevant family members, including their husband or their in-laws. We elaborated that the conversation would be confidential, anonymized and not shared outside of the research team.

The interviews were conducted in a private room by a female researcher fluent in the local language. Insight from these interviews included sociocultural factors like their and their spouse's education, age at marriage, and traditional household and gender roles where women handled small amounts of money.

What You'll Need to Conduct Interviews

- ▶ A semi-structured interview guide.
- ▶ 40-60 minutes of time.
- ▶ A trained interviewer who can navigate the conversation.
- ▶ An audio recording device.
- ▶ A notepad to take notes.
- ▶ Access to a private setting.

Interview Recommendations

Consider including multiple interviewers on your team. During the analysis phase, it is helpful to get everyone who conducted the interviews together for a discussion of key findings and insights. This potentially surfaces differences based on the participant, potential methodological variations, or even the setting in which data was collected. The team can then analyze whether

the findings align or meaningfully diverge, regardless of who conducted the interview.

Focus Group Discussions

A focus group is a moderated small group discussion among participants with similar backgrounds or experiences. The aim of focus groups is to surface and discuss individual and group level experiences, attitudes, beliefs, or opinions that people can reach consensus on. Given its group nature, topics that are sensitive or personal need to be limited or asked in a way that makes the participants feel comfortable.

Focus groups are an important tool when you want feedback on a certain topic, and when you feel you need to understand the “why” and the “how” of specific components. You want to allow for participants to differ in their feedback and generate a breadth of knowledge. If relevant, you can ask the participants to vote or rank the emerging themes to decide on the next steps. You’ll find UX or user experience research that aims to learn what users of a platform or product require or want often will use focus groups.

Focus groups are useful if you want to:

- ▶ Define a problem.
- ▶ Pre-test ideas to sharpen research questions.
- ▶ Identify program strengths and weaknesses.
- ▶ Generate new ideas.
- ▶ Interpret quantitative findings in mixed method studies.

It is crucial to engage skilled facilitators. Group dynamics may overpower opinions of subdued participants. Setting group rules and giving everyone an opportunity to speak up allows varying opinions to emerge. A good facilitator aims to appear

neutral and be aware of biases they might introduce, which can drastically influence the trajectory of discussions.

An Example of a Focus Group

To understand perceptions of social distancing and usage of public toilet facilities in Tamil Nadu, we conducted focus group discussions among men and women in urban cities. We included same-sex and similar aged participants in group discussions to better understand concerns of using shared facilities during high perceived risk of COVID-19. Insights included concerns about hygiene, risk of transmission and fear of spreading disease to older family members. In ranking exercises, we found that fear of spreading diseases to other family members restricted their overall use of enclosed public toilets.

What You'll Need to Conduct Focus Groups

- ▶ A semi-structured discussion guide.
- ▶ 60-90 minutes of time.
- ▶ A trained interviewer who can navigate the discussion.
- ▶ An audio recording device.
- ▶ A notetaker to take notes.
- ▶ At least 8-10 participants.

Observations

There are insights that are subject to biases such as social desirability bias, where respondents answer with what they think is the “right” answer instead of truthfully. For example, if you ask teenagers if they smoke, they might under-report their behavior. Or if you ask people if they take their medicine as prescribed,

they might misreport their actual behavior. In settings like these where self-reported metrics are unreliable, observations as a qualitative tool are available. You can look, see, and collect data in a semi-structured or even unstructured manner, depending on the research topic.

Observations allow researchers to gather an unbiased account and understanding of a phenomenon or a behavior in its naturalistic setting. This “real life” setting allows study participants to be and act as they would outside of formal research settings. You can use these at the initial phases of research, where you generate exploratory insights by observing relevant actors, relevant factors, works, expressions and situations. You can think about using this behavior when studying undesirable behaviors.

This method is cheap and requires minimal training. The cons include the fact that it’s time consuming. The observer can introduce a high degree of bias depending on the object of observation. It is hard to get an accurate understanding of a balanced demographic sample. However, it allows researchers to gather relevant data to inform other research methods used later.

What is the Role of Biases in Qualitative Research?

The most important tool and source of bias in qualitative research is the researcher themselves. We all have different socio-cultural backgrounds, life experiences, and perspectives. Often the research participants may hold beliefs and attitudes that are distinctly different from the researchers collecting the data. Because of the inter-personal modality of the data collection, it is important to ensure that the researchers appear neutral during the interaction.

Critically examine your own role in the research setting and the potential biases you might hold by doing the following:

- ▶ Reflect how biases might affect the sample selection, location, study procedures, and even data interpretation.
- ▶ Consider drawing from aspects related to diversity, equity, and inclusion (DEI).
- ▶ When collecting data, reflect on and understand your identity and position within the context.
- ▶ Respect local customs and culture, keeping in mind that acknowledging them will enhance trust and improve any conversation and knowledge sharing from the participants.
- ▶ If possible, ask local stakeholders to provide input on your research approach and tools. This can address implicit assumptions and create space for additional dimensions of inquiry.

It is also important to cultivate empathy and be curious when probing for further details. Often in qualitative research, you'll find the need to be flexible and creative, especially when conversations diverge into different paths. Work with co-workers who can give you feedback and reflect on questions where you might have led or guided the participant's responses.

Make Time to Test Your Tools

We construct tools and guides sitting in our office, but often in field experiments or data collection, we find things need to change. Make time to pilot or test your tools with test subjects who are like your study population *before* taking it to the field. Usability testing is a humbling exercise where you can improve the flow of the components and address key issues before you expose them to the larger audience. Consider drawing on the expertise of your field research

assistants to improve your tools, because qualitative tools are strengthened by contextualizing it to the target population.

Use of Qualitative Research Tools in Human Centric Design

You may have heard about *human centric design*. It is an approach that uses an iterative pathway to design programs, applications, service, or an intervention. This is a technique used across multiple fields, including technology and social sciences. It optimizes benefits to the user or the human being interacting with it by including their perspectives and experiences early in the development stage.

This user-oriented approach uses qualitative methods, especially in-depth interviews and focus group discussions to surface critical insights of using a product or interacting with an intervention. The researcher invites the respondent to be as honest and transparent as possible in their feedback to improve the product across dimensions, such as ease of use, acceptability of content, and feasibility of delivery.

Researchers also use non-verbal data collection methods, such as observing the user interact with the product to record the human role in the interaction areas of difficulty or reactions to the content. In my global health work, qualitative approaches were used to study how young mothers used micronutrient packets to supplement complementary food. It allowed us to revise the design of the sachet, the frequency of use, and include behavioral suggestions to prevent older children from eating it as well. It helped us design a contextually appropriate nutrition intervention that optimized both the mother's and the child's feeding experience. Keep the qualitative tools in mind while designing your intervention!

Conclusion & Key Points

In this chapter, we have introduced you to some key tools you can use to conduct qualitative research. However, there are many other research methods that use participatory research methods and relevant insights from anthropology and sociology. Many researchers now value mixed method research strategies where qualitative research is done as the first phase to inform key aspects of the study, followed by surveys or quantitative experimentation, and then confirmed through qualitative research again. Think through your research question and cultivate your curiosity to incorporate impactful qualitative research.

In the next chapter we review the use, implementation and evaluation of quantitative methods for behavioral interventions with experts from the World Bank eMBed group.

6.

Quantitative Data and Methods for Behavioral Interventions

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Most development institutions have a strong focus on evidence-based policy making. This means their work combines an operational focus (designing and implementing policies) with a knowledge focus. Data, research and innovation, and policy evaluation are at the core of these activities. As the applied behavioral science unit at the World Bank, the Mind, Behavior, and Development Unit (eMBeD) is part of an institution that uses evidence for effective policy design and implementation with the goals to end extreme poverty and promote shared prosperity by increasing the incomes of those at the bottom of the income distribution.

Data use and evidence generation in an institution that has a strong policy focus for and in development face similar

challenges, whether for behavioral interventions or standard structural ones. We work in data scarce-environments, where capacity and funding for research are sometimes limited, and where the needs and demands from government action are manifold.

While we have a strong focus on rigorous approaches, compromises are needed. Sometimes this comes in the form of the amount of exploratory research we can conduct to inform a project, or the extent of monitoring and impact measurement we can undertake. Hence, we see the use and production of data for behavioral science in development to be one focused on practicable, context-sensitive solutions that can help policymakers refine an existing policy, design a new one, counteract their own biases, or a combination of them.

Deliberately Digging for Diverse Data

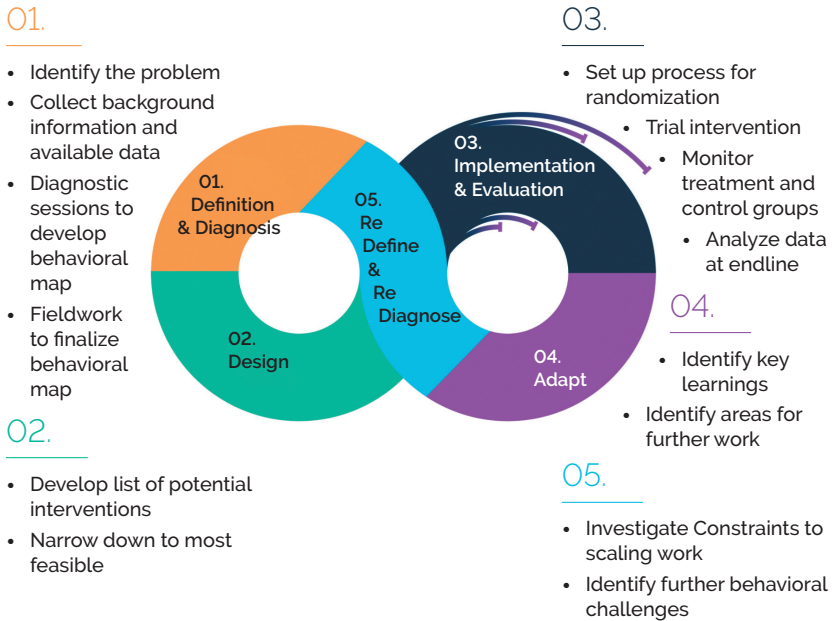
While the focus of this chapter is quantitative methods and quantitative data, these are not a replacement of a well-rounded approach to evidence, ideally one that combines qualitative data, documentary data, and other sources of information that allow for a systematic and well-informed approach to a problem, particularly when approaching it for the first time.³⁴

As noted by Datta and Mullainathan (2014), one of the first ways behavioral economics (and behavioral science more broadly) can affect program design is by changing how we diagnose problems. Switching the focus from the desired outcome to look at how that outcome happens shifts the attention from data on outcome trends – and the implicit knowledge we have on why things are not happening – to an exploratory

³⁴ For more information about our approach to behavioral diagnostics, see Castaneda et al. (2022).

Figure 1. A systematic approach for applied behavioral science projects

Implementing a Behavioral approach



context-sensitive diagnosis that relies on multiple sources of data. We must diagnose, not suppose.

This approach can include expanding from descriptive outcome data (e.g., take-up of a program), to data on costs, processes, times, profiles of potential beneficiaries, and more. Here data analysis is used to:

- ▶ Define the behavior of interest.
- ▶ Measure its prevalence.
- ▶ Describe the context in which it takes place (e.g., target population, region of interest).

- ▶ Explore factors associated with the attainment of the behavior.
- ▶ Start identifying possible profiles or personas that can represent the different experiences and barriers different individuals or groups might experience.³⁵

The outcome of these diagnostics can take different forms. We often construct a behavior map for the target group, including environmental, social, personal, and economic factors influencing the target behavior.³⁶

We try to use all the data relevant to the respective diagnostic, whether it relates to core behavioral or contextual elements. In the diagnostic phase, data analysis is a tool for exploring, describing, and (when we have them) testing hypotheses using either existing or new quantifiable data (e.g., administrative data, surveys, census data).

We approach issues, depending on their complexity and our previous experience with them, through a combination of unsupervised and supervised data analyzes. This analysis is exploratory and conducted with few pre-existing labels or rules. This decreases the impact of our own mental models and biases and allows data to reveal unexpected patterns and insights. Supervised analyzes, in contrast, are focused on predicting the outcome or target behavior. Here, data is used to confirm/reject pre-existing hypotheses or expected patterns of behavior.

For example, we wanted to expand our knowledge of fathers' priorities and worries for an intervention in India intended to increase the involvement of fathers in child nutrition. While

³⁵ Personas are concrete and fictional representations of target groups as rounded human beings that designers can use to characterize target populations and develop, and design tailored products or services. For more discussion on personas use in behavioral science see Wendel (2020).

³⁶ For examples of behavioral process maps, see our website: <https://www.worldbank.org/en/programs/embed/our-work>

we had some qualitative data, this was not sufficient. We used artificial intelligence to assess what was said online, and by whom, regarding child nutrition. The questions were broad: what did mothers and fathers search for, care and talk about? What patterns could be identified? These data were then used to create a series of profiles, who then were targeted via an online intervention (Kundu et al., forthcoming) where specific messages focused on three personas of parents (traditional providers, uninformed, modern caregivers).³⁷

Similarly, in Iraq, for a social cohesion focused project, we developed personas based on the data gathered on target populations during the mixed-methods behavioral diagnostic. Here, we targeted policymakers' biases by 'humanizing' the data and analysis so that we could better understand the constraints faced by different individuals. We could then discuss potential policy solutions that would enhance wellbeing in the region.

Supervised analyzes are useful when both the data to be collected and analyzed are aimed at revealing systemic challenges, such as pressure points or frictions that are pre-identified as coming from the setup of a system or program. This, for example, has been the focus of much of the work our team has led in the tax space, where we have learned that most barriers result from the tax filing systems. In these cases, looking at both the processes and the administrative data available (e.g. whether some people or businesses file on time) allow for a better diagnostic of the different processes (Dalton et al., 2021).

³⁷ For more detail see: <https://blogs.worldbank.org/digital-development/leveraging-artificial-intelligence-reach-fathers-and-support-child-nutrition>

Maximizing Impact in the Design Stage

Quantitative tools and methods can better tailor the treatment, tenor, and timing of behavioral interventions to the target group or certain contextual and environmental factors. By utilizing data, we can:

- ▶ Model intervention designs so they can better reach the intended target group.
- ▶ Test multiple intervention ideas and be iterative to arrive at the best one (A/B test or prototyping).
- ▶ Include both delivery and evaluation features as part of the intervention design.

Following a diagnostic, we focus on designing interventions that are aware of the individuals in our target group.

For example, we collected a large amount of information regarding vaccine perceptions, individual characteristics, etc. when addressing COVID-19 vaccine hesitancy (Bidani et al., 2022). We use that information to design different interventions focused on communications, and to conduct nimble trials of comparatively simple A/B testing to understand which interventions were most effective.

In Zambia (Hoy et al., 2022) we combined a nationally representative, in-person household survey with randomized experiments sending online (Facebook) and SMS messages to survey participants to test how different information about the benefits from receiving a COVID-19 vaccine affected their willingness to get vaccinated, and how these effects were shaped by their household members' willingness to do so.

The study found that information was more important for those who lived in households where there were divergent views about vaccination. The information impacted their own intention to get a vaccine and made them more likely to encourage other household members to get a COVID-19 vaccine.

Part of the design stage involves deciding how to proceed with the delivery of an intervention, and whether the delivery method will be effective. When delivering a training to job seekers in Turkey (Cemalcilar et al., 2021), we identified that changes to how we invited and reminded participants of the training changed take-up rates. Iterating during the first weeks of delivery, including live monitoring and data analysis of take-up (from sign-in to attendance), allowed us to identify the invitation modality that got the most responses and intended attendance.

Despite all these efforts, only a fraction (21%) of those who confirmed participation actually attended the training. It was an important lesson for government counterparts seeking to identify mechanisms to increase attendance or use of services.

At the design stage, data can help answer important questions like:

- ▶ Is randomization possible?
- ▶ Is direct data collection possible?
- ▶ Is it feasible within time and budget limits?
- ▶ Are there other data sources we can use?

When seeking to measure behaviors, we know self-reports are not always accurate, more so when they are about behaviors that are habitual or subject to social sanctioning.³⁸

Implementation and Evaluation

Once a design is clear, and our assumptions and predictions explicit, we can begin the intervention implementation and

³⁸ Hence we've collected 'direct' data, for example on recycling (see: <https://blogs.worldbank.org/voices/how-collect-waste-behavioral-scientist>). On self-reported data biases, see for example (Baker et al., 2004; Fisher and Katz, 2000)

evaluate the causal effects of such intervention. This is the phase where we plan, roll out, and test the intervention.³⁹

At this stage, we draw on a portfolio of tools to better gauge monitoring data, collect output and outcome measures, and assess the effects of a policy or intervention. We decide on the type of evaluation design, which can be:

- ▶ Experimental
- ▶ Quasi-experimental
- ▶ Process
- ▶ Another type unique to the intervention.

We also identify the:

- ▶ Randomization procedure
- ▶ Trial design
- ▶ Treatment structure
- ▶ Data collection strategies.

We quantitatively evaluate almost all our projects. So, we design and implement them with this goal in mind. However, policy goals and expectations (and available resources) are as important in defining what, when, and how would be measured.

To measure the effectiveness and/or efficiency of an intervention that often affects several psychological, behavioral, and economic outcomes at once, it is crucial to collect apt data on relevant groups at the right time. This is no easy feat given the challenges we try to address and the varying contexts we operate in. Thus, as with the diagnostic, eMBED collects data from a wide set of sources, including:

- ▶ Administrative records
- ▶ Open-source data banks

³⁹ As many, we have started pre-registering analyzes to keep us accountable to our assumptions, test our predictions, and report both wins and failures.

- ▶ Surveys
- ▶ Experimental games
- ▶ Other, less common sources.

We also monitor implementation fidelity, or how closely the real-world implementation adheres to the plans we made. This happens using administrative data (e.g., teacher's reports, records of SMS or letters sent on time or received, frontline workers reports). Sometimes we implement random monitoring check-ins, such as household visits, calls, or chatbot rapid surveys. We implement both standard impact evaluations, but also mechanism evaluations (Ludwill et al., 2011), and nimble evaluations that come at a lower cost, provide results for iteration, and avoid the sunk cost fallacy we can all fall prey to.

Crucially, we believe interventions should be designed with an evaluation in mind, that such evaluations should have a credible counterfactual through randomizations or quasi-experimental methods. We also work to minimize data collection expenses. We follow the standards of a robust impact evaluation design by (Gertler et al., 2016):

- ▶ Trying to ensure randomization
- ▶ Tracking and minimizing attrition
- ▶ Measuring effects on the control group(s)
- ▶ Attempting to measure the fidelity of the intervention implementation and potential covariates

A perfect design that is costly to implement, implies a large change in existing policies, and requires the mobilization of large external field implementation teams is likely a design that will not scale, irrespective of how impactful it will be. Between perfect and useful, we err on the useful size even if that means sacrificing potential effect sizes.

Take, for example, two of our education interventions to increase socioemotional skills among students. Both involved

large-scale experimental randomized control trials. Whether in Peru (800 schools, Outes-Leon et al., 2020) or North Macedonia (nationwide, covering 350 schools; Santos et al., 2022), the interventions and evaluations were designed to adapt to the school's curriculum and delivery systems while minimizing disruptions in the classroom.

Implementation was left to the teachers who received limited instruction on the content and delivery, but adequate and clear materials to use in the classroom. Schools and classrooms were randomized to eliminate noise and challenges to the schools themselves (i.e., randomize full schools instead of classroom or students within classrooms), monitoring data was kept to receipts of materials by the school, and simple pictures of completed materials or just collection of them accompanied a one-page day report. Evaluations relied mostly on student performance. Both evaluations showed that, with all their caveats, the interventions had a positive impact on grades.

In Indonesia a similar intervention was tested in a quasi-experimental design (World Bank, 2020). An ex-post difference analysis was conducted that compared the changes of scores between control and treatment schools over two years. Direct data collection of the impact of the intervention was not feasible, and we did not have access to administrative data. Similarly, the child nutrition intervention targeted at fathers in India mentioned before also used a difference-in-difference design.

In other cases where experimental research designs aren't feasible, data science is of value. Non-randomized experimental designs such as propensity score matching rely on rich data so that a strong and representative control can step in as a counterfactual.

If quasi-experimental approaches aren't feasible either, we find alternative ways to test how successful an intervention is and why. For instance, we often measure a broader set of outcome variables, proxies, and contextual factors which enable

us to see how the intervention affected the target behavior and other causal links throughout our theory of change.

To identify key measures of the target behavior (or relevant proxies) we aim to measure a set of relevant variables along the theorized causal chain which links intervention components to larger effects. This helps identify how an intervention changed behavior or which of the myriad causal links hampered its outcomes. Getting data at different points in time can shed further light on how the intervention functions, how lasting its effects are, and which second-order effects might boost or undermine our objective. Where feasible, we also try to collect data on potential positive and negative second-order outcomes to create a holistic picture of the interventions overall impact.

Data analysis uses many common tools that are likely familiar to most researchers doing quantitative analysis. However, depending on the outcomes at hand and the data available, we blend tools more common in psychology with standard econometric analysis tools. More recently, we have used machine learning tools such as random forest or LASSO regressions that allow for improving the outcomes of our analysis (see, for example, Gauri et al., 2021, where LASSO is used to test the external validity of the intervention).⁴⁰

We have also used text-as-data tools including sentiment analysis to further expand the lessons from our qualitative data (e.g., Gauri et al., 2019). A robust and tested result is important. Where possible, we want to capitalize on our findings of the trial, using them to inform how the intervention is scaled up. In doing so, we follow a clear path:

- ▶ **Step 1:** Test
- ▶ **Step 2:** Learn
- ▶ **Step 3:** Adapt
- ▶ **Step 4:** Re-Test

⁴⁰ For more information see, for example, Turgeon and Lanovaz (2020).

Adaptation and Scaling

During the *adaptation phase*, we make sense of the data we gathered to identify the most important lessons. Almost all our projects are large in scale. We plan many of them as pilot tests to be further scaled to other socio-demographic groups, regions, or countries. This includes thinking about:

- ▶ How results will replicate with different groups or in different contexts
- ▶ Whether a larger scale intervention will generate equal effects
- ▶ Whether cost adjustments for scale or institutional constraints as delivery conditions become more heterogeneous will affect the intervention and its impact.⁴¹

Preparing and adapting for scale might include a diversity of smaller pilots rolled out in different conditions with a nimble evaluation to assess how well the intervention adapts to different conditions. It might include a progressive (staggered) rollout of elements or population groups, or others.

A clear identification of the active intervention components that “do the changing” brings us back to diagnostic before scaling. This includes analyzing how reliant they are on specific external conditions and how easy or difficult those conditions are to maintain if the intervention is rolled out more widely. This means that for some projects destined to apply to different contexts, we *re-define and re-diagnose* based on what we learned to overcome barriers and constraints to scaling it.

⁴¹ Two useful discussions of these issues include DellaVinga and Linos (2022) analysis of behavioral interventions by government Nudge units; Paluck et al. (2021) review of prejudice reduction interventions; and Vivalt (2020) on impact evaluations more broadly.

Conclusion & Key Points

Using quantitative approaches to design, deliver and evaluate behavioral interventions in international development requires a series of trade-offs, adaptations, flexibility, and attention to factors that go beyond the project's confines. While research ethics are part of all (or most) quantitative analysis and data efforts, they are critical for an institution like the World Bank.

When it comes to the security, protection, and ethical use of data, our institution must align with government policies. We also have our own regulations regarding the collection, storage, and analysis of data. We hold a special responsibility to pay attention to the risks of data leaks, breaches, or illegal practices – whether for our clients or partner countries, the World Bank, or, most importantly, the people we strive to support. We also try to raise these issues with our partners, whether they are governments or other entities holding rich bodies of data.

Another ethical concern associated with the use of quantitative methods and data relates to the risk of biased data and/or models that could exacerbate rather than ease existing inequalities. Predictive models often cement the status quo or reinforce existing biases by replicating patterns in the data that might (or might not) mirror real biases. We aim to use data in ways that are both fair and representative of the relevant populations, and where possible we collaborate with the people we work with to get a good understanding about the actual conditions of data collection.

We challenge ourselves regarding whether differential treatment groups are warranted by underlying factors or whether we can ameliorate fairness concerns, such as comparing various plausibly effective treatments, using stepped-wedge designs, or later scaling the most effective treatment to all treatment groups.

Implementing experimental methods often raises ethical concerns, including whether a randomization of a treatment we expect to be effective to only a select group of people is unethical. Researchers and practitioners alike have covered these concerns exhaustively. We take these concerns seriously and usually plan to extend an intervention (or its most effective version tested) to a wider group of people where possible. Similarly, with interventions, we must consider the implications on particularly vulnerable populations.⁴²

Another ethical concern pertains to data collection efforts that become quasi-interventions themselves. This often happens during our diagnostic phase, particularly in more open-ended or qualitative data gathering methods such as interviews or focus groups when we prompt people to think about their motivations, aspirations, mental health status or even racial, religious, or sociodemographic biases. To mitigate these concerns, we work closely with regional experts and community members to identify problematic questions and seek ethical approval prior to all our surveys and interviews.

One key limitation we face in our work involves the inevitable shortcomings of any data we might wring out of the messy fabric of reality. To begin with, data are limited, very specific reflections of reality, and dependent on the context in which they were collected – as well as by whom and by what method. Most data are historic, which inherently limits their ability to predict outcomes and behaviors. Data often are shaped by other, external contextual factors, which are difficult to account for in any model. While we rigorously check both the explanatory power and robustness of our analytical models, we are not free of potential biases.

⁴² Other international organizations, such as UNICEF have started outlining ethical considerations for behavioral interventions, for example, for children (see: Tindal et al., 2021)

Against our best intentions, we are influenced by our mind-sets, as well as institutional and social settings. Thus, throughout our work, we use data to counter our own assumptions, to plan for error and failure, to fail intelligently, designing trials in an iterative, pre-planned process that ensures we will understand the reason that an intervention did not work. We prioritize deliberate experimentation and adapt interventions according to what we learn. We don't see mishaps, mistakes, and failures as a bug, but as an essential feature of innovative policymaking.

Finally, we remind ourselves not everyone is a data expert and can understand caveats, nuances, or read a regression table. While evidence matters, it matters more how it is presented. Evidence-based policy decision-making must meet the policy-makers where they are, and this might take many forms, from simple infographics to presentations or policy notes. The soundest data analysis is rendered useless if the intended user cannot understand or use it for good.

**In the next chapter we learn more
about behavioral mapping.**

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7.

Behavior Mapping

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We implement behavioral interventions by changing the way an organization delivers an experience to end-users. Understanding not only the subject's journey but also the organizational context unveils what levers would need to be pulled, and where, to make an intervention happen. Those who own certain pieces of a product or service experience may be too close to the product or service to recognize and highlight structural barriers and organizational limitations off the bat. So, how can you set yourself up for success? Make a map!⁴³

Maps are an exceptional way to visualize information. They have the potential to break down siloed thinking, allow viewers to grasp interlocking relationships, and mobilize teams towards

⁴³ While I firmly believe these mapping methods can be applied in international development, the example included here is from my time working at the Common Cents Lab trying to increase the number of credit union members who signed up to round up their auto loans into a savings account. This example, combined with the additional resources in this book, should give you a sense of how to apply this in your own work. If you do use Behavioral Maps based on what you read in this chapter, please reach out to me and tell me how it went. I'd love to hear about it.

a shared purpose (Kalbach, 2016). Within applied behavioral science, behavior maps do an excellent job of drawing on existing research to highlight potential barriers and opportunities within a process. Within service design, blueprints add unique value by illuminating what's going on underneath the surface to bring a process to life.

In this chapter, I'll explore both behavioral and service design mapping processes and lay out a concept for a single, comprehensive map that combines the strengths of each. This combined map sheds light on customer experience, business process, and how the science of human behavior influences each, enabling more informed problem framing. It can serve as a valuable tool for anyone seeking to apply behavioral science insights to improve their products and services.

Behavior Mapping

Within applied behavioral science, the first phase of a project is typically conducting a behavioral diagnosis, often referred to as a behavioral audit. A behavioral diagnosis is an approach to problem identification that involves deeply understanding the environment in which an individual performs a target behavior, as well as understanding the cognitive biases that could influence the way the individual interacts with the environment. During a behavioral diagnosis, some organizations use a method called behavior mapping, which Ideas42 describes as “a systematic approach to identifying candidate bottlenecks... each of which is a possible intervention point.” (Datta & Mullainathan, 2014).

Organizations that create behavior maps generally follow common steps in creating these tools:

- **Step 1: Define a target behavior.**

Define a specific and measurable action you want a

user or customer to perform, such as signing up for round-up savings.

► **Step 2: Pick your starting point.**

Pick a starting point that is early enough in the process that you don't miss barriers that affect downstream choices, but not one that is so far back that you're mapping steps outside of your scope of influence. With savings, this could be when a member opens the digital banking app prior to learning about round-up savings, not when the member first wakes up on a given morning.

► **Step 3: Choose your target user.**

While users or customers can be segmented according to many characteristics, what's most important for a behavior map is to segment users on dimensions that affect how they behave when interacting with your product or service. For example, this might be an existing credit union member who regularly uses the mobile banking app but is not yet enrolled in round-up savings.

► **Step 4: Map the steps.**

Capture the end-to-end steps an individual would go through in an optimal world to exhibit that behavior – from starting point to target behavior (e.g. “member opens app” through “member is signed up for round-up savings”).

► **Step 5: Layer on the barriers and opportunities.**

This is the point at which you can identify hypotheses about how to improve the product or service through the application of behavioral science. Behavioral

science typically buckets improvement hypotheses into two categories:

- ▷ **Barriers:** What are the real or potential friction points that could keep a customer from performing the desired behavior? If you find it helpful, we can further break these down into **structural** (having to do with how the decision-making environment is structured), and **psychological** (how the individual is responding to the environment).

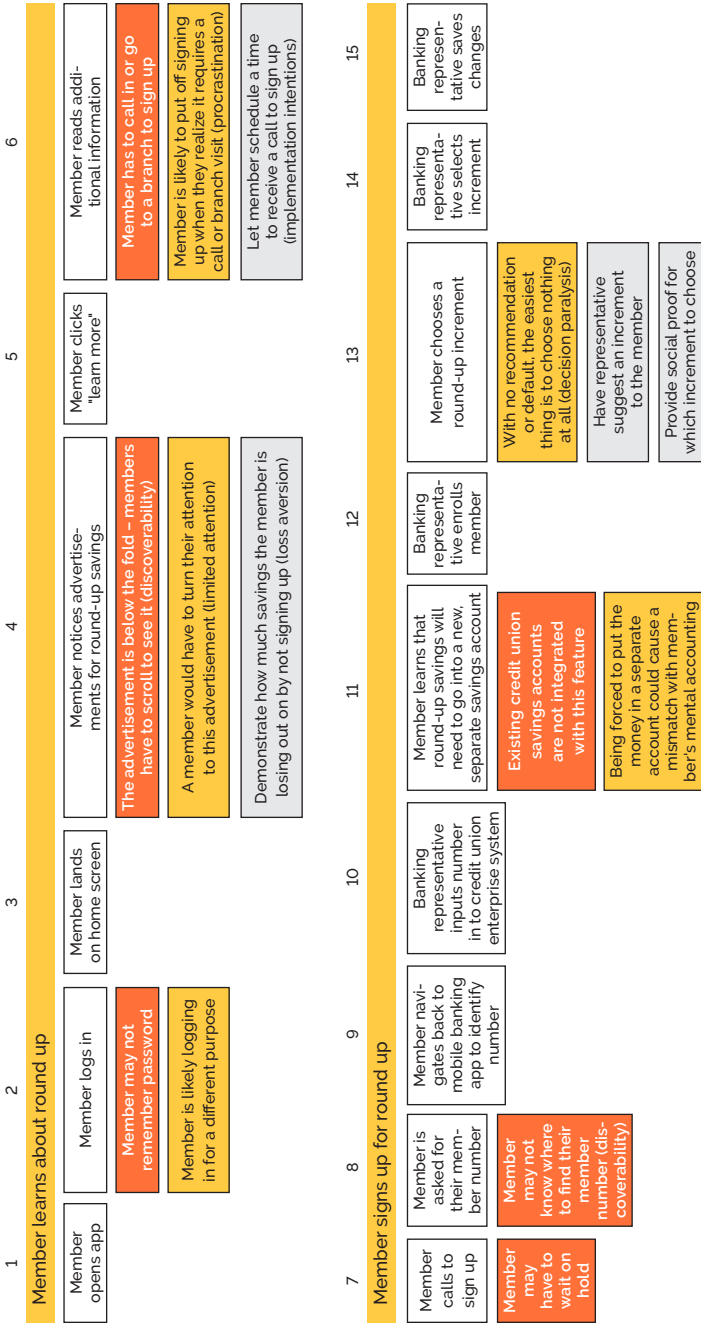
Examples of structural barriers are the discoverability of the ad about the round-up savings program, as well as friction points like needing to call or go to the credit union to sign up. Examples of psychological barriers are limited attention (a member would need to turn their attention to the ad) and decision paralysis (how do they decide how much to round up?).

- ▷ **Opportunities:** What opportunities exist to increase the customer's motivation to perform the desired behavior? These are typically rooted in cognitive biases. For this round-up savings example, we could leverage loss aversion, implementation intentions, or defaults to increase the performance of the target behavior.

A complete behavior map could look like this:



SCENARIO: A credit union member learns about round-up savings through a mobile banking ad and calls in to sign up.



Once the behavior map is finished, you need to prioritize barriers and opportunities. Ideally, you could also perform additional research to help you prioritize the most important barrier to remove or the best opportunity to increase motivation. Here are some helpful questions that might help you prioritize your options:

- ▶ Which ideas are associated with more structural interventions? Structural interventions versus light-lift nudges usually have larger effect sizes.
- ▶ Which ideas align with your organization's strategy? Ideas that already have support may be likeliest to make it into the field.
- ▶ Which ideas do you have the strongest hypothesis about, based on what you know of both existing research and the individuals you're designing for?
- ▶ Where, on the map, is the biggest drop-off happening? If you know, it's probably most worth your while to design an intervention for that step.
- ▶ If you don't know where the drop-off is happening, typically intervening further upstream, versus further downstream, can impact more individuals, because fewer people have dropped out of the process at that point.
- ▶ Do you have enough resources to pursue each idea?

Once you've prioritized your barrier(s) or opportunities, it's time to move on to prototyping, which you can read about in the next chapter.

Behavior maps are a valuable tool for performing a behavioral diagnosis, but they also have some shortcomings. They don't do a great job capturing the behind-the-scenes systems, actors, touchpoints, and policies that are invisible to the customer, but that produce the customer's decision-making environment.

Capturing these details is important for applying behavioral science to complex environments where multiple products or services or multiple organizations are involved in delivering a single experience. These details also help when it's unclear which pieces of the experience are owned by whom.

Behavior mapping can add additional value by borrowing a page from the discipline of service design. Specifically, from service blueprints.

Service Blueprinting

Service design is a human-centered design approach that places equal value on the user experience and the business process, aiming to create quality user experiences and seamless service delivery (Practical by Design, n.d.).⁴⁴ First described by G. Lynn Shostack in 1984, a service blueprint is the most powerful tool in a service designer's toolkit. Managing services effectively, he argued, requires better service design. The workflow design, PERT/GANTT charting, and quality control methods used on the operations side of the service were not adequate and failed to capture the customer's relationship to and interaction with the service (Shostack, 1984).

Since Shostack's original development of the concept, there have been many iterations and additions to service blueprints, but fast-forward to 2015 when Erik Flowers and Megan Erin Miller founded Practical Service Design (which has since been rebranded to Practical by Design) to integrate service design concepts into practical work contexts. Practical service design

⁴⁴ For more information see: <https://www.practicalbydesign.co/service-design-101>. Retrieved October 5, 2022

clearly described the lines of visibility through the metaphor of a theater production:⁴⁵

- ▶ The **front stage** is where the action happens and what the audience can see.
- ▶ The **backstage** is where all the support processes live that produce the front stage, like the lights, the sets, and the crew. It's the organization and all the things we do to make that front stage happen.
- ▶ Then there is **behind-the-scenes**, where all the intangible things that the organization must do to make both the front and backstage possible. That includes rules, regulations, policies, budgets, and all the things that aren't really a part of either the front or backstage.

Miller writes, “There are huge complexities that go unseen that are the support structures beneath every journey — the responsibilities of the internal actors, the systems that support those actors, all the processes and policies that dictate what can and cannot be done. Service blueprinting shows you a picture that not only includes the breadth of what happens along the journey, but all the depth that makes up the substance that the journey traverses across.”.

Rather than focusing only on the end-to-end user experience from the user's point of view, service blueprints map out a process both from end-to-end and from surface-to-core. This is the term Flowers and Miller used to discuss blueprints' ability to capture the layers of relevant information one could see if one were to pull back the curtain. Service blueprints add unique value in the depth of layers they bring to mapping out an experience.

But what exactly do those layers look like?

⁴⁵ For more information see: <https://blog.practicalservicedesign.com/the-difference-between-a-journey-map-and-a-service-blueprint-31a6e24c4a6c>. Retrieved October 5, 2022.

The Anatomy

Service blueprints can map out a current state process or define a future state experience that does not yet exist. Because behavior maps are typically used to map out the current state, we will focus there.

The initial layers of a current state blueprint seek to capture relevant details about the current state scenario and include the following (Practical by Design, n.d.):⁴⁶

- ▶ **Step definition:** Define what is happening in that step, including steps that are visible to the subject/participant/customer and those happening behind the scenes.
- ▶ **Touchpoint:** A point of interaction between the subject/participant/customer and the service.
- ▶ **Actor:** A person in the service scenario. This includes either the subject/participant/customer, service staff, or 3rd parties involved in service delivery.
- ▶ **System:** Technology systems that support the service.
- ▶ **Policy:** Rules or regulations that apply to aspects of the service.
- ▶ **Information:** Facts or observations relevant to the step.
- ▶ **Metric:** Data or statistics about this step.

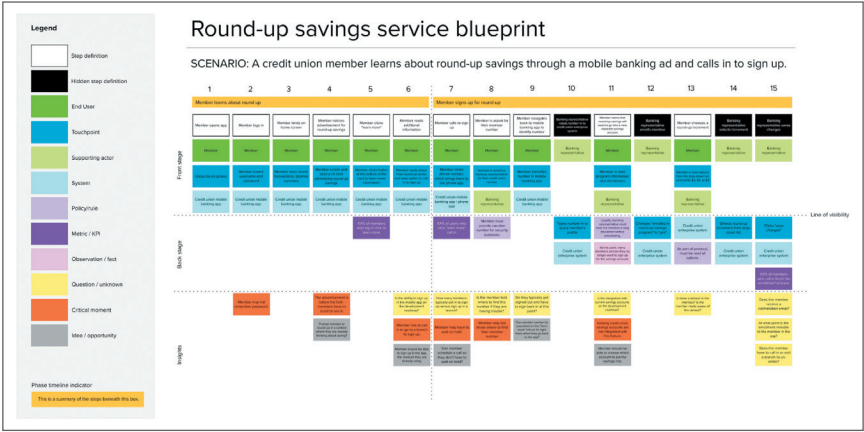
It also includes a section to capture critical information:

- ▶ **Questions:** Specifically, these are questions that you have about the step that needs to be followed up on.
- ▶ **Critical moment:** This is where pain occurs and breakdowns or potential breakdowns happen. This is the step where we identify what can go wrong.

⁴⁶ Retrieved October 5, 2022 from <https://www.practicalbydesign.co/the-guide>

- **Opportunity:** These are “Ah-ha” realizations, ideas, or opportunities to improve or fix for a broader impact.

The layers serve as a checklist to make sure the team is both discussing and documenting the reality and depth of what is happening at each step. A completed current state service blueprint could look something like this:



Behavioral Blueprinting

Behavioral scientists can gain an understanding of the system beyond the intervention by leveraging a service blueprints’ organized approach to capturing information not just from end-to-end, but also from surface-to-core. This is helpful both to inform the intervention design, including understanding why some steps are set in stone because of rules or policies, and where the most promising opportunities for intervention exist. It can also provide insight into what needs to be done to support that intervention.

What would it look like to combine these two methods? The steps are quite like behavior mapping, with a few differences:

- ▶ When **mapping the scenario steps**, map the individual steps a participant must take to achieve that target behavior within that scenario. Rather than just listing user-performed actions, list actions performed behind the scenes (by either supporting actors or systems).
- ▶ **Fill in the blueprint layers.** Use them as a checklist to add the surface-to-core detail to each step, before adding in the barriers and opportunities.

A behavioral blueprint provides a team with a complete picture of which pieces of the product or service experience are fairly set-in-stone because of actors or systems outside of their control, technical constraints, or policy constraints. It provides a sense of which steps in the process are the biggest problem areas based on funnel metrics (when available), and it documents hypotheses about bottlenecks.

Equipped with a shared understanding and comprehensive picture of the experience, the team can move forward with validating barriers and opportunities with additional data if necessary, framing the problem, and exploring potential solutions. The comprehensive view provided by a blueprint should reduce the number of surprises, misaligned assumptions, and implementation hiccups experienced by the team in subsequent stages of the project.

Do I Really Need All These Layers of Information?

If the delivery of the experience you're modifying is simple, it may not always be worth the time to map out the surface-to-core detail behind each step. However, I will share an example of a time when the layers really would have made a difference.

The Common Cents Lab worked with a credit union to design a two-condition experiment to reduce the number of

members opting out of an auto loan round-up savings program. The experiment was fairly successful, with enrollment rates in the round-up savings program increasing from 28% to 36% by highlighting the “loss” of not having savings at the end of the loan.

The round-up savings program, along with the embedded experiment, was rolled out as a pilot. Based on a traditional behavior map, there were no indications that the pilot would be difficult to scale. However, qualitative interviews with loan officers conducted after the pilot revealed some new information.

Car Loan Payment Schedule

Transfer Amounts	
Monthly Transfer Amount:	\$ -
Loan Principal & Interest	\$ -
+ Savings	\$ -

☐ I don't want to save as I pay off my loan

Car Loan Payment Schedule

Transfer Amounts	
Monthly Transfer Amount:	\$ -
Loan Principal & Interest	\$ -
+ Savings	\$ -

☐ I don't want to have \$ 640 - in savings by the end of my loan payment period

The intervention design had loan officers open a special purpose savings account the same day they closed the loan, but these savings accounts closed automatically if they did not contain a minimum of \$10. This became problematic because members did not make their first deposit until the first loan payment was due, but the 3rd party system used by the credit union barred them from removing this \$10 account minimum. During the pilot, loan officers went in and manually overrode this minimum for each member who signed up for the savings program, but this workaround was not feasible at scale.

The blueprint layers serve as a checklist to gather this kind of behind-the-scenes information. Taking the time to understand the surface to core detail may have revealed the special purpose savings account minimum, the inability to customize the system easily, the need to create a workaround, and ultimately, that the intervention design was not scalable.

Conclusion & Key Points

A behavioral blueprint leverages the information visualization practices of behavior maps and service blueprints to organize information with more clarity and depth. The tool paints a fuller picture of the context in question and provides a more comprehensive approach to identifying opportunities to apply behavioral science to improve an experience. However, a behavioral blueprint is just one tool within a larger toolkit.

“For many behaviors – controlling finances, exercising, getting politically active, decreasing energy usage – the product [or service] only interacts with part of the story, but all of it determines whether users will actually change their behavior or not,” says Stephen Wendel in *Designing for Behavior Change*.

Other methods, such as contextual inquiry, in-depth interviews, literature reviews, and surveys, ought to be leveraged prior

to the creation of a behavioral blueprint to identify the mediums and processes in which individuals most naturally perform a target behavior. Only then can we know whether the opportunity spaces and scenarios we blueprint have the potential to affect actual behavior change.

If you're interested in learning more, here are a few resources that may be helpful:

- ▶ <https://www.habitweekly.com/behavioral-mapping-pro>
- ▶ <https://advanced-hindsight.com/wp-content/uploads/2021/06/Behavioral-Mapping-and-Blueprinting-Cheat-Sheet.pdf>
- ▶ <https://actiondesignradio.libsyn.com/allison-white-behavior-mapping>

In the next chapter we learn more about prototyping in conflict zones from researchers working at the International Rescue Committee.

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8.

Prototyping in Conflict and Vulnerable Settings

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***Tahirat Omolara Eniola**, International Rescue Committee*

There are over 103 million forcibly displaced people around the world, 340 million requiring aid, and the need for humanitarian assistance is only increasing.⁴⁷ In Northeast Nigeria, after over ten years of conflict, 8.3 million people (over half the population) need humanitarian help (The International Rescue Committee (2022)).

We urgently need more high-impact solutions that can reach people in need. As these numbers continue to rise, while resources continue to decline, those solutions also need to be evidence-based and cost-effective, such as those afforded through behavioral science. Behavioral science has now been used in countless countries to achieve consistent impacts in most social areas, from education to health, and affecting the lives of millions around the world at a low cost.

⁴⁷ See UNHCR (2022) and The International Rescue Committee (2022).

Necessity is the Mother of Invention

The same behavioral solutions or approaches used in stable contexts cannot be implemented in the same way for these populations. While challenges implementing behavioral science projects exist in contexts everywhere, challenges like access to populations or instability become acute in humanitarian settings. Extreme constraints highlight the problems and opportunities for expanding applied behavioral science to new contexts, like populations experiencing displacement.

It is commonly said that necessity is the mother of invention, and there's no shortage of need when it comes to the application of behavioral science in contexts where the population continues to suffer the effects of conflict, crisis, and displacement. Since core strategies to the behavioral science field can't be applied as is because of contextual misalignment, different and more innovative solutions are being tested. This not only directly benefits populations but has the potential to offer innovative strategies to the behavioral science field and the expanding diversity of communities we serve.

For example, text message reminders are one of the most used solutions to remind populations to engage in a new behavior in the global north and more in low and low-middle income countries (LMICs). However, how would you provide helpful weekly regular reminders to a population in Tigray, Ethiopia who doesn't have a phone signal and is inaccessible due to conflict? (Zelalem, 2022). How about a population of women in rural Mali who don't have access to their own phones, are illiterate, and may even have different cultural conceptions of timekeeping?

This chapter will answer the last question and argue that to do so, our field and the populations we serve would benefit from more rapid, community-focused, and iterative testing approaches, such as **prototyping**.

Challenges and Opportunities

The replication crisis in the behavioral science field has highlighted the challenge of trying to use a one-size-fits-all approach to behavioral solutions (Frias-Navarro et al., 2020). However, contextualized interventions are even more critical for populations whose realities have not been reflected in the evidence-base, such as populations who are displaced and are experiencing the psychological effects of conflict or crisis.

While behavioral science is increasingly being applied across the Global South, most behavioral science research has been conducted in Western, Educated, Industrialized, Rich and Democratic (WEIRD) contexts (Silan et al., 2021). Over 80% of behavioral science teams are still in the Global North (Behavioral Teams Directory. Bescy, n.d.). This discrepancy is more problematic in humanitarian settings, as solutions have rarely been rigorously tested with populations experiencing conflict and displacement.

One option could be to generate this missing evidence in these locations using experimental methods, including typical methods leveraged in behavioral science such as randomized controlled trials (RCTs), which in recent years have been run at an accelerated rate in low-resource contexts around the world.

However, moving straight from solution ideation to evaluating those ideas with RCTs might not prove useful in these contexts. While RCTs and experiments are good at measuring targeted, isolated changes in behavior, traditionally they may not consider the importance of the complexities of context, nor allow for iteration, improvement or adaptation in response to changing conditions.

The challenges that behavioral science teams work on in humanitarian settings may extend beyond tweaking or optimizing specific behaviors by a known set of actors. These challenges

often develop in response to complex issues that have changing dynamics or layers of behaviors at different stakeholder levels of the system. For example, this could involve an interdisciplinary team aiming to improve the diagnosis and treatment of severe acute malnutrition in children in a fragile setting. Alternatively, a team may work to improve social and emotional skills in children who are experiencing the effects of toxic-stress in an ongoing conflict.

Given that behavioral solutions don't currently exist in the evidence space to address complex problems like these, teams need to co-create new behavioral solutions with communities and test them with end-users in context, iterating and adapting them as they learn in response to changing or complex environments which play an outsized role in these settings.

Prototyping

One way of rapidly iteratively testing and adapting solutions around users and their lived reality is called prototyping. Prototyping allows a project team to test various assumptions about what will work directly with the people who will use the solution, sometimes referred to as “end users.” At its core it is simply a way of turning solution ideas into tangible “mockups” and sharing those concrete ideas with end users and other stakeholders to engage with. Creating a prototype also provides an opportunity for end users of the solution to react to or suggest changes and co-design improvements with the project team.

Through observing clients reacting to mockups of behavioral solution ideas in their lived reality, teams can learn even more about their behaviors, values, and drivers of behavior. This improves the applicability and desirability of a solution and may even lead to the design of new ideas.

Through two case studies, this chapter will show that prototyping can be used in the design phase of a project to:

- ▶ **Contextualize** behavioral insights to ensure they are desirable and feasible for communities.
- ▶ **Iterate and improve** on those ideas before testing for impact and scaling to higher numbers of people.

This type of iterative prototyping helps the sector develop new solutions that go beyond the tried and tested nudges we have used or replicated in the past.

Prototyping, Co-Designing and Iterating Solutions in Humanitarian Contexts

In one of the International Rescue Committee's (IRC) projects in Mali, the team aimed to find an effective way to remind women to screen their own children for malnutrition (Lopez, Titus & Diarra, 2022).⁴⁸ Globally, the identification and referral of cases of child malnutrition to treatment remains the biggest gap in improving access to nutrition treatment. As a result, a program that asks mothers to screen their own children for malnutrition at least once a month, called "Family MUAC", has been increasingly implemented by the humanitarian community.

However, the IRC has seen low uptake of regular screening by mothers, which limits progress: 2020 data showed that, in the Nara district of Mali where 90% of women had been trained to conduct MUAC assessments, only 20% of treatment admissions resulted from family referrals. During formative research, women told the team that they wanted to conduct the screenings, but one of the main reasons for low uptake was that

⁴⁸ See https://www.ennonline.net/attachments/4651/FEX-68-FINAL_23-25.pdf

they often forgot to do it because of their competing workloads, childcare and household responsibilities.

Remembering to do something once a month is very difficult. How many of us would remember to brush our teeth if we just had to do it at some point once a month without setting any type of reminder?

A behavioral science evidence review of global literature provided the team with several effective interventions from other contexts, including:

- ▶ SMS reminders (Armanasco, et al., 2017)
- ▶ Digital alerts and alarms⁴⁹
- ▶ Mass media reminders (Wakefield, et al., 2010)
- ▶ Using radio or loudspeakers
- ▶ Other, less common methods of communication.

Because Malian women rarely had their own phones and had limited literacy, the team created “low fidelity” paper mockups of each potential idea. The team simply printed a photo of each idea and showed it to Malian women to solicit feedback and suggestions for improvement.

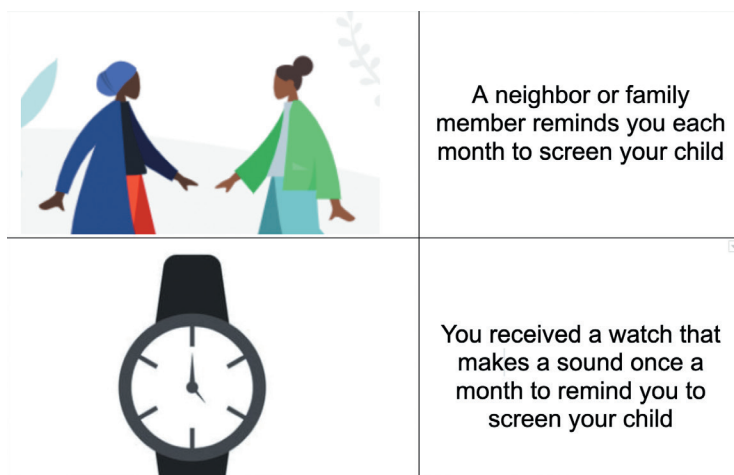
It can be difficult for people to feel comfortable providing direct, honest or constructive feedback to a humanitarian organization or any organization providing critical services. In these settings, organizations like the IRC provide important services supporting peoples’ livelihoods and wellbeing. Therefore, teams like IRC need to be aware of how power dynamics between provider and client can lead to understandable fears of offending and explore the use of different activities to gather indirect

⁴⁹ See more at THe Decision Lab. How digital reminders reduced workplace tardiness by 21%. Retrieved February 28, 2023, from <https://thedecisionlab.com/intervention/how-digital-reminders-reduced-workplace-tardiness-by-21>.

feedback about ideas that help participants feel safe to share their inputs.

The team in Mali asked women to rank the paper prototypes from the most useful to the least useful in their communities. Women told us, among many other things, that the phone messages and physical alarms were not practical because they didn't have their own phones, couldn't read, and wouldn't be able to get batteries or charge the devices. Also, women said they did not always go to the market regularly, and it also turns out that marketplace loudspeakers are used for many campaigns, so messages, literally and figuratively, get lost in the noise.

Image 1: Simple paper prototypes used in Mali to solicit feedback and use feedback to co-design new solutions



In the end, the highest ranked ideas were those showing a person in their community *socially* providing the reminder. The person was ideally a neighbor, female leader, family-member, or trusted loved one. Further probing about how other people

in the community could remind them revealed that most of them were already part of women's savings groups run by trusted female leaders. In fact, we learned that they typically attended these social gatherings at least every month, which is also the required frequency for malnutrition screenings.

Because of these findings, the team came up with a fresh approach that trained savings group leaders (women chiefs), religious leaders, and husbands to act as social reminders for mothers. This not only provided a regular reminder but also got the support of the local community and provided social proof for the new behavior as women could see and experience others engaging in screening throughout the community. It has also "piggy backed" on an activity that already happened: attendance at savings group meetings for women every month.

These are insights and solutions that the team would not have come up with had the team implemented and evaluated interventions from the existing behavioral science evidence base. In fact, based on these insights, some of the most cited behavioral interventions aimed at reminding populations would have failed, or could have even backfired.

Since prototyping this package of social reminders, a small Pilot Study with over 350 people has shown promising initial results. A pre-post test conducted with 387 people found that there was a 33% increase in women feeling "confident" in their ability to use the MUAC tape independently and an 11% increase in women feeling "very motivated" to use the tape regularly after 2 months. Qualitative results found that even women who were not part of the savings groups have learned about it through their networks and have come to the meetings to screen children. Further research is currently being planned.

This way of co-designing our solutions with affected populations ensures the team is promoting client-focused solutions that promote dignity and local ownership. It is also one of the

most effective ways to learn what is feasible and desired by users as well as create new solutions together with populations before investing time and resources in experimental studies, RCTs or evaluations.

Do as I Do, Not Do as I Say

Once a team has tested low fidelity, rough prototypes like the “paper prototypes” outlined in the previous case study, it’s important to test refined solutions again in real-world settings to see if assumptions hold up. The rationale for this type of iterative testing should not be new to behavioral scientists. Our field is full of studies showing that even if people say one thing, they might still do another.

The importance of iterative testing was clear in a recent education project in North East Nigeria, where the IRC was seeking to encourage teachers to use new teaching methods in their classroom (Bailey, et al., 2021). This method is called Social and Emotional Learning (SEL). The term describes an evidence-based approach to teaching children how to manage their emotions, improve social relationships and accomplish goals, among other skills. Projects in the past that had tried to implement SEL programs in places like Northeast Nigeria, where such skills are needed, had found limited or even no impact on children’s learning outcomes (Kim, et al., 2019 and Weiss-Yagoda, et al., 2021). It was hypothesized that these results were likely due to low rates of implementation by teachers and lack of contextualization or familiarity with these activities.

Prototyping provided an opportunity to consider the specific needs of the communities in Northeast Nigeria, earlier rather than later in the project life cycle. We tested assumptions about what strategies would promote teachers’ uptake of the activities

and gathered feedback in real time while course-correcting throughout.

As is typical with behavioral science projects, for this project in Northeast Nigeria, the team started with a behavioral science evidence review to learn what has been effective in other contexts and followed this with a behavioral mapping exercise to understand the specific drivers and barriers to behavior change. The behavioral science evidence showed that:

- ▶ SMS reminders can be effective when forgetfulness is a barrier.
- ▶ Chunking or simplification can make information easier to digest and act upon when receiving information about new education activities (Cortes, et al., 2021).

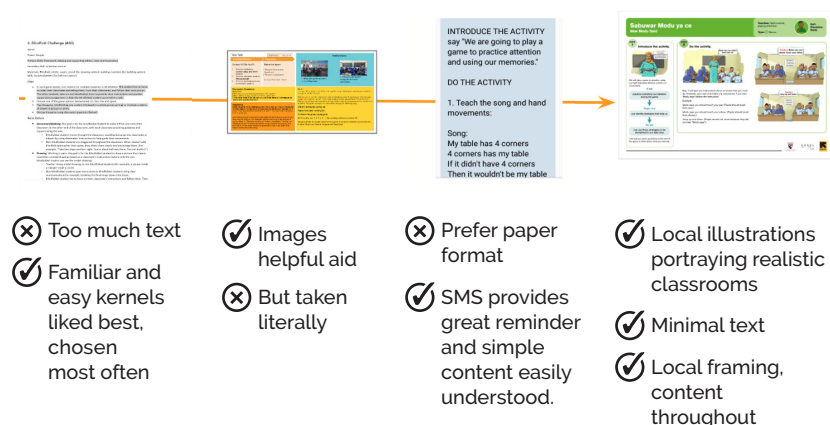
When the team showed teachers very low fidelity paper prototypes of this idea, they were excited about the prospect of receiving them. The team co-designed the message content with the teachers to make sure it was easy to understand, read and act on.

However, after the team tested the SMS messages over a period of weeks and asked for their feedback, the findings were not as expected. One teacher told us they had spent time every night writing out each SMS message by hand on paper. Another told us they were looking for a print shop to have them printed out on paper. It turned out they still needed the paper copies to refer to in classrooms and to use at the front of the class.

It was highly important for the teachers' identity as educators to have a paper book or card to teach from. That is what Northeast Nigerian teachers do. While they still wanted SMS reminders as support and reminder, the team ended up focusing the core package of interventions on physical paper cards. During implementation teachers shared which of the newly

designed SEL materials were better able to meet their needs. We learned that the more of their feedback and local input we incorporated into the solutions, the more uptake we saw of the SEL materials in the classroom.

Image 2: the process of iterative testing different prototypes in Nigeria



Throughout this project, the team worked closely with a core group of 12 teachers who served as collaborators in designing the activities their community would ultimately use. During the first weeks of prototyping, some of the feedback the team received about ideas and solutions were vague, broad and mostly positive. Because the IRC is a service-providing organization within the community, asking for honest and constructive feedback from those who receive those services can be challenging.

The team worked to improve trust and communication over time with teachers to ensure they understood their role as advisors and future users of the materials. This meant giving them the power to decide what gets included in the materials.

Demonstrating the adaptations that were being made to the materials based on the feedback they provided was critical to building trust. This process led to:

- ▶ More insightful feedback from the group.
- ▶ A sense of ownership.
- ▶ Materials that were designed and tailored to meet their needs.

The end package of solutions was entirely contextualized and integrated. It included localized and evidence-backed content and framing. It also featured targeted behavioral solutions. The team developed a gamified paper checklist for teachers to track progress and score “badges” as they advance, goal-setting prompts and SMS reminders to encourage teachers to develop new classroom habits, among others.

Ultimately, through this process of iteration and adaptation, teachers were better able to help students build SEL skills appropriate for Northeast Nigeria. After the prototyping was completed, a 6-month pilot study with 224 teachers found that after being exposed to the final package of interventions:

- ▶ Teachers used the new activities for an average of 19 minutes a day.
- ▶ Over 70% of teachers implemented the activities with high levels of quality and fidelity.
- ▶ The program cost only \$7 USD per student. Previous SEL programs which have not affected student outcomes have cost between \$67 and \$117 USD per student.

Conclusion & Key Points

Behavioral science projects typically:

- ▶ Define the problem
- ▶ Define target behaviors that need to occur
- ▶ Consult the evidence base to come up with behavioral solutions to encourage those target behaviors
- ▶ Design and run an experiment

These projects benefit from the addition of co-designing nudges with communities using prototypes, and iteratively testing refined and improved ideas (see images below). The program teams have used this approach to achieve higher confidence in what works at improving outcomes in a specific context, before trying to design a Pilot Study or RCT.

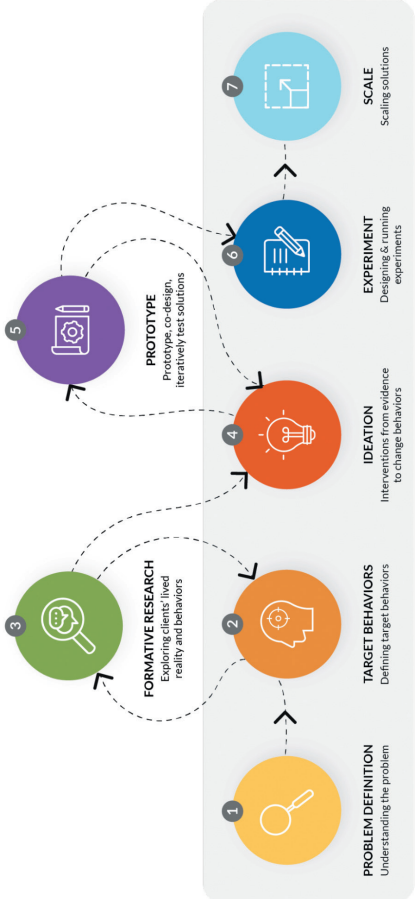
Since much of the evidence from behavioral science cannot be ‘cut and pasted’ to humanitarian contexts in a top-down manner, rapid testing solutions directly with the population can contextualize intervention options quickly and increase our confidence that the solutions will work for intended outcomes.

Iteratively testing a concept before moving to experimentation is essential, since RCTs are extremely difficult to run in these contexts and should only be used once the team has high confidence in the solutions being tested. As populations are on the move, access routes are often shut down or can become insecure within a moment’s notice, preventing access by the research team to populations for indefinite periods of time.

Image 3: proposed steps for a behavioral insight project



(Image 3: simplified steps for a "traditional" behavioral insight project)



The focus of applied behavioral insights should always be the person's life that the project is trying to support. We must meet people where they are. If everything developed in a project is guided by the person it is being designed for, we can expect it to lead to better results. When users share the language, delivery format, and framing they want to be communicated with, the chances of achieving a positive impact on outcomes increase. This will prevent situations where a team sends out SMS messages only to find that teachers are only going to use paper copies.

One participant we spoke with summarized it perfectly by quoting the saying, “There is nothing for us without us.”

In the next chapter we review ways to overcome challenges when scaling-up interventions in the Global South. We will learn from researchers working at the Center for Social and Behaviour Change, India.

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9.

Challenges to Scaling Interventions in the Global South

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The Centre for Social and Behaviour Change at Ashoka University is uniquely positioned to liaise with the central and state governments of India, the largest scale-up organizations in the world, for operationalizing evidence-based behavioral interventions. Even with rigorous causal evidence in their favor, we've seen interventions fail to translate into a program or a set of job responsibilities that the supply side of policy can easily deliver. In this chapter, we delve into the behavioral barriers that impede last-mile scale-up in India, specifically:

- There is a job responsibility overload that does not align with frontline policy implementers' expectations. The same implementer is expected to deliver multiple interventions to a variety of beneficiaries,

and transitions between job responsibilities lack streamlining.

- ▶ An effective incentive structure for implementers is not in place. They are not paid a sufficient salary that would increase extrinsic job motivation, and there may not exist the appropriate institutional mechanisms to sustain intrinsic motivation.
- ▶ The onus of filling the gaps in translating how interventions were delivered by trained professionals during an RCT and how they will be delivered at scale falls upon untrained frontline implementers, who are left to make an educated guess as to what they need to do.

We will subsequently recommend the following applications of behavioral science to change the behavior of supply-side agents:

- ▶ When we work on scaling an intervention, we must acknowledge that we are trying to break a routine and install a new one. Incorporating implementers' learning and habit-changing costs into intervention testing could have a significant impact on their scalability.
- ▶ Behavioral diagnostic research is required to understand the incentives that are currently driving implementers' job motivation, and whether there are better rewards to consider. For instance, the opportunity for upward mobility and job security through public sector jobs is an important incentive. We describe an intervention designed and tested by CSBC to increase job motivation (and subsequently job performance) of frontline policy implementers in Bihar, India.
- ▶ Interventions must be bootstrapped down to a set of clear, concise instructions for implementers to follow. The cognitive load of translating interventions

to programs needs to be reduced. We elaborate on CSBC's scalability-testing phase with a case study of interventions designed to increase iron and folic acid supplementation in India.

Psychological Contract Breach and Job Responsibility Overload Among Frontline Policy Implementers

The behavior of public policy implementers in India is well-encapsulated through the lens of psychological contract breach and cognitive dissonance (Rousseau, 1998). When signing up for a policy implementer's job, an individual carries preconceived and established beliefs about the reciprocal obligations, rights, and benefits the job entails. These beliefs form the implementer's psychological contract (PC) with their implementation organization (including, but not limited to, the local government). The PC of a typical policy implementer in India would include the following core beliefs:

- ▶ Stipulated working hours, tasks, and effort expectations
- ▶ Expectations of a safe income and post-retirement pension
- ▶ Expectations of respect and privilege from their network and beneficiaries

Formed in the initial few months of their job, this PC serves as a mental model for the implementer to frame their behavior within and even outside of their job. The individual and organizational factors shaping the PC of a typical policy implementer in India would include:

- ▶ **Individual factors:** Predispositions about the aspirational nature of governmental jobs, reference points for upward mobility set by peers or role models in their

network, and one's personal prosociality-based motivation to work for the public good. We discuss in section 2 how policy implementers' motivation is threatened and suggest interventions to address this problem.

- **Organizational factors:** Initial information shared by employers, coworkers, and supervisors that could lack clarity regarding the socialization and promotion process. In Abdel-All et al., 2019, policy implementers express a pre-established preference for the possibility of career progression and training leading to promotion, along with fixed incentives. As we will discuss later, these expectations are not met.

Importantly, when an individual faces discrepancies in their expectations and the actual state of affairs, they may make adjustments to their beliefs and behavior. This relates to Festinger's (1962) cognitive dissonance theory. Robinson (1996) argues that "flexible, dynamic, and changing modern work environments increase the probability of psychological contract breach". A government-dependent job would easily fulfill these criteria. Local governments routinely change. So do politicians, bureaucrats, and the policies that they must prioritize and implement.

Perceived contract breach (PCB) can occur when the policy implementer is arbitrarily asked to halt their routine of delivering policy A, learn a new set of implementation guidelines for policy B, and start delivering it. Or they may be asked to stack the implementation of policy B on policy A for the same beneficiaries. In both these cases, from the implementer's perspective, PCB occurs, as their employer has failed to adequately fulfill their obligations of having pre-decided tasks and effort expectations. Unless empathetic steps were taken to facilitate this transition by the supervisor, the hassle and cognitive load of breaking old routines to learn new ones add to the PCB.

For this chapter, let us consider the example of one of India's key types of frontline policy implementers, the Accredited Social

Health Activist (ASHA). ASHAs are women who belong to the same community where they deliver policy benefits, making the role an essential last-mile link between governments and citizens. An ASHA's PC would include the implementation of various health programs across various villages, with multiple health indicators they must improve (for example, immunization rates, maternal mortality, and infant mortality).

As per the National Rural Health Mission's guidelines, ASHAs are technically volunteers who choose to help deliver policy benefits, primarily driven by intrinsic motivation. Being an ASHA is supposed to be a part-time, unpaid supplement to an individual's primary livelihood. The time commitment outlined as an initial obligation is two to three hours daily for four days a week. These components are key components of an individual's PC when signing up for the role of an ASHA.

However, various studies have documented ASHAs working for considerably longer hours than stipulated in the PC, especially during and after the COVID-19 pandemic (Khandre et al., 2022; Bajpai & Dholakia, 2011). Additional pandemic duties were offloaded to ASHAs, constituting a PCB in terms of the tasks they must perform. Kawade et al. (2021) qualitative inquiry with ASHAs in the Pune district of Maharashtra revealed that 58 out of 67 ASHAs in their sample had been tasked with at least 10 additional responsibilities (including conducting surveys and facilitating financial inclusion by asking people to open bank accounts, the latter notably is not related to health policy). The authors note a prevalent willingness to take up these additional tasks in their sample which relates to the importance of prosocial motivation we discuss in the next section.

An ASHA stated they demanded extra compensation for new workloads, arguably to justify their additional effort through extrinsic motivation, since prosocial motivation may be a limited resource. ASHAs clearly and repeatedly demand higher financial incentives for their work and claim that they would put in

greater effort, even work full-time, in return for commensurate compensation.⁵⁰

The consequences of PCB that signal the breach occurred and has affected job performance are clear in studies of ASHAs' job performance. Post-pandemic, many instances of ASHAs going on strike across states to demand higher pay, regularization of work, and social protection were covered by the national news (Ambast, 2021). In the absence of organizational commitment and trust, policy implementers may express high intention to leave, and job performance may worsen.

Su-Fen Chiu and Jei-Chen Peng (2008) effectively sum up the literature on the effect of PCB on job performance — employees exhibit dissatisfaction with their job, frustration, and anger, discretionary absenteeism (Deery, Iverson, & Walsh, 2006), and job neglect (Lo & Aryee, 2003). Mondal and Murhekar (2018) found quantitatively that ASHAs in the Howrah district of West Bengal with relatively lower job performance scores were more likely to serve a larger population than that stipulated in their job role (1,200), or they were lacking supervisory support from ANMs.

In the context of policy implementers, these negative job behaviors imply enormous operational costs for governments, loss of working days because of absenteeism, and greater churn among implementers.

Gautam Patel, CSBC's Policy Lead at the NITI Behavioral Insights Unit (BIU), the Uttar Pradesh BIU, and the Bihar BIU, quotes an educational intervention that was intended to be scaled through governmental institutions/schools as a case study in job responsibility overload affecting scalability. An

⁵⁰ In Tripathy et al. (2016), a majority of the sample reported having worked as a community health worker for several years without being promoted. Cases of more experienced health workers earning roughly the same as those who started the job recently were prevalent. An ANM reported that they had no appropriate authority for voicing these concerns.

educational intervention to improve student learning of two specific subjects was tested with non-governmental teachers who were dedicated to testing the intervention.

The intervention, which was proven effective in the non-governmental context, struggled to float within governmental institutions, primarily because of their myriad distractions. Government school teachers were roped into national and state-level campaigns and missions such as the Swachh Bharat Abhiyaan and elections-related activities. Their non-teaching-related work rendered the scaling of a straightforward two-hour teaching intervention difficult.

When a non-governmental organization conducts an evaluation, we see impact, but that impact often fails to replicate when piloted by government staff. Governments setting up dedicated workforces to implement specific policies may not be the best answer either. In CSBC's experience, despite a government agreeing to set up a new dedicated workforce for a policy's implementation, it was impossible to prevent the new staff from being pulled into other work. This systemic inclination to be distracted could be addressed through "campaign-mode" or "drive-mode" phases—short bursts of time wherein staff is dedicated to a singular high-priority task.

Job Motivation and Performance

In a motivational assessment administered with various types of community health workers, including ASHAs, in the Ambala district of Haryana, low levels of job motivation were correlated with job burnout, job insecurity, lack of career progression opportunities, and poor supervisory support (Tripathy et al., 2016).

What are the factors that motivate frontline policy implementers? We considered two key elements:

- **Extrinsic motivation:** We reference Deci and Ryan's (2004) self-determination theory, whereby employees engage in their work to achieve a goal that is distinct from the inherent satisfaction of the work. We focus on the extrinsic motivation that is externally regulated; i.e. behavior driven by external rewards and punishments. Some of these rewards are embedded in the implementers' PC—the possibility of career progression and promotion, job security, and fixed monetary incentives.

Along with the earlier reference to the demands of ASHAs for promotions and increments (Tripathy et al., 2016), through an experimental approach, Abdel-All et al. (2019) observed that the possibility of promotions is an important factor driving employee motivation in Community Health Work (CHW). Over 85% of the participating ASHAs were willing to switch to a CHW job with the prospect of career progression that offered them a monthly salary that was Rs. 2,530 less than a job without career progression.

Therefore, implementers currently operate in a working environment that lacks satisfactory extrinsic motivators.

The Janani Suraksha Yojana (JSY) is an example of how extrinsic incentive structures could improve implementers' job performance. JSY introduced a set of financial incentives for mothers and community health workers to promote prenatal care, institutional childbirth, and postnatal care. ASHAs receive 250 Rupees for transporting a woman to a medical center for delivery, and another 150 Rupees immediately after the delivery. Another 200 Rupees is given to the ASHA after the newborn receives the appropriate

immunizations within two weeks of delivery (MoHFW, n.d.). There is no separate incentive for ensuring adequate postnatal care for the mother. ASHAs receive the post-delivery incentive of 400 Rupees regardless of whether they had scheduled antenatal care visits. ASHAs receive the post-delivery incentive in cash immediately at the hospital itself, while availing the postnatal care incentive requires the ASHA to track the beneficiary down again and schedule a visit.

Therefore, ASHAs may prefer to not bother with postnatal care visits and focus instead on maximizing their institutional deliveries. This behavioral pathway explains Joshi's (2014) observation—while the rate of institutional deliveries has increased post-JSY, prenatal and postnatal care rates remain largely unchanged. Behaviorally auditing extrinsic incentive structures and re-aligning them to better influence behavior can therefore prevent such behavioral fallout.

- **Intrinsic and prosocial motivation:** Both these types of motivation are relevant to understanding frontline policy implementers' behavior in the absence of adequate extrinsic motivation.

Intrinsic motivation comes from within, and implementers enjoy their work for its own sake. Prosocial motivation requires conscious effort and self-control. To maintain prosocial motivation, implementers need regular positive feedback from others.

Khandre et al. (2022) confirm this motivational pathway by observing that positive feedback from beneficiaries helped induce a sense of responsibility within ASHAs to work further for the public good. However, the lack of positive feedback from both beneficiaries and supervisors led to a decline in their

job satisfaction, indicating the strength of the pro-social motivation component. In particular, some ASHAs reported a non-supportive dynamic with supervisors; for instance, there was supervisory pressure to complete policy targets along with the hassle of navigating sludge in getting documents signed by supervisors to complete their work.

When intrinsically motivated, employees are process focused—they see the work as an end in and of itself (Amabile, 1993). When prosocially motivated, employees are outcome focused—they see the work as a means to the end goal of benefiting others (Grant, 2008).

Kawade et al. (2021) find that ASHAs derived self-esteem through their work, and even use that sentiment to offset the inadequate financial compensation. ASHAs carefully observe when and how their community health work moves indicators—one agent reported pride in having encouraged more women to opt for institutional deliveries (Kawade et al., 2021).

How do we ensure policy implementers remain motivated, especially when career progression and other extrinsically-motivating incentive structures are inadequate? Building on the employee motivation literature (Dowling & Sayles, 1978, p. 16) (Latham & Pinder, 2005) (Amabile, 1993), in 2022, we tested the efficacy of Facebook group-based job training modules for public policy implementers working under the Government of Bihar's Rural Livelihoods Promotion Society (JEEViKA).

We tested whether fostering dialogue across hierarchies of cadres, including work-related discussions and problem-solving, increases their intrinsic and prosocial motivation to deliver policy benefits, and subsequently leads to an increase in job performance. We posited that a Facebook-based peer group would heighten JEEViKA cadres' positive feedback with each

other and to their work, while simultaneously delivering training modules that improve their competence. Results show a positive and small though statistically insignificant effect.

In its work with JEEViKA, CSBC has also observed how crisis periods strongly influence and sustain prosocial motivation and behavior. Immediately after the devastating “Delta variant wave” of COVID-19 in India, JEEViKA cadres overcame the hesitation of working during the pandemic in favor of saving their community through immunization efforts.

Making Interventions Scalable for Existing Policy Delivery Infrastructure

Who delivers the intervention when implementing RCTs? Trained enumerators, paid more than policy implementers, bringing an entirely distinct PC to their work, as compared to typical policy implementers. To step closer to a ‘perfect’ imitation of the policy during the testing phase as it would perform during the scale-up phase, it is imperative that we somehow replicate the PC of policy implementers within the experiment, along with other barriers such as potential cognitive load and dissonance brought on by the cost of learning how to deliver the new policy.

Involving intervention designers and prototyping teams in the experiment design stage may induce a shift in how we create interventions as well, by taking cognizance of how they will be tested, and whether their appearance during the tests will closely reflect their scale-up version or not. For more on this, see the chapter on Prototyping.

There are three questions to consider in determining which interventions can be made “more scalable” for implementation:

- How can we replicate the PC of policy implementers within enumerators?

- ▶ How can we design the intervention to minimize the likelihood of PCB among policy implementers?
- ▶ Is it necessary to have a different evaluation — a scalability test — that specifically measures how easy it is for the policy implementers at the grassroots level to deliver the intervention we designed?

At CSBC, we involve bureaucrats, policy implementers, and other non-governmental scale-up organizations during stakeholder workshops wherein the interventions are designed. The prototypes we develop are empathetic of the implementer PC, and we usually intend for our interventions to act as job aids with minimal invasiveness on an implementer's routine implementers simply carry the CSBC-designed job aid along with them during their routine work.

Since 2018, CSBC has been working on tackling anemia among pregnant women by enhancing the consumption of iron and folic acid tablets. Through extensive diagnostic research and behaviorally-informed prototyping, we designed two job aids intended for incorporation into the routine tasks of ANMs (Auxiliary Nurse Midwives) and ASHAs. Anemia is highly prevalent among pregnant women in India.⁵¹ Strengthening iron and folic acid (IFA) supplementation is imperative for preventing and treating anemia (WHO, 2012). IFA pills are distributed under the Anemia Mukh Bharat (AMB) policy led by the Ministry of Health and Family Welfare of the central Indian government, but only 26% of pregnant women consume IFA tablets for the recommended number of days (NFHS-5, 2019-20).

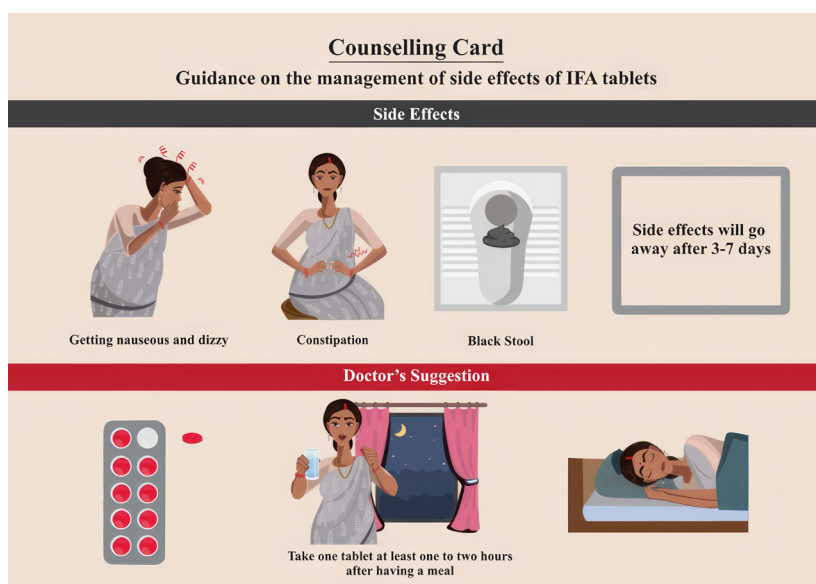
IFA supplementation is evidently a demand-side problem in India. CSBC's behavioral audit identified various behavioral barriers preventing pregnant women from routinely consuming IFA tablets, two of the primary ones being:

⁵¹ About 52% of pregnant women in India are anemic (NFHS-5, 2019-20).

- **Barrier A: Lack of adherence** — The side effects of IFA tablets were deterring pregnant women from consuming them.
- **Barrier B: Intangible payoffs of consuming the tablets** — The benefits of consuming IFA tablets were not clear, leading to their de-prioritization.

To tackle barrier A, CSBC designed a counseling card that made salient the actual side effects of consuming IFA tablets and how to deal with them (see image 1 below). This card was intended as a job aid to be carried by ANMs and ASHAs during their counseling sessions with pregnant women.

Image 1: Counseling Card job aid designed by CSBC and Busara



For barrier B, CSBC designed a goal-tracking calendar that provides the user with a reminder to consume the tablets while making salient the benefits of consuming them (by demonstrating the visual progress of happy children) (see image 2 below).

We subsequently conducted an RCT with a sample of 1,200 pregnant women to test the efficacy of these interventions in increasing IFA tablet consumption. Both interventions were successful to increase the adherence to IFA tablets, as compared to the control group.

Our key “scalability” arguments are relevant here, in the post-RCT and pre-scaling-up phase. For this purpose, CSBC undertook a three-pronged approach to understand the most efficient way to integrate these interventions in the existing health-policy machinery of Anemia Mukh Bharat (AMB) at the grassroots level, without disturbing pre-existing dynamics. We:

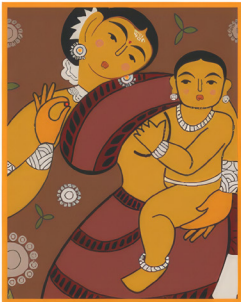
- ▶ Behaviorally audited existing policy documents on the distribution of IFA tablets, counseling, and anemia-related IEC materials under AMB.
- ▶ Consulted with Frontline Health Workers (ANMs and ASHAs), pregnant women, and block and district-level officials to understand the existing processes under the AMB strategy.
- ▶ Consulted with relevant stakeholders in NITI Aayog, the Department of Public Health and Family Welfare, and the Ministry of Health and Family Welfare to validate the findings of the policy review and qualitative study and refine the operational guidelines.

These consultations formed the first phase of the evaluation of these interventions’ scalability. We conducted a non-experimental, observational evaluation of the interventions’ performance in the scale-up phase by asking ANMs to deploy them during their routine counseling sessions and observing their performance in health centers. Feedback was subsequently collected

Image 2: Progress-Tracking Calendar designed by CSBC and Busara

Name of pregnant woman: _____
Start date of IFA tablet consumption: _____

1st Month



When the pregnant mother consumes IFA tablets
she gives birth to a healthy child

How to use the calendar




Everytime you consume an IFA tablet, scratch-off one box

Demo box

 1	 2	 3	 4	 5	 6
 7	 8	 9	 10	 11	 12
 13	 14	 15	 16	 17	 18
 19	 20	 21	 22	 23	 24
 25	 26	 27	 28	 29	 30

from the ANMs and incorporated into our scale-up framework for efficient integration of the interventions in existing delivery platforms.

The scalability evaluation phase resulted in two sets of operational guidelines being drafted to recommend implementation mechanisms for the counseling card and the goal-tracking calendar. These operational guidelines provided clear and concise steps for the relevant central, state, district, and village-level officials to follow to deliver these interventions through their existing health-policy delivery infrastructure. The concerns we have elaborated on in the previous sections — policy implementers' PCB, cognitive load, and cost of breaking routines — were empathetically incorporated into these operational guidelines.

The guidelines suggest modifiable and non-modifiable features for the design and delivery of the interventions. This allows implementers to make context-specific changes while retaining the behavioral features of the interventions, given India's extensive social and cultural diversity. The guidelines include a short and interactive training guide that may be integrated into the existing anemia-related training materials to prevent policy implementers from being cognitively overloaded by new tasks.

To prevent the costs and hassle of breaking existing routines, the guidelines suggest roles and responsibilities for stakeholders at various stages, including estimation, printing, storage, delivery, training, implementation, and monitoring. Finally, to ensure the intervention iterates upon itself, the guidelines suggest the Responsive Feedback approach, which involves taking regular feedback from grassroots policy implementers to make continuous and timely improvements to the interventions and their delivery.

Conclusion & Key Points

The authors posit that applying behavioral science insights to understand frontline policy implementers' job systems and adapting policy interventions to their idiosyncrasies can improve scalability in the Global South. Policy interventions that are transformational in their implementation plan, those that do not seamlessly integrate into existing policy implementation infrastructure, are bound to induce psychological contract breaches within implementers. This can lower job satisfaction and performance. Before systemic changes that deal with PCB can be implemented, such as better onboarding processes, regular job expectations evaluations and goal-setting exercises, and conscious efforts by supervisors to minimize job responsibility overload, interventions must be designed with the intention to minimize PCB.

CSBC shows how to design such an intervention during its work on IFA supplementation in India. Thorough behavioral audits to understand how to seamlessly fit interventions into existing routines, designing simple interventions that reduce cognitive load, and engaging with diverse stakeholders during the scalability research process are imperative to ensuring the success of large-scale policy interventions. New scalability-specific research must be conducted with implementers to observe how well the intervention performs, with an emphasis on novel measures of hassle aversion, cognitive load, and willingness to implement.

Finally, we observe that certain policies have extrinsic incentive structures that disproportionately favor specific components of a policy by influencing implementers' behavior. To address the larger systemic barriers and facilitators of extrinsic, intrinsic, and prosocial motivation, thorough behavioral audits must be conducted to trace how existing incentive structures influence

policy implementation. New policy interventions should attempt to regularly update their incentive structures according to the progress of the interventions' outcome indicators.

In the next chapter we learn more about the need for dignity in applied research.

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10.

Dignity in Applied Research

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“We will learn to live together, cooperate with one another, and recognize the dignity of others, or we will perish.”

That’s Barack Obama’s stark message from his memoir, *A Promised Land*. Esther Duflo and Abhijit Banerjee would agree, in their more measured way. In their 2019 book *Good Economics for Hard Times* they write that, “restoring human dignity to its central place sets off a profound rethinking of economic priorities.”

That echoes the message put plainly by a human rights activist in the informal settlement of Mathare, Nairobi, who told our interviewer, “We would prefer to have respect...Unfortunately these things only happen to the rich people.”

Dignity is one of the great human themes. It underpins all rights, according to the Universal Declaration on Human Rights. Yet it is easier to vaguely invoke dignity than it is to actually realize it. In our research practices, and in development

processes, we make an implicit promise of “respect for persons.” We make a commitment to treat people in a way that respects their dignity. Yet all too often, we break that promise. People come away from interactions with researchers, charities, governments, and businesses feeling used and bruised, exploited and extracted from.

That sense of extraction happens when:

- ▶ People give consent to vague and misleading study descriptions and are shocked to suddenly answer the most intimate questions about their lives and morals.
- ▶ Disorganized researchers keep busy people waiting.
- ▶ Research projects cease after months of furious and inscrutable activity, and the participants never hear another word about what was learned.
- ▶ Institutional review boards are more concerned with mitigating legal risk than whether a consent form is actually legible or appropriate to the people and context where it is being used.
- ▶ Overworked enumerators are too focused on a daily target to give respondents a moment to think.

When we do these things, we break the researcher’s code of respect for persons. And when research is not conducted, or is done badly, it leads to badly designed programs.

A Better Way

In other areas of life, we see similar bad practices. Reporting and photography exposes people in their worst moments. Exploitative businesses manipulate us into paying more than we should. Instead of asking what people need, charities hand out their donors’ idea of what a community ought to want. Police exert unaccountable power over the lives of young men. Border

guards coldly obstruct those with weaker passports from the trip of a lifetime. These are examples of institutions disregarding the dignity of people and doing harm. They are examples of failures to design processes that support those with the least power, and to listen to those people about what they want from those processes.

This does not have to be the case. There are practical measures we can take to build dignity into our lives, our work, and our advice. Good design and good research can see and serve the full complex of humanity. Yet our disciplines can easily fall into manipulative and agency-diminishing nudges, designs that serve clients more than users, and research practices that result in disillusionment and anger.

But when we get it right, we see much more positive consequences.

In this chapter, we'll explore those positive consequences we've seen in our research at IDinsight's Dignity Initiative and the wider literature. We'll then look at how this links to other efforts to improve the world, including behavioral science and design. Finally, we will give you some quick tools for implementing dignity in your own work.

Defining Dignity

First, though, we need to briefly deal with the big conceptual issue: how do we define dignity? There are lots of ways to approach dignity. It has been important to virtually all cultures and philosophical traditions. No one definition can capture the full variety of meanings, but we've attempted what we call a "workhorse" definition in the box below. It's not perfect, but we have found it practical in our own work.

Dignity is the moral worth or status of all persons. We can elaborate to say that it is “a trait universal to all humans, which is inalienable, inherent, and unearned.

Recognizing the dignity of a person requires us to treat them in a way that respects their dignity. When we fail to show that respect for dignity, the disrespected individual can appeal to the wider society for redress.” (Wein, Lanthorn & Fischer, 2022)

There are a lot of concepts packed into that definition, but the key feature is that dignity doesn't go up or down. It is neither earned nor stripped away by our lives and choices. Instead, what goes up and down is whether someone feels their dignity has been respected. That has important implications for how we design interventions and measures.

Fully respecting people's dignity requires listening to what they are telling you about their lives, situations, and the experiences they have in encountering your service, process or product. That's the sort of thing lots of behavioral scientists are good at, especially when they draw on ethnographic and thoughtful qualitative techniques.

There are countless hypotheses about what might be respectful of people's dignity, and the right answer will vary by culture, person and situation. However, in our research we have seen that there are three pathways that seem to recur across multiple traditions of dignity (Lamberton et al., forthcoming). That these are relevant has been born out in our research with American citizens (Lamberton et al., forthcoming), and with South Sudanese refugees in Uganda (Kahura et al., 2022).

For each of these pathways, we ask ourselves two questions:

- **Recognition:** Do they feel seen by, and see themselves in, the institution that they are interacting with?

- ▶ **Agency:** Do they have choices, and a meaningful chance to consent, around this interaction?
- ▶ **Equality:** Do they feel treated as fundamentally equal, and were genuine efforts made to reduce power asymmetries?

These three pathways are not fully validated, and they will not capture every tradition of dignity. That is why we do not include them in the formal definition. However, we have found them to be a practical starting point for brainstorming ways to better respect the dignity of others, and for structuring our thinking and our research around this topic.

Consequences of Dignity

Why must we treat people in a way that respects their dignity? The overwhelming, urgent answer is because it is intrinsically morally good to do so, and a fulfillment of the values we commit to in our fields.

There are other powerful arguments for treating people in the way they would want to be treated. One of these is the democratic argument. People across the world tell us they value dignity, and they frequently feel treated with disrespect. Respecting people's dignity also engenders many positive consequences for wellbeing, health, service uptake, and other social outcomes.

The following points summarize the general literature around the area:

- ▶ **Well-being:** Studies have shown that dignity experiences can have important impacts on wellbeing in multiple parts of the world. An increase in experienced disrespect was associated with lower wellbeing and empowerment (Wein et al., 2022). Experiencing disrespect was associated with a wide range of negative emotions and some self-reported mental health

conditions (Lamberton et al., forthcoming). Analysis of a recent online experiment in the US also shows that writing about examples of dignity denial causes people to feel angrier, less hopeful, less trusting of others, and lower in status (Lamberton et al., forthcoming).

Research also shows that self-affirmation interventions to counter dignity threats can affect wellbeing, other cognitive outcomes, intentions and (health) behaviors (Cohen & Sherman, 2014). Research has shown that dignity messaging reduces the perceived stigma of receiving charity (Thomas et al., 2020), and is also highly valued in the workplace because it affects wellbeing (Sayer, 2007; Wein et al., 2022).

- **Health:** Healthcare is one field that has gone furthest in studying dignity, because there is such potential for failures in this exceptionally intimate domain of human life. Being treated with dignity is associated with higher satisfaction, adherence, and receiving optimal preventive care (Beach et al., 2005). Unsurprisingly, respectful patient care is also associated with higher trust in information provided by healthcare professionals (Duke et al., 2017) and in healthcare providers overall, which contributes to more open sharing of health issues (Lamberton et al., forthcoming). It boosts satisfaction, sense of meaning among patients and families (Fitchett et al., 2015).

In a recent study in Nepal, the Philippines and Kenya, respectful treatment by medical professionals is associated with small but significant increases in covid vaccine uptake (Busara, forthcoming). These lessons are important for health but may also imply lessons for related categories of service delivery.

- **Service uptake:** All kinds of service interactions, bureaucracies and private sector enterprises can benefit from considering dignity, with evidence suggesting that respectful service boosts uptake. Dignity increases the desire to build skills (Thomas et al., 2020), and willingness to take part in future research (Lamberton et al., forthcoming). Disrespectful financial service interactions leave people unable to access formal financial services and in greater debt (Wherry et al., 2019). Dignity-focused advertisements are rated as more persuasive (Lamberton et al., forthcoming). Fear of dignity denial causes people to be less likely to recommend a health service (Lamberton et al., forthcoming). Disrespectful maternity care is one of the most important drivers of lower uptake of formal health services (Bohren et al., 2014; Jungari et al., 2021; Minckas et al., 2021).

Disrespectful experiences have led to people taking more time to research consumption options, while respectful experiences led to reported customer loyalty and positive reciprocity, such as positive reviews and larger tips (Lamberton et al., forthcoming).

- **Other social returns:** There are a range of other benefits to the functioning of societies, with reduced intergroup conflict and better democratic outcomes. Treatment that respects people's dignity leads to more ingroup cooperation (DeCremer, 2002; Sleebos et al., 2006), positive intergroup actions (Lalljee et al., 2009), greater punishment of moral transgressions (lab, UK, Laham et al., 2010), more empathy (Lamberton et al., forthcoming), reduced interethnic conflict (Huo & Molina, 2006), reduced conflict between sports players (Prestwich & Lalljee, 2009), reduced political

tribalism (Lalljee et al., 2013; Voltmer & Lalljee, 2007), higher civic engagement (Boeckmann & Tyler, 2002), greater tolerance (Simon et al., 2015), and better functioning democracies (Paller, 2019).

Cash transfers, which were considered respectful, were linked by participants to reduced intercommunal tension (Kahura et al., 2022). Post-study feedback led to greater reported empowerment (Schilling et al., 2021). Disrespectful experiences had led to people breaking off friendships (Lamberton et al., forthcoming).

Dignity matters. When we treat people in the way they want to be treated, it has real impacts on the world. This conclusion, however, is still based on a collage of qualitative, laboratory and online experiments. Those are powerful tools, but we are still waiting on the “slam dunk” causal inference for this claim in a real-world context.

We hope to see well-designed randomized trials in the coming years that more clearly establish which actions best increase perceptions of respectfulness, and what the consequences of that are. We lack research from many locations, with the literature assuming its typical WEIRD bias. We should especially look to learn more in Latin America, East and Southeast Asia, and West and Central Africa. Still, we feel confident in the value of dignity to people worldwide.

Tools for Dignity

Let’s get practical. One tool we can offer readers to implement ideas of dignity is a validated measurement scale. Efforts to improve our ability to respect the dignity of others must be informed by a credible understanding of how our actions, programs, and processes are experienced by others, *especially* in interactions of power asymmetry.

However, for the reasons mentioned earlier, dignity can be difficult to define, and therefore hard to measure, but there is a tool you can use. The “felt respect for dignity” scale comprises five items for surveys and measures whether people experienced an interaction with an organization as respectful of their dignity.

It has been shown to perform well across three countries that have very different traditions of dignity: the US, Morocco and China. It performs well across three varied interactions: financial services, policing, and healthcare. This is true across the demographic spectrum, with the scale performing well with people of different genders, ages, income levels and educational backgrounds. (Details about the development and performance of the scale are available in Wein et al., 2022).

Felt Respect for Dignity Scale

Thinking about the interaction you just described, please rate your experience on the statements below:

The organization treated me with dignity.

1. The organization representative listened to my requests.
2. I felt respected by the organization.
3. I felt valued by the organization.
4. I felt supported by the organization.

(Strongly disagree / Disagree / Neither / Agree / Strongly agree)

A second tool an organization may employ is a dignity workshop. Organizations need to start their dignity journey somewhere. That’s why we have found thoughtful workshops are an essential part of the toolkit. They help everyone on a team get to grips with some hard concepts and hear from their colleagues what an important value this is, and how often they may feel their personal dignity is not upheld. They allow us to generate initial ideas for addressing dignity failures in the organization’s

work. At their best, they build a coalition for change, allowing us to take on more ambitious work with a group that is ready to see and hear the unique people their work serves.

A third tool is what we call a Dignity Audit. These may help organizations in the social sector to examine whether they are respecting the dignity of those they serve. This tool looks at the processes, decisions, and behaviors that impact dignity outcomes at every stage of the program, from initial conception and design to implementation and evaluation.

The Dignity Audit we developed at IDinsight is a modular tool that attempts to do this work by paying attention to whose voice gets included and elevated. We work to ensure that the Dignity Audit is less a judgment of an organization's work and a more collaborative effort to align actions with aspirations. It uses the felt respect measure shared above, but also builds on the work by other thinkers across disciplines to ensure that issues are discovered and organizations are empowered with the information they need to take corrective action.

Conclusion & Key Points

There are many parallel efforts to understand and improve the world. International Development aims for:

- ▶ Empowerment
- ▶ Agency
- ▶ Accountability
- ▶ Localization

The entire world is wrestling with the imperatives towards racial and gender justice and ridding ourselves of the lasting effects of empire. These are projects we value and support.

We hope that dignity equips them in additional armor. We hope our work towards dignity can help us see more clearly the people, in all their complexity, at the end of ideas and processes

of development. That hope comes in part from the particularly comprehensive intellectual framework that dignity offers, in combination with the practical implications and recommendations that it has yielded in our research so far.

The practical work of building a dignity agenda also seems to parallel many of the disciplines of the members of the Bescy. Behavioral science, human-centered design, user experience, testing, feedback and all kinds of research can all be fruitful ways of approaching a more respectful set of processes.

Yet, those are all also tools that can easily be turned towards manipulation, extraction and the service of an institutional client at the expense of the good of the end user. Our dignity agenda seeks to collect perspectives on why and how we can use these tools for the good of humanity.

Much of this has implications for research ethics, where we borrow the term “respect for persons.” We think discussions of research ethics have often been overly focused on processes and pronouncements by august thinkers and have too rarely included the voices of those who are harmed when research goes wrong (Mumo, Owsley & Wein, 2021). Dignity prompts us to refocus on the priorities of research participants, most importantly of which are processes of consent and post-study feedback.

Dignity is easily promised, but harder to deliver. We challenge all our readers to think about how they can uphold the promise to respect the full humanity of those they work with and those they seek to serve. We hope that the work of the IDinsight Dignity Initiative, and the discussion and tools presented here have both persuaded you of why that is so valuable a path to tread and given you a map to take your first few steps along it.

**Next we explore Part III to learn more
about the landscape of behavioral science
in international development.**

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Part III:

LANDSCAPE OF BEHAVIORAL SCIENCE IN INTERNATIONAL DEVELOPMENT

The application of behavioral science has swiftly permeated many fields of international development and regions around the world. Practitioners and spectators seeking to understand the landscape of its actors and organizations must first understand that the landscape is always in flux. As funding, values, missions, and goals shift according to new insights and strategies, so too do the organizations that engage with its tools.

This context offers natural questions:

- ▶ Which actors are involved and where are they located?
- ▶ What is their work and how do they accomplish it?
- ▶ What lessons can we learn from them?

In this section we attempt to answer these and similar questions. We present a series of chapters aiming to identify the landscape of development, how to build a global team, how to integrate behavioral science within the context of the organizations, and how we can bridge knowledge built in academia, government and the private sector.

11.

Landscape of Development Professionals

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Luis Artavia-Mora, *Independent consultant and Bescy*

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The body of evidence connecting international development and behavioral science has skyrocketed over the past decades. With every new discovery, there is novel evidence about how and where we can implement behavioral tools and policies in a given context.

This growing enthusiasm about the evolution of the field has promoted the creation of many new teams, units, centers and many players employing behavioral science in their work. While they may be unique in their efforts, they all aim to design products, services, communications, and policies that promote positive behavior change. This introductory chapter traces the landscape of players to contextualize the growing body of work in this evolving field.

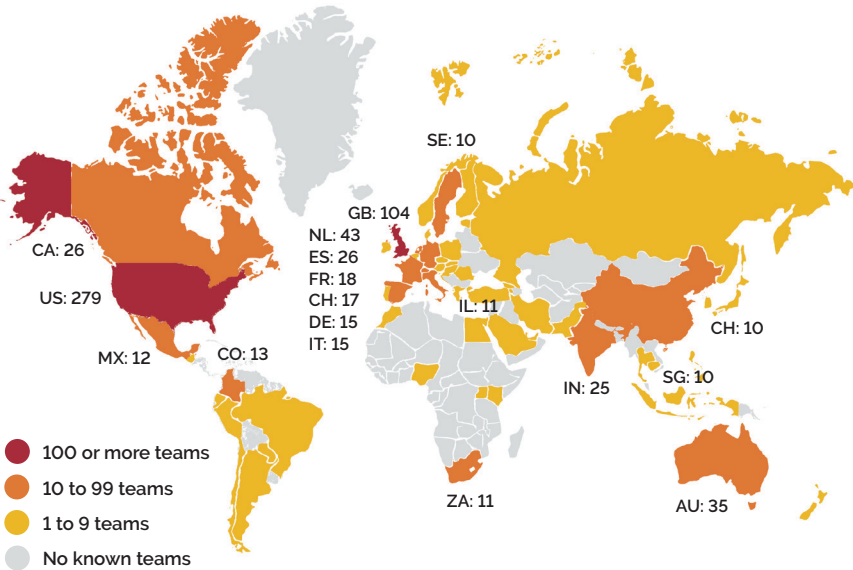
Two resources provide a useful overview to help understand the current landscape of groups, their scope and location. Bescy

in partnership with Busara Center, Applied Behavioral Science Association and Habit Weekly began compiling a directory of behavioral science organizations in 2020. As of July 2023, this directory contains a master list of 820 actors, and somewhat detailed information from over two hundred teams (see Figure 1 below).⁵² Another key resource is the OECD report on Behavioral Insights and Public Policy.

Image 1: BehavioralTeams.com free map of behavioral science & design teams around the World

BehavioralTeams.com

BehavioralTeams.com hosts a public, free map and database of behavioral science & design teams around the world, developed by Busara and the action Design Network Data shown as of 18 July 2023



⁵² The information and map are available at: <https://www.behavioralteams.com/teams/>.

This literature provides an initial list of organizations which was reviewed and amended by professionals and academics working at the intersection of international development and behavioral insights. These efforts resulted in a final list of 219 groups or organizations (bodies) which are involved in applied behavioral science, behavioral public policy, and/or developmental efforts.⁵³ These two lists show a wide variety of active organizations working in many countries around the world.

The evidence in these lists guide us to distinguish seven categories of agents:

1. **Funding Organizations**

These organizations typically provide funding and can be housed in or associated with governments or philanthropies interested in or supporting behavioral teams. They sponsor international development projects and may have their own team dedicated to supporting projects, aggregating insights, building networks, and producing content that helps shape the field.

The Mind, Behavior and Development Unit at the World Bank, The Behavioral Economics Group at the Inter-American Development Bank, and the United States Agency for International Development (USAID) are three examples in this category. Organizations in this class can intersect with related structures, including Non-Governmental Organizations and Intergovernmental Organizations.

2. **Nonprofits and Non-Governmental Organizations (NGOs)**

These organizations achieve humanitarian, social, or political goals, integrate nonprofit structures, and

⁵³ The OECD's report is available at: <https://www.oecd.org/gov/regulatory-policy/behavioral-insights-and-public-policy-9789264270480-en.htm>

are outside of government control. They are often classified by their orientation (i.e., type of activity such as human rights, environmental protection, health or education) and level in the scale of their operation (i.e. local, regional, national or even international). In this sense, these organizations host teams of behavioral scientists to help accomplish their goals. Some examples of nonprofits and NGOs hosting behavioral experts include Save the Children, and The Brookings Institute.

3. **Inter-Governmental Organizations (IGOs)**

An agreement between members of a group creates IGOs through a common treaty or legal instrument that acts as the rules that govern them. IGOs may vary in membership criteria, function, goals and missions. These are often described in the core treaty among members. IGOs with a behavioral group include the Behavioral Insights Unit at the World Health Organization, the Behavioral Unit at the Organization for Economic Co-operation and Development (OECD), and the United Nations Behavioral Science Group.

4. **Consulting Organizations**

Broadly speaking, these organizations receive external funding to conduct research, design interventions, and provide measurement support to behavioral science projects in international development. It can include consulting and private sector companies that either specialize in behavioral science for development or house a behavioral group within a much larger organization matrix. These organizations have various structures and can be for-profit or

nonprofit and can be big or small. Examples include Innovations for Poverty Action, Busara, Ideas42, Rare, Evidence Action, and the Behavioral Insights Team.

5. Government Teams

These can be both funders and recipients of funding. However, groups in this category are typically funded by, and act as, official bodies of a government. Often, they create their own internal behavioral science teams or sponsor behavioral science work in the field that is directly associated with a government department or unit. A famous example in the Global South is the Indian NITI Aayog founded in 2015. Similar efforts exist in Argentina, South Africa, Brazil, Lebanon and Israel.

6. Academic Research Centers

Their work specializes in generating and publishing novel academic behavioral and social insights including new techniques for their improved application and measurement. Indicative examples include the MIT Abdul Latif Jamaal Poverty Action Lab, the Chicago Booth Center for Decision Research, Duke University's Center for Advanced Hindsight, and the UCL Center for Behavioral Change.

7. Other Mission-Driven Enterprises

This category distinguishes a few other mission-driven organizations not fully grasped within the categories above. They include those entities that aim to make the world a better place within a specific niche of work but do not house a specific group or do

not explicitly lead a behavioral effort on the matter. Opus Insights is an example focusing on agricultural issues and overlapping work on environmental, business efficiency, international development, and behavioral science. As the field of applied behavioral science within international development evolves, aspect-themes may appear among these organizations which allow more precise categorization.

Organizations in each of these seven categories have distinct roles to play in international development and take different approaches to their work. The following chapters provide evidence and lessons from each of these entities to help current and future practitioners to navigate their own work.

In the next chapter we learn more about the challenges, opportunities and ways to build a global team. Experts working at the Busara Center will share their experience and knowledge.

12.

Building a Global Team

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Stanley Ngugi, Busara

Deqa Aden, Research and Public Policy Professional

What is a global team? One way to define it could be saying it is a team that works globally—all across the planet—or that the team represents the globe by comprising people from many corners of the world.

Surely, that sums up what a global team is. But does it?

For us, that answer does not cover it. Building a global team is much more complicated than just hiring people with different passports to work in different countries.

In this chapter, we speak from the perspective of our experience at Busara. Busara thinks of itself as a behavioral research organization with a Global South identity: we work and primarily live in the Global South. We aim to produce research informed by behavioral approaches that reflect deep contextual knowledge. Our goal is for that knowledge to lead to better policies and programs that fall under the broad heading of international development.

Building a global team requires appreciating the subtext of

terms such as “global” and “international,” as well as the origins of dominant ways of thinking in this sector. It requires revisiting what categories an organization uses—such as looking at nationality or living location—to create diversity and inclusion.

Often underappreciated in this categorization exercise is that the location of where someone received tertiary education might be a crucial determinant in how they experience their role in a knowledge creation effort. We also suggest that building a global team needs to focus on “boring” stuff, including policies, processes and the everyday practice of inclusion and diversity.

Loaded Terminology

In international development, the term “global” carries subtext, as does the term “international.” Not everyone is part of a big global family. Some internationals are more international than others.

For example, some funders do not consider an organization “international” unless it is headquartered in Europe or North America. Everyone else is “regional,” no matter how far and wide their offices are spread across the globe (in our case India, Kenya, Nigeria, Tanzania, Uganda). This points to the need to unpack some power dynamics that underpin the notion of globality and internationalism.

Recently, scholars have suggested moving away from the descriptor of “international development” (which suggests that development is most needed in a set of previously called ‘developing countries’) to using terminology of “global development” to denote that inequalities are a problem not just between countries, but within countries—including those that were previously considered the set of “developed”

countries (Horner & Hulme, 2017). This shift in terminology might be helpful in foregrounding the shared challenge.

Yet, for the purpose of highlighting deeply historical inequities, the term “global” has also been qualified in other ways. Terminology of Global North and Global South draws a dividing line not along the equator, but along the historical role played by a country. The Global North, in this definition, represents the countries whose histories are those of conquering and imperialism and which today sit on the upper end of the Human Development Index (HDI). That is why, for example, New Zealand, usually in the southernmost corner of the world map, is part of the Global North.

The Global South comprises the countries populating the lower ranks on the HDI: the countries that were conquered and explored for the development of others. That is why Haiti, in the northern hemisphere, in this classification is part of the Global South. There is no even split between the imaginary hemispheres: 85% of the world’s population lives in the Global South, making this the majority category (World Population Review, 2023).

So, is a global team one that represents people from either imagined hemisphere? Or is there more we need to consider?

There is more.

The term “global” is deceptive. It suggests an equal coexistence on the same planet. Yet it is not a unifying adjective. While the terminology Global North/ Global South is now widely used, it is often done without unpacking the implications of it—such as the fact that it hides inequalities within these categories or carries a subtext (Haug et al., 2021).

For us, to build a global team means to speak openly about power, about representation, about knowledge hierarchies and

how they influence the research work we do — and about how to counter the status quo of these through deliberate processes. Rather than leveling differences, the descriptor “global” can also highlight them. We need to bring these differences into the foreground when building global teams.

Why Global Development Struggles to Build Global Teams

Development should be inherently global, inclusive and international. In recent years, there has been increased emphasis on it being locally led (Smith, M., 2022). This is easier said than done.⁵⁴ What exactly development is or what is beneficial for people is still primarily decided in countries that are not the target of development programs. Research for international development continues to be commissioned through funding mechanisms that prioritize international organizations over the so-called regional ones.⁵⁵

In this hierarchy and resource pyramid, it is still commonly the case that research steps are split up in a specific way. An international team (with a headquarter in the Global North) designs the research and owns the analysis, while data collection is done by a team national to the Global South country in which the research is conducted.

⁵⁴ See, for example, Ingram (2022), and Craney & Hudson (2020).

⁵⁵ This is exacerbated by the fact that national spending on research differs drastically: Global spending on Research and Development (R&D) was around \$2.7 trillion in 2020, but only ten Global North countries dominated the knowledge production, accounting for 85% of the global total (Congressional Research Service, 2022). The majority of African countries spent less than 1% of their gross domestic product on R&D (World Bank, not dated). While this does not mean that the research conducted under Global North auspices was conducted elsewhere, this nonetheless skews what is considered acceptable knowledge and research approaches, as it is so driven by Global North perspectives.

Of course, there can be nationals of different countries on either team, but this does not challenge the dynamic. While there is both an articulated and unspoken consensus that this is not right, the reality has remained unchanged.⁵⁶

Why?

One reason might be because research efforts or other development work are currently not supported by genuine efforts to build global teams. Building global teams goes beyond balancing between the national and international staff categories. It must start by analyzing how power manifests itself in hiring for international development work. It must also recognize which identity categories are relevant for the diversity the team needs to represent.

Right now, global teams are often characterized through crude categorizations. If you are local to a Global South country, you are adding to diversity within an international team. But this kind of diversity thinking does not shift the underpinning power dynamics with which development struggles: Global South and Global North team members hold different kinds of power. Right now, Global North knowledge interests continue to rule because they support maintaining ways of working and funding in international development that uphold the existing power dynamics.

The international development sector, while full of well-intentioned people, continues to be quite homogenous in whom it represents in its team and the beliefs they uphold, including what it considers rigorous and viable ways of knowledge creation. The roots of this go deep and can be found in entrenched knowledge hierarchies.

⁵⁶ See, for example, Olufadewa et al. (2020) and, Reidpath & Allotey (2019).

The Role of Knowledge Hierarchies

International development might be uncomfortable with its imperial background. But tackling knowledge hierarchies requires acknowledging that bettering people and unpacking their lives through research has deep colonial roots, (Smith, L.T., 1999) and is also anchored in a mental model that prioritizes transactions and economic growth, which reflects capitalist and usually neoliberal approaches (Schomerus, 2023). Knowledge production continues to celebrate particular ways of knowing or solving a development problem that follow Western or Global North traditions, which then reinforce global power relations (Grewal, 2001).

University cultures in the Global North dominate definitions of what makes up knowledge and good research. These definitions often exclude the narratives or ways of knowing of scholars (i.e. other manifestations of what constitutes research) or ordinary people from the Global South (Medin et al., 2017). In fact, the dominance of ideas is so strong that it can seem impossible for those from the Global North to imagine that there are different ways of knowing, of being, or of asking and answering questions than the one they favor.

Many people working in development are deeply conditioned to think the world works only in one particular way because of who they are. Often from a background that Henrich et al. (2010) have so memorably called WEIRD, they are from countries that are Western, Educated, Industrialized, Rich, and Democratic (WEIRD) (Henrich et al., 2010).

Coming from this background, argues Henrich, means carrying a legacy of centuries of rule and institution-building that has deep influences on social structures and economic systems, but was not mirrored in non-WEIRD countries, which often continue to be kinship-based (Henrich, 2020). The notion of

the WEIRD person primarily teaches us how deeply our origins condition us, which translates into the role a person plays in a global team.

However, there is an imbalance here: while WEIRD perspectives dominate global narratives and mental models of international development, numerically, WEIRD people are the minority. This is why we must prioritize hiring with an acute awareness of backgrounds. We must ask ourselves: what has shaped a person? Are they attractive to the global team because they are from the place in which development work is to be implemented *and* because they are fluent in WEIRD work cultures and mental models of international engagement, thus making them an easier team member for colleagues from Global North countries?

This is reminiscent of Fanon's (1967) argument about Africans learning another language (in Fanon's argument, French) "because it is the key that can open doors" but is also "evidence of dislocation, a separation" — but a separation that serves the purpose of Global North-driven development narratives (Fanon, 2016).

Is the purpose to hire someone who can seem representative, or to hire someone who can push development practitioners to make their work context specific and locally owned? Someone who can challenge knowledge production in, for example, Africa with queries about what research requiring context seriously means? Someone who recognizes the need for a new contextually relevant type of research, rather than building capacity to produce research in the image of traditions from elsewhere?

Answering these questions requires rethinking diversity towards establishing relevant (and maybe not obvious) categories.

Rethink Diversity Along Relevant Categories to Create a Global Team

In hiring for international development research, the language of diversity and inclusion usually represents an attempt to acknowledge differences and the need to create a team that has different kinds of connections to the people being researched.

A prominent manifestation of diversity approaches is thus to hire local people (whose nationality matches their workplace) and have them represent different categories like gender, race, age, or ethnic background. However, this crude categorization (which is replicated in how the international development community targets its so-called beneficiaries) comes with various definitional and managerial challenges (Schomerus, 2023). Teams that are put together based on these categories might not represent the work they are supposed to do.

We thus have been rethinking diversity and inclusion as relevant to the work we aim to do: produce knowledge that is behaviorally informed, contextual, localized and relevant to the lived experience of the people being studied (who are in the Global South), while still usable for global development (which works off of ideas rooted in the Global North).

Building a global team requires being completely transparent about these differences. This has to start with an operationally relevant definition of what diversity is for an organization that wants to conduct research in context-specific and appropriate ways while still creating deep knowledge. But it requires moving beyond categorizing people by their geographical origin to also recognize the particular path that an individual has taken.

Sometimes, this path represents considered choices. Sometimes this path means taking whatever opportunities were available. Having been able to make a considered choice is a marker of privilege, and it is a privilege most often held by Global North staff.

Within a global team, Global North staff usually have options. Based on their backgrounds, they made a conscious decision to work in a global context (usually meaning away from their home country) and to apply their skills in an environment foreign to them. Being global with this background is often a value-driven lifestyle choice.

Staff from the Global South, however, become part of a global team as a recipient of the globality of the organization. They are the local hires that are expected to bring the credibility of local knowledge. They have less of a choice of being a global team. They are just hired into one that thinks of itself in this way. Being in a global team with this background is often a livelihood-driven matter, which creates a unique relationship to the work.

This crucial difference means that within one global team, people are on very different footing. This difference needs to be spelled out and its implications must be understood.

Identity and Backgrounds Matter

During the hiring process, several identity factors play a role when seeking to build a global team, including:

- ▶ Demographic identities such as country of origin, race, gender, and even religion
- ▶ Educational background, including the type of degree obtained and at which educational institutions
- ▶ Experience or the number of years working in the relevant field for the position

These are all important elements, as staff will bring their demographic identities, their educational background and their experience into the work.

This is, however, where we have learned to question our assumptions about broad-brushed categories and to realize

that there are other parts to the identity of team members that require attention when building a global team. In an internal exploration of our organizational identity, we realized that beyond country of origin, race, gender, age, ethnicity, or experience, there is another marker that deeply matters; where you were educated. This intersects with geographical origin in ways that deeply influence a person's perspective on their work and how to conduct it.

In an internal explorative research project, we looked at how the makeup of the team at Busara supported or hindered the ability to produce contextualized research. This question led to the examination of the meaning of the term Global South as a foundation of Busara's identity and how this shapes the team and their view of the work they do.

We realized we needed to look at how to introduce the categories of Global North and Global South into our thinking on diversity and global teams. In this examination, we categorized people into Global North (GN) and Global South (GS) – meaning where they were born and grew up as a relatively average member of that society.

A striking difference was in how people thought of themselves in relation to their work. GN staff members described themselves through their values and educational accomplishments. They work in international development research because they believe in the mission and thus made a choice to do so.

However, GS staff broadly used their ethnic identity, country of citizenship, and gender to describe themselves and highlighted that their origin made development challenges personal to them. They think of their work in terms of context, social norms, family traditions, and their own culture and habits. The value-based identity of GN seems rooted within an abstraction of the problems their research seeks to address, while the location-based identity of GS staff has developed from lived experience.

We further looked at what we considered a crucial intersection: whether someone is GN or GS and whether someone was educated (primarily in tertiary education) in the Global North or the Global South. This helped us create three identity categories that we came to recognize as crucial when build a global team:

1. Global North staff educated in the Global North (GNGN)
2. Global South staff educated in the Global North (GSGN)
3. Global South staff educated in the Global South (GSGS).

In a clear nod to global power dynamics and prioritization of certain types of knowledge, we did not have anyone who fit the category of Global North educated in the Global South (GNGS).

We learned that people from the three intersectional categories approach their work with different intentions, values and even understandings of their own identity and contribution to knowledge creation. GSGS use their academic training to create contextualized knowledge, while GNGN use their academic accomplishments to ensure technical rigor in research.

Though both points of references are needed for good research, a global team requires valorization of both contributions in equal ways. This does not happen automatically because of the existing power dynamics that think of expertise as something that is learned in educational establishments, while local knowledge is gained by being from a particular place and thus somewhat anecdotal.

The implication is that GSGS staff members experienced their localized educational background being deemed less relevant for decisions on research designs, where technical rigor

imparted through education in the Global North ultimately won out. GSGN often sit in the middle. They are viewed by outsiders as a local diversity hire, but are deeply rooted in their GS context, while having been shaped by a GN education and particular version of values and beliefs.

How might this play out in practice?

Imagine a research team working in a Global South country is tasked with research that informs a new program to empower women by getting them employed in jobs. What perspectives might the different researchers bring to this somewhat reductive scenario? GNGN might be so focused on the technical questions of the research design that questioning the underlying mental model does not occur. Plus, they might be more aware of the growing interest in studying gender empowerment. The World Bank, the largest supplier of development capital, commits millions per year to projects that aim to close this gender gap, so studying it seems to be supported by more than just intellectual trends (World Bank, 2013).

GSGN might have experienced the contradictions between how they experience life at home and what they were taught in Global North universities. They might recognize that the underlying assumption of the proposed empowerment program is informed by a notion of the family structure that is central to capitalist means of production, where women's empowerment is imagined through levels of education and participation in the labor market (Radhakrishnan & Solari, 2015).

GSGS might know this is unlikely to actually empower women because they have never seen that the concept of employment-driven empowerment has meaning in, for example, most African countries that are not as industrialized as America and Europe. They might have a hunch that the capitalist Western ideology of empowerment takes a space away from African narratives of empowerment that have little to do with the labor market.

In this example, international development research suffers from a lack of genuinely seeking to make research and solutions to problems locally owned.

Inclusion and diversity in a truly global team must mean that these dynamics are brought out in the open, including the culturally driven behavior that comes with these categories. In our organization, which was founded after GN models and with GN founders, certain characteristics, such as being outspoken, are valued. These are often not that easy to fulfill for GSGS, whose cultures place more value on respectful behavior and less value on dissent and debate, and the nuances in which debates and discussions occur in different cultures.

Within this set-up, often GSGN are intrinsically fulfilling the role of translators, implicitly knowing how open feedback and outspoken critique are atypical and often highly uncomfortable in a GS context but having honed their communication skills through a GN education, thus widening the channels for discussion and generating verbal contributions. However, it is a role that can be exhausting — and not necessarily strategic — to play.

These elements are important when building a global team. Behavioral scientists know the impact of people feeling alienated.

Staying within a GN outlook, we can see the issue that arises when these dynamics are not brought out into the open and then addressed as a productivity challenge. Those dynamics include:

- ▶ Feeling alienated.
- ▶ Feeling as if one is working under an inappropriate ideology.
- ▶ Failing to recognize that there are different identity-based categories of valorization that can make people less effective, efficient, or interested in their work.

Not putting in the effort to understand what makes a genuinely exclusive global team creates productivity issues for the team.

Conclusion & Key Points

Building a global team might sound like quite a flashy undertaking. That flashy image is fueled by the imagination of what international means that calls up GN-influenced images of different countries and travel. In this imagery, the hiring process is the end point of the attempt to build a diverse team.

Once we have enough women, we are diverse. Once we have enough people from the place where we are working, we are contextual. Once we have enough technical experts that bring rigor, we can produce good research. Positive answers to this often come at the end of the team-building exercise.

But to create diverse, inclusive teams, the hiring process should be the starting point of the practice of global team building. A truly global team cannot start from the international imagery. Rather, to build a global (and with that diverse) team requires all the things that on paper look boring, like:

- ▶ Asking questions
- ▶ Understanding how background influences choice and thus the relationship to work
- ▶ Understanding how identity (and not feeling recognized) can alienate people and diminish certain voices while working together

Diversity is a never-ending process, and it requires commitment to intentionally and consistently shape an inclusive environment that acknowledges the power dynamics between GN and GS as well as the more nuanced GNGN, GSGN, and GSGS categories.

Each identity might have different limitations. GNGN might mostly operate from an academic, theoretical perspective. GSGS

might offer the richest contribution in terms of context and lived experiences when producing knowledge on and in the Global South. Achieving diversity comes from everybody acknowledging the contribution of the other. The presence of all categories produces quality, contextually knowledgeable research.

Below are suggestions for practices that could be transformative when building global teams to conduct research for international development. They point towards human resources policies and bureaucratic decisions, but they are only seemingly boring. They are the scaffolding on which a genuine commitment to global teams hangs.

► **Hiring practices.**

How do people actually get hired for a job? What is the role of membership of specific networks and how does one gain access to these networks? Who gets to jump several steps in the hiring process simply because of who they are, who they know, or how they expect the hiring process to go, based on their backgrounds?

► **Pay practices.**

Teams with people from different origins tend to somehow acknowledge that difference through pay. In a truly global team, what does equal pay mean for GN and GS staff, or for staff in different regions or countries within the same hemisphere?

► **Benefit practices.**

What work benefits are open for GN and GS staff without reducing their work value? How do different living situations create different needs? For example, what staff members would like to see an organization spend money on a Christmas party versus paying a bonus (however small) that can be used for family Christmas expenses?

► **Organizational cultural practices.**

How are differences spoken about in the organization? What does language and settling on how terminology is used across an organization mean? How are the boring procedural elements that create the possibility of diversity and inclusion lived every day? Are mentoring systems in place? Do they allow mentoring of staff members in different perspectives? For example, can a junior GSGS staff mentor a senior GNGN staff, or are there unspoken power differences that stand in the way of learning?

Frank and continuous reflection on these questions goes a long way towards understanding what it will take to build truly global teams. Such a team does not simply take a differentiating notion of global as a given and does not think of the term “international” as glitzy and thus requiring little reflection. A global team does not dwell on passports or simplistic identity categories but pays attention to the paths of life that brought someone to the team.

In the next chapter we learn more about starting, integrating and leading behavioral science projects at an NGO. We will learn from experts working at CUBIC, Save the Children.

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13.

Integrating Behavioral Science at an NGO

Allison Zelkowitz, CUBIC, Save the Children

Jimena Llopis, CUBIC, Save the Children

The Center for Utilizing Behavioral Insights for Children (CUBIC) is the world's first applied behavioral science team focusing on the rights and welfare of the most marginalized children. Hosted by the Save the Children International Asia Regional Office, we function as a centralized internal team that partners with Save the Children's offices across the globe. CUBIC is a very recent development in Save the Children's long history. Founded in 1919, Save the Children is one of the world's oldest international aid organizations, now working in 118 countries with approximately 25,000 staff.

When CUBIC launched in April 2020, one of our core objectives was to introduce the Save the Children movement to the applied behavioral science approach. Introducing this approach to such a large and complex organization is an ongoing and ambitious task. Having worked towards this goal for over two years, we are using this chapter to look back at what we have

learned from the process so far and reflect on what is needed to support other organizations with similar ambitions.

Create a Team and Process

As of 2022, the CUBIC team included 13 full and part-time staff based on four continents. Most of our team members have a background in economics, psychology or another area of behavioral science and they bring strong qualitative or quantitative analysis skills – or both. Equally important, however, is the ability to lead and *drive* projects, and to build positive and enduring working relationships with our partners on the ground (e.g. our Save the Children colleagues based in Indonesia, Philippines, Cambodia, Kenya, and Cote d’Ivoire, among others).

However, we were a much leaner and less experienced team when CUBIC launched. Our remote team of 4 members included our Director/Founder (Allison Zelkowitz, one of this chapter’s authors), who had been working in international development for 14 years, plus a recent behavioral science master’s graduate, a researcher with some trial experience, and a part-time graphic designer. While we had a range of necessary practical experience and subject matter expertise, none of us had actually implemented a real, end-to-end behavioral science project.

From the outset, we knew we wanted to focus our time and energy on external, program-based work, rather than just internal organizational effectiveness projects. This focus emerged through the ideation and pitching process, as the idea for CUBIC took shape and won internal innovation funding. We then spent our first year defining our method, which is a combination of the OECD’s BASIC Framework, the Behavioral Insights Team’s TESTS (Target, Explore, Solution, Trial, Scale) process, and elements of design-thinking (i.e. primarily prototyping and user testing). We also brought on two part-time

secondes from the Busara, who helped us better understand and actually apply this step-by-step process.

Alongside developing our specific behavioral science approach, we have learned that conducting research in the context of Save the Children rests on effective collaboration. Today, project management, adaptability, problem-solving and interpersonal communication skills are as critical to our work as understanding the latest developments in behavioral science. We have found that success requires recognition of the extremely dynamic nature of our programs, where the external context and landscape change often and staff are working on multiple humanitarian or development programs simultaneously. We can take nothing for granted when introducing novel approaches to applied behavioral science into such settings.

Galvanize Interest in Behavioral Science

CUBIC works to build awareness, appetite, and application of behavioral science within Save the Children through three primary methods, chosen specifically to suit the organization. These include:

► **Bitesize Learning**

We try to make learning about behavioral science as easy as possible for Save the Children staff around the world through our weekly *In a Nutshell* email. *In a Nutshell* explains one behavioral science principle or tool in just 150 words, and highlights how our colleagues could apply it in their life or work. Sent to over 1000 colleagues in English, Spanish or French, *In a Nutshell* has been one of our most effective means of introducing behavioral insights and building awareness of CUBIC. We recently launched a webpage

that includes over 100 of these *Nutshells*: <https://sci.shorthandstories.com/in-a-nutshell/index.html>.

► **Bespoke Training**

In our first year, CUBIC created our own training course specifically designed for Save the Children staff, so they can learn the basics of behavioral science theory and process. Through 8 remote sessions held over 6 months, teams refine their own behavioral challenge, conduct rapid formative research, and prototype behavioral interventions. They also learn about the importance of testing and experimentation, the latter of which is quite new to them, as it diverges from the typical international development evaluation method (i.e. baseline/endline assessments).

This course has now been run for teams from 24 countries in Asia, the Pacific, Africa and Latin America. In addition, CUBIC designs on-demand short bespoke courses for different programs and teams, such as applying behavioral science to support youth campaigning or early childhood development. We also provide training through partnership, including by organizing the BVA Nudge Unit's *B.E. Wiser* course for individual Save the Children staff across the globe.⁵⁷

► **Research Collaboration**

Our Save the Children partners are fundamental to the success of our projects. Without their expertise and deep understanding of contexts, we could not access, identify or articulate the specific challenges faced. We work with Save the Children teams around

⁵⁷ Available at <https://www.bvanudgeconsulting.com/b-e-wiser/>

the world to tackle challenging issues such as child labor in Indonesia, teacher absenteeism in South Sudan, violence against children in Cambodia, or childhood vaccination in the Philippines. We're currently working on 20 projects in over a dozen countries in Asia, Africa and Europe.

During these projects, we collaborate with our teams on the ground to narrowly define the behavior they want to target, conduct formative research and analyze the results, co-create behaviorally informed solutions that address key barriers and enablers, and prototype and test these solutions, ideally through randomized controlled trials where they are feasible. Collaborating with Save the Children country teams as implementing partners also allows us to benefit from the organization's established presence, reputation and infrastructure in different locations.

Build an Advisory Board (and Deal with Detractors)

One secret to CUBIC's successful launch was the creation of our advisory board, which comprises a Strategy Group of five Save the Children leaders and a Technical Advisory Group of six Behavioral Science experts (both practitioners and professors). This board has helped us review and enhance our processes and tools, refine our business model and strategic goals, and identify fundraising and communications strategies. Importantly, they also provide encouragement and support when we encounter thorny issues.

One such issue has involved the introduction of applied behavioral science as a relatively fresh approach to an agency with a history and expertise in other Social and Behavior Change

approaches, such as Risk Communication and Community Engagement (RCCE) and Community Capacity Strengthening (CCS). While there are similarities among such approaches, some key differences can cause tensions. For example, many Social and Behavior Change programs implemented by international development organizations focus on community-led initiatives.

While applied behavioral science processes *involve* communities in formative research, prototyping, user testing, and evaluation, they are not necessarily community-led. In addition, while the applied behavioral science approach focuses on experimentation, and ideally randomized controlled trials in order to determine causality, this method of evaluation is not only complex, time-consuming, and expensive, but can also be perceived as too austere and “top down,” and not sensitive or adaptable enough for some of the challenging contexts we work in.

While there is no simple solution that easily resolves these tensions, we have learned that a combination of humility – including affirming that behavioral science is never a “silver bullet” – combined with a focus on relationship building and producing innovative, quality work, can help grow more space for and acceptance of this approach as one tool in the broad “Social and Behavior Change toolkit.”

Establish Core Values to Drive Your Work

Our experience to date has uncovered differences that exist between assumptions inherent in scientific approaches and those that underpin operational approaches. In our first year of operation, CUBIC defined our culture and principles, reflecting that we have a foot in both scientific and operational fields:

- ***We are scientific:*** all of our work rests on the principles of rigorous research, experimentation, and transparent sharing of results.

- ▶ ***We focus on quality and impact:*** we have limited resources and know small things can have a big impact. We use this knowledge to choose projects wisely and ensure quality wherever we invest resources.
- ▶ ***We empower team members to deliver quality work:*** we seek honest feedback and constantly work to improve our team's skills and capacities.
- ▶ ***We are bold:*** we tackle challenging issues through our projects and research. We accommodate the risk of failure in pursuing evidence.
- ▶ ***We are good partners:*** we care about our collaborators' experiences, and try to make working with CUBIC interesting, productive, and fun.
- ▶ ***We value ours and others' time:*** we communicate effectively and concisely within our team and with our stakeholders across the world.

In striving to embody these values, we have experienced challenges and tensions between them. For example, the precision planning and implementation required for experimental research can clash with the more dynamic, problem-solving approach to planning often employed by our Save the Children partners. In one project, a required piece of equipment – essential to the trial design – was not provided by a supplier, and this was only discovered after the trial had begun and couldn't be paused. Such issues have required us to rapidly problem solve and pivot, though we are not always able to fully compensate for mistakes or omissions.

We have also experienced the need to re-design interventions when plans failed to produce the engagement required for a trial to go ahead. For example, on one occasion, we had initially designed an experiment with 3 arms where caregivers from 4 different communes were randomly assigned to one treatment and were then grouped in classes of 15 caregivers to attend positive

parenting sessions. Although we planned to provide a transportation stipend for them to join the sessions, to our surprise, less than 30% of caregivers attended the first session.

We quickly learned that the communes were too far from each other. We had to restart the intervention and redesign the groups based on location. While less than ideal from an experimental point of view, these pragmatic adaptations are part of the environment we work in. Getting better at horizon scanning and contingency planning is a priority for adapting our methods to the operational contexts of Save the Children.

Another frequent challenge we face is the lack of accurate existing data, which affects quality, impact and our desire to be effective with our time. In one project focusing on encouraging COVID-19 vaccination of teachers in Uganda (the country with the longest pandemic school closure in the world), not only was there no centralized database of teachers' vaccination status, but there was also no consolidated record of teachers' contact information.

All the data collection for this project was therefore human resource-intensive. First, we called school head teachers (i.e. principals) to gather teacher contact info. We then called each individual teacher to collect both demographic and vaccination related information. Most of our projects require this kind of manual data collection two or more times during the project life cycle. Addressing data quality and data collection is another priority for our future development.

Conclusion & Key Points

Although our team's process framework includes a "scaling" phase, to date our funding has been based on quick project timelines. Planning and supporting scaling of activities has been difficult to facilitate. In this line, we could not observe or support a project scale-up at CUBIC. In theory, once we have

proven a solution to be successful, government counterparts or Save the Children country offices will bring the solution to scale. Securing longer term funding for projects with an explicit goal on scale is among CUBICs ambitions for the future.

The field of applied behavioral science not only needs more successful stories of scale-up but also evidence that interventions work in the long-run. Like many other project-based interventions in development and humanitarian work, fixed timelines, funding constraints, and limited secondary data mean it is very challenging to observe or measure potential longer-term impacts of the work we do. Again, finding appropriate funding and effective approaches for long-term impact analysis is a key priority for CUBIC.

Along with tackling the challenges of scale and long-term evidence, another area that needs attention is around how research is documented and disseminated and for which purposes. CUBIC wants to help answer specific program and policy questions that can help improve the lives of children. We also want to share the knowledge and experience we generate with other practitioners to help improve program quality.

Finally, we want to contribute to international discourse and support advocacy for policy changes around the globe as part of Save the Children. Finding our niche and the most effective ways of working within networks of applied behavioral science practitioners, academic researchers and international development practitioners is critical to ensuring that applied behavioral science can reach its full potential in the international development sector.

In the next chapter we learn more about bridging and expanding behavioral science channels in development. We will learn from the experience of the Inter-American Development Bank.

14.

Bridging and Expanding Behavioral Science Channels

Carlos Scartascini, IDB Behavioral Economics Group

Deborah Martinez, IDB Behavioral Economics Group

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The Behavioral Economics Group (BE Group) at the Inter-American Development Bank Group (IDBG) has promoted behavioral science to boost the organization's development impact for over a decade. To do this, it developed a strategy that started by training and building capacity in behavioral science among economists inside the institution and policymakers and private sector actors across Latin America and the Caribbean (LAC). The training has allowed many IDBG specialists and government officials to implement better public policies in the region in education, gender, health, social security and pension, taxes, public administration, and small and medium enterprises.

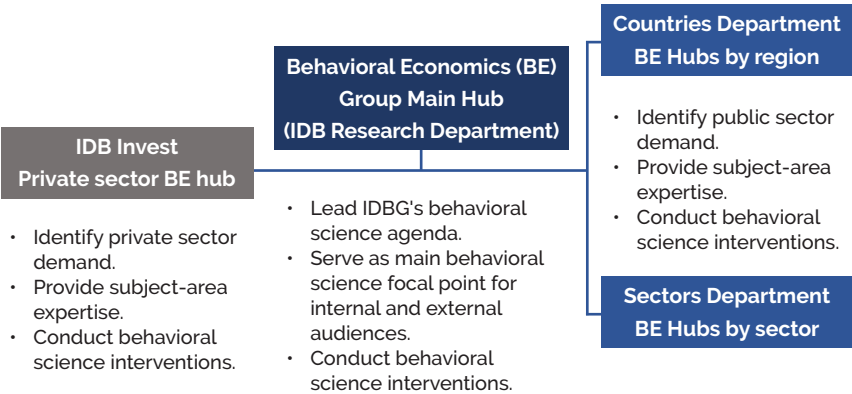
To maximize the impact of this work, the BE Group invests time and resources to disseminate behavioral insights within the Bank and the region. The solid institutional presence of the

IDBG in the region and its involvement in all the development sectors puts the BE Group in a great position to support the private sector and innovation ecosystems in LAC. This chapter summarizes the work done by the BE Group to bridge and expand Behavioral Science channels.

Operationalizing Behavioral Economics Within the IDBG

The Behavioral Economics Group is a centralized body formally established in 2018 by a small group of behavioral scientists working on the intersection between behavioral science and development economics. The group comprises a core team based in the Bank’s Research Department and a network of behavioral specialists across area sectors and departments, and includes behavioral scientists, communications experts, and researchers, including senior specialists, and visiting scholars (see Figure 1).

Figure 1. IDBG Behavioral Economics Group Organigram



The BE Group attempts to carry out most of the work in-house by contracting behavioral fellows that are part of the department or work at country departments. The widespread nature of members results in a decentralized and efficient model for business development and a diverse method for developing behavioral projects. To date, seventy projects at the IDBG have used behavioral science insights.⁵⁸

The IDBG BE Group in Action

The BE Group puts training and capacity building at the core of its strategy. It continues to be the case that this is where projects get imagined. The success of the BE Group, however, relies on the projects that continue to show results and the ability to disseminate those projects across the region.

To absorb the increased demand from the public and private sectors for behavioral science insights, the group provides a mix of individual support and capacity building to other departments' teams. A better grasp of the topic by specialists at the Bank allows for the timely identification of opportunities to apply behavioral science to their projects and to move forward at a speedier pace to design and implement interventions. Because of the significant number of projects implemented by the Bank, the BE Group created a repository of best practices and insights to support practitioners of behavioral economics inside and outside the Bank.

Regarding regional capacity building, around 6,000 policy-makers have benefited from a free online behavioral economics course hosted at the IDBG Academy. Created by online education experts and supported by many champions of behavioral economics at the IDBG, the course launched in 2020 as a free,

⁵⁸ See the repository of projects here: <https://behavioral.iadb.org/en/our-projects>

self-paced, and interactive curriculum. To those new to the area, it teaches basic economic principles, and it has several real-life cases that allow policymakers to imagine applying the concepts to their own work. Twenty-five editions in three languages (Spanish, English, and Portuguese) have been launched in the past four years and are planned to continue running.

The BE Group's small in-person workshops are an essential bridge between theory and practice for the BE Group. Practitioners bring into the workshops a specific policy problem that can be alleviated with behavioral science, and together with behavioral economics experts at the Bank they develop ideas that can be tested in the field.

The demand for this training has grown, leading to collaborations between departments at the Bank and regional organizations. The demand of private sector clients of the Bank to apply behavioral insights that boost their development impact has also increased. Thus, the BE Group is developing a training program to build capacity and expand behavioral science's delivery channels for the common good that include counterparts in the region's private sector.

Because of the greater understanding of behavioral science and the cost-effectiveness of behavioral interventions, the BE Group has executed many interventions. To follow the standards of the field, the BE Group developed a method based on standard practices of the area that focuses on the testing part of interventions. The priority is to run Randomized Controlled Trials (RCTs) so advice can confidently be developed and communicated to governments and private sector clients with sound results derived from testing. Some projects come directly from Bank operations. Some are direct collaborations with government agencies or the private sector.

During the past ten years, a hybrid of the Research Department in-house team and the behavioral department specialists has succeeded in responding to the goals of the Bank and the

member country's needs. Below we summarize several projects that showcase the bridging between experimentation and insight creation to design better policies that BE Group does continuously.

Projects Implemented by the BE Group

HPV Vaccination Rates in Bogota, Colombia.

The project began because of the interest of a team member of the American Cancer Society that reached out to the BE Group after taking the free behavioral economics online course. Their interest was to use behavioral science to help improve the problem of low HPV vaccination rates in Bogota, Colombia. The BE Group, in coordination with the Secretariat of Health of Bogota, the Ministry of Health of Colombia, the American Cancer Society (ACS), the *Liga Colombiana Contra el Cancer*, and the Behavioral Government Lab at Universidad del Rosario ran six experiments that tested SMS messages designed with behavioral economics principles.

In 2021, we sent about 1.5 million text messages to parents of girls between 9 and 17 that lacked vaccination. The complete intervention targeted 174,181 parents. The most impactful message increased the vaccination rates (first dose) by 32% more than the rate of those parents who did not receive an SMS. For those parents whose kids lacked the second HPV vaccine dose, the best reminder increased vaccinations by 63% compared to the control group that didn't receive an SMS.

The BE Group is currently using the insights from this project to run a similar project with the Health Secretariat in Cali, Colombia. This project is the perfect example of how the free online course can spark the interest of practitioners and how the group works alongside governments and private actors to implement effective projects.

Tax Registration and Compliance in Different Countries.

The BE Group provided support to different governments in the region by building the capacity to improve tax collection by using behavioral science.

The first example comes from the government of Fortaleza, Brazil (Eguino, et al., 2020). In collaboration with the BE Group, the tax administration of Fortaleza ran a project focused on behaviorally inspired, low-cost strategies to update registry information through the online platform. We designed two different email communications, both of which provided a morally persuasive message and a clear deadline, and, in one case, participation in a lottery for the taxpayers who registered. These messages increased enrollment between 2.5-3.5 times compared to a control group that received no email. Based on the study results, the BE group helped the tax administration in Fortaleza to continue using a layered approach that considers the beliefs and differences across taxpayers to increase registration cost-effectively.

The second example comes from the Government of the city of Santa Fe, Argentina. For this study, the BE Group embedded behaviorally informed messages directly into tax bills (Castro, & Scartascini, 2019), where taxpayers were nudged by one of three different messages:

- ▶ **The first message** stated the penalties associated with not paying taxes
- ▶ **The second message** described the public works made possible by taxes
- ▶ **The third message** addressed the moral obligation to pay taxes

The most effective message was associated with penalties,

achieving a 9% increase in compliance. Here, the support of the BE Group had significant policy implications, as the most effective message was scaled and used as part of traditional tax bills sent to all taxpayers.

The third example comes from the Colombia's National Tax Authority (DIAN), as the BE Group advised the government to test different communications methods to promote tax compliance (Ortega & Scartascini, 2020). The national tax authority sent messages with the same content through letters, emails, phone calls and personal visits. The personal methods (visits and phone calls) were more effective than the impersonal ones; among the more impersonal ones, the email was more effective than the letter.

However, an essential lesson for the government came from the cost-benefit analysis of these methods, the impersonal ones being the most cost-effective and scalable. This exercise provided substantial evidence for the Tax Authority to allocate its resources better to achieve impact more efficiently.

Micro and Small Entrepreneurs.

The private sector also faces challenges in increasing its development impact. Here, the BE Group worked with the commercial bank Banco Pichincha in Ecuador, a client of IDB Invest, the private sector arm of the IDBG. The team developed a heuristics-based business training aimed at micro-entrepreneurs and small businesses (Arraiz et al., 2019).

We tested two types of training. The first was based on a traditional approach, with structured content and classes. The other was based on heuristics and rules of thumb, employing a more simplified method to reduce the cognitive load.

The heuristics-based training had a significant positive impact on sales and profits compared to the control group, while the

traditional training did not show any statistical difference. A cost-benefit analysis reveals that the heuristics training's rate of return is high, estimated as an average extra profit per month per participant of US \$104, and it showed a more positive impact for women.

This example is relevant as it shows the capacity-building efforts of BE Group at bridging behavioral science applications for development impact beyond its more traditional applications in the public sector.

Greater Impact by the IDBG BE Group

Funding and helping set up nudge units within Latin American governments.

As part of the commitment to install capacity within public institutions, identify opportunities to use behavioral science, and internally conduct and evaluate interventions to boost their development impact, the BE Group has been involved in structuring the Behavioral Science and Public Policy Unit of the Argentinean National Government.⁵⁹ The mission of this unit is to advance behavioral science intervention across diverse areas within the government and to inform public policies to increase their effectiveness and improve the lives of the citizens.

Among the examples of interventions that the unit has been implementing, it is worth noting an intervention to reduce a particular drug prescription among medical staff by using social norms messaging and an intervention to promote effective recycling in Patagonia by using informational prompts. Although these interventions have not been conducted with the support of the IDBG, the genuine success relies on the fact that the

⁵⁹ More information available at <https://www.argentina.gob.ar/consejo/unidad-ciencias-del-comportamiento>

government now owns the capacity to conduct its behavioral interventions and successfully scale them up.

Advisory Services and Support

Departments inside the Bank, governments, and private sector companies are increasingly asking for advice because of the BE Group's capacity-building efforts. The BE Group has supported many projects inside the IDBG in tasks ranging from auditing communications prototypes to designing communications, designing experiments and evaluation protocols, implementing the interventions, and helping them with the analysis of the intervention.

Regarding national governments, the BE Group has supported government officials in Argentina, Colombia, Jamaica, Peru, Chile, Trinidad and Tobago, and Mexico. At IDB Invest, the group is working to develop a capacity-building program for private sector companies in the region that are looking to use behavioral science to boost their development impact, which is in line with the IDBG's mission of promoting development through both the public and private sector.

Dissemination

A crucial part of the BE Group's mission is disseminating knowledge continuously created in behavioral economics. The website exists summarizing, in an easy-to-understand language, around 50 behavioral economics projects. Every summary teaches key behavioral economics concepts and translates the academic findings into policy implications.

We have summaries of behavioral economics experiments implemented in: Covid-19, gender issues, governance, the private sector, education, health, pensions, savings, and taxes.

Soon, that list will include agriculture, water, and sanitation.

The BE Group has published diverse publications on the application of behavioral science, including a compilation of all the work that the Bank has been doing for the past ten years in the intersection of behavioral science and development economics.^{60, 61}

Barriers

Since the BE Group's projects are carried out with government agencies or private sector clients, they face many restrictions in designing and carrying out interventions as they are not entirely under the BE Group's control. Here are some of the key lessons we have experienced while collaborating with our counterparts that might be of relevance to other teams:

► **Political barriers.**

Particularly relevant for LAC, there are political barriers to what governments may be willing or unwilling to do. Coordination between government agencies or a company's departments may be less than optimal, and an intervention carried out by one agency may undermine another's.

To overcome or anticipate these challenges, it is important to have a clear view of the political conditions of each country by the time the experiment is expected to be implemented. For instance, understanding the electoral cycle is key as governments with less time in power will often want immediate results, even if that implies investing less in the diagnostics and design stages. Newer administrations may have more time to

⁶⁰ Available at <https://behavioral.iadb.org/en/videos>

⁶¹ More information available at <https://publications.iadb.org/en/nudging-latin-america-and-caribbean-decade-improving-public-policies-through-behavioral-economics>

invest on the different stages of the intervention and might also be more willing to scale its results.

► **Getting buy-in.**

Convincing counterparts, both in the public and private sector, about randomization may also prove problematic. Once the partner agrees to undertake an intervention it believes will have an impact, it prefers to pursue universal coverage and forget the randomization.

When working with private sector companies, it becomes key to align the motivation of a behavioral intervention to improve their business results and increase the development impact they can achieve, and how we can only ensure this if the results are robust (i.e., there could be potential losses for the company if an ineffective intervention is scaled up).

Similarly, for the public sector, it is key to align the behavioral intervention with the positive outcomes for the citizens and improved public perception about their administration. However, it is important to always highlight that interventions might prove to produce positive, negative, or neutral results, and finding that is key to avoid investing resources on strategies that do not work.

► **Technical barriers.**

There can also be different technical barriers that need to be addressed on the way. For instance, regulatory barriers regarding what governments can and cannot do (e.g., the government may be legally limited to communicating formally with citizens using digital channels), or a company may face low technological advances, restricting intervention delivery methods

and data collection. Data access can also be restricted and varies from country to country, limiting the type of outcomes that can be measured.

To mitigate these challenges, it is important to be as specific as possible with the public or private sector counterpart at the beginning regarding the minimal technical requirements to implement the intervention. This might involve asking about the type of administrative data they already collect and their data privacy protocols. It is also useful to think about backup plans, such as partnering with local data collection firms or offering capacity building efforts to strengthen the counterpart's technical teams.

Conclusion & Key Points

In contrast with developed regions that have been widely using behavioral science in many sectors, LAC has recently experienced a sharp increase in the demand of behavioral science solutions, particularly to promote development. In this sense, there are few specialized units within public and private sector entities that address this demand and, thus, an important emphasis has been placed on international development actors such as the IDBG's BE Group. However, resources in terms of human capital with expertise in behavioral science is also limited to attend LAC's increasing, and very promising, demand.

To further expand behavioral science applications in the region, one strategy that has been key is building partnerships with the more extensive behavioral science consulting community, as reflected in our work with private sector clients (e.g., partnering with Busara) and with governments (e.g., partnering with the Behavioral Insights Team).

Another important strategy is to build connections and alliances with individual expert researchers that are interested in

applying their behavioral science expertise on a specific sector and jointly produce knowledge. Both approaches have proven highly effective for attending the growing demand for behavioral science interventions while also maintaining the highest quality and academic standards.

In the next chapter we learn more about creating an academic research center from The Center for Advanced Hindsight (CAH) at Duke University.

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15.

Creating an Academic Research Center

Mariel Beasley, Duke University

Every year, there are thousands of peer-reviewed articles published in academic journals about behavioral science or behavioral economics. These papers are wide-ranging, including:

- ▶ Laying out new frameworks for the field
- ▶ Showcasing empirical evidence of a newly discovered cognitive bias
- ▶ Identifying boundaries and nuances of well-known biases
- ▶ Examining how a behaviorally informed tool works in a particular context

Many are traditional lab experiments, some are mathematical proofs or theoretical papers, and a growing number are field experiments.

Many of these articles will never be read beyond a handful of highly specialized academics or PhD students. They often sit, collecting digital dust, in the annals of Google Scholar.

Concurrently, there is an increasing demand for insights

from behavioral science in business, policy, charity, and our own everyday activities. Many companies have added a Chief Behavioral Scientist to their C-Suite. Governments have created their own “nudge units”, and behavioral scientists are in high demand across industries.

However, these internal job functions typically focus on the immediate needs of the organization and face pressure to merely translate existing findings for their respective organizations or to run quick, low-cost tests that immediately impact the bottom-line. It is only in rare cases that these lessons advance the field. It is likely that dozens of organizations have run almost the same experiment and found it did not work. Unfortunately, no one knows anyone else’s results, leaving future organizations doomed to repeat the same failed experiment.

An academic research center that focuses on applied research, like the Center for Advanced Hindsight (CAH) at Duke University, attempts to bridge the gap between the ivory tower and the practitioners implementing designs, policies, and procedures.

As part of an academic institution, CAH adheres to three primary tenets:

- ▶ We conduct ethical and rigorous research
- ▶ We share findings to improve the public good
- ▶ We educate the public and the next generation of behavioral scientists

Whether you are an academic looking to start your own applied research center, a researcher at an applied center, or someone looking to join one, here is how we approached key decisions and some lessons we have learned along the way.

Outlining A Vision

CAH was founded in 2010 by Professor Dan Ariely, the James B. Duke Professor of Behavioral Economics at Duke University in Durham, North Carolina, USA. Early on, Dan and CAH focused on advancing the field of behavioral science primarily by applying the methods and insights to practical settings.

We take on real-world problems that are ripe for behavioral science interventions. Our mission, to use behavioral science to help people be happier, healthier, and wealthier, focuses us on research topics related to consumer behavior, medication adherence, financial decision-making, social interactions, transportation demand management, vaccination rates, voter turnout, exercise, and similar topics.

For these, we can define specific behaviors that, if better adhered to, would benefit both society and the individuals themselves.

In a departure from traditional academic expectations, we measure our work by its impact on actual people and behaviors, rather than its presence in publications. And we prioritize research deliverables that are developed with the rigor of an academic institution but geared towards scaling and adoption by a broad practitioner audience.

For example, we went to Uganda to explore how social norms and limited attention could influence the purchase and consumption of bio-fortified, orange sweet potatoes that are rich in vitamin A.⁶² HarvestPlus developed these sweet potatoes to improve the nutritional value of traditional staples. We found that, contrary to expectations, people enjoyed the vitamin A sweet potatoes.

However, they could not reliably purchase them because not all sellers in the market reliably had them and, on the outside,

⁶² More information available at <http://www.bioline.org.br/pdf?nd17039>

the orange sweet potatoes were indistinguishable from traditional sweet potatoes. Buyers were missing market signals to remind them when and where they could purchase these particular sweet potatoes.

This project culminated in a report summarizing key insights from user surveys, interviews with sellers, and field observations. The report also quickly provided concrete recommendations to increase visibility of biofortified sweet potatoes in the market, like giving sellers an orange umbrella or orange tarp to show that they had orange sweet potatoes.

The report did not focus on mechanisms, p-values, or novelty. It focused on what mattered most to HarvestPlus to improve access and consumption. This prioritization of impact over publication allowed HarvestPlus to immediately and directly implement recommendations from the findings.

We know that our success depends on collaborative partnerships. CAH works with local partners, including government teams, researchers, local businesses, and development organizations to test and scale interventions in context. By collaborating with local organizations, CAH can gain a deep understanding of the cultural, social, and economic context in which interventions will be implemented, and to design interventions that are tailored to the needs of specific communities.

These partnerships also allow us to use administrative data to evaluate the effectiveness of interventions rather than having to go door-to-door to collect data directly. This use of administrative data from local partners dramatically decreases the cost and time to run randomized controlled trials.

Local organizations also benefit from the expertise and resources provided by CAH, which can help them design and implement more effective programs. For an in-depth look at this process and examples of the work, see the chapter on Financial Resilience in Latin America.

Lastly, as an academic institution, education and training remain a key component of our work. Besides research and applied work, CAH also offers a variety of learning opportunities for students, researchers, and professionals interested in behavioral science. These include workshops, “office hours,” courses, public dissemination of research findings, and internships.

Building the Right Team

At CAH, we ensure we have the right people involved in each project. The team includes:

- ▶ The internal project team which handles the day-to-day management of the research.
- ▶ The larger lab staff who can provide ad hoc support and feedback at critical points.
- ▶ The local partners who may be local researchers and/or organizations implementing the experiment and sharing their administrative data.
- ▶ Other academic researchers.

Across each of these groups, it is essential to include people with the right skill sets, backgrounds, and perspectives.

Internal project teams at CAH typically comprise three to five researchers:

- ▶ An advisor
- ▶ A lead
- ▶ One to three supporting staff.

The staff are grouped such that each team has a balance of essential skills, including project management, oral and written communication in the local language, research design, and data analysis.

Across the lab, CAH actively seeks staff with diverse training,

backgrounds, and perspectives. Past and current staff include PhDs in Social Psychology, UX designers, Master of Public Policy degree holders, economists, former consultants, qualitative researchers, data scientists, and more.

They represent about a dozen countries around the world and share a deep curiosity, a love of science and data, and a desire to make the world a better place.

Finding the right partners within the local organizations can make-or-break a field experiment. An enthusiastic partner who knows how to navigate their own organization can take a mediocre research project and transform it into an exciting and novel contribution. And vice versa: an exciting project can become mundane and irrelevant if the partner is not quite the right person.

Besides being a champion within their organization, local partners are invaluable for helping the research team understand the context and needs of communities to ensure the work is relevant and meaningful. Similarly, collaborating with local researchers can be especially helpful if you need to collect data in local dialects or if local customs and social norms differ from those of the internal research team.

We also often collaborate with other academic researchers. Through these collaborations, CAH can gain access to the latest cutting-edge research and experimental designs, and to test and validate theories in real-world settings. Academic researchers benefit from the practical insights and applications that emerge from working with CAH.

Across these groups, the team should be able to bring diverse perspectives to the table and be open to questioning and revising approaches and beliefs.

Picking Projects

In the last ten years, CAH has conducted over 150 field experiments around the world and launched countless surveys and online experiments. Our field experiments typically follow a similar cycle:

- ▶ **Stage 1:** Conduct diagnostic research.
- ▶ **Stage 2:** Develop interventions.
- ▶ **Stage 3:** Launch field study.
- ▶ **Stage 4:** Translate insights into practice.

CAH adheres to the rigor and review that is typical of a university. All studies go through Duke's Institutional Review Board (IRB) to ensure the research does not cause undue harm to the participants. This means that as we evaluate prospective projects, we consider the ethical implications of the research, making sure that the benefits outweigh any potential harm. We also pre-register our studies, including our hypotheses and analysis plans, using Open Science principles.

This adds to the rigor and reduces the likelihood of inadvertently finding effects that may not be real. This also means that we aim to have the research designs completed at least 30 days before launching the intervention. This is sometimes challenging for local organizational partners who may be more accustomed to rolling out a pilot or changes to an existing program as soon as feasible. These situations require transparent dialogue between CAH and the IRB, as well as CAH and the local partners.

CAH uses robust evaluation methods, particularly randomized controlled trials. These are considered the gold standard for being able to say that an intervention caused an outcome. There are two essential components to a randomized controlled trial and they are right in the name: your study population must be divided *at random* into either receiving the intervention(s) or

into the *control* group, who continues with “business-as-usual.” This means we prioritize research projects with large sample sizes. The sample size is the number of people taking part in the intervention, across the control group and the treatment group.

A large sample size allows us to be confident that the results that we are seeing in the data reflect the truth and not a weird bit of random chance. What constitutes a “large sample” can vary significantly between projects, as it depends on factors like how big of an effect you expect your intervention to have and what the baseline rate is on that outcome. A general rule of thumb is that the bigger the effect you expect, the smaller your sample can be.

Finally, most CAH studies use administrative data to measure the impact of an intervention. This means that we are measuring actual savings deposits, rather than household’s self-reported savings amounts. Using administrative data can dramatically reduce the time and costs for data collection, but it requires a partner to be accurately collecting the right data and be able and willing to share that data with the university. Not all organizations collect all relevant data to answer a research question.

For example, we partnered with a FinTech designing automatic savings solutions in Mexico. We tested whether a smart disclosure would increase the percent of users accepting the automatic savings contract compared to the standard disclosure language. We found that the smart disclosure, even though it added friction to the process, increased sign-ups by 10-percent-age points.

However, we could not see how that translated into actual savings behavior or if increased automatic savings led to increases in debt.⁶³ Similarly, not all organizations can enter a Data Transfer Agreement with CAH, making it near impossible for CAH to work with those partners.

⁶³ Case study forthcoming.

Obtaining Funding

Much of CAH's work is funded by philanthropic and mission-driven organizations, which provide the resources needed to conduct research and design interventions that can scale for the public good. This funding typically allows us the flexibility to identify good field partners with aligned research goals without necessitating that the field partner also cover our costs.

We work closely with our funding partners to ensure that research questions are aligned with their priorities and that interventions are tailored to the needs of the communities they serve.

Importantly, as an academic institution, all funding agreements must have a research and/or education component to them. We are researchers, not contractors. In adherence to the University's mission, we strive to "advance the frontiers of knowledge and contribute boldly to the international community of scholarship."⁶⁴ This means that we share the results of the work publicly, regardless of the findings. Uniquely, we do not enter funding agreements that restrict our rights to publication.

We have found that multi-year grant agreements that support both field studies and lab studies are the most effective for an applied academic research center. The longer time horizon allows for university processes, including reviewing Data Transfer Agreements and getting IRB approval, to complete prior to studies being launched. It also allows for the research team to fully use the pipeline of lab studies informing field studies or vice versa.

For example, we conducted a series of lab studies that showed that framing earnings in a yearly salary was more effective at pulling people into a long-term mindset than framing earnings at an equivalent hourly rate. This long-term mindset translated

⁶⁴ Mission Statement available at <https://trustees.duke.edu/governing-documents/mission-statement>

into a greater willingness to save for their future. After conducting several lab studies to explore this, we partnered with Payable, a payment platform for contract workers, and found that by framing their income from contract work as per year versus per job, contractors were more likely to save part of their earnings.⁶⁵

Conclusion & Key Points

Over the last decade, we have also learned a good deal about what does not work as well for an applied academic research center.

First, don't take on too much at once. Prioritization ensures you can effectively carry out projects, meet deadlines, and see meaningful impact. This is true in an academic research center when staff are often transient. Over-committing to projects and then having research staff head to graduate school or student research assistants head out for summer vacation can leave you without the capacity to absorb outstanding tasks.

Second, don't assume you can replicate a model from one context to another. It's important to do an in-depth understanding of the context before designing any interventions. In a project related to increasing trust in elections, we found that the most effective intervention varied by country. In each case, the most effective intervention directly addressed the very specific, vocalized concerns that were unique to that country. Context matters.

Finally, don't forget to remain humble and open to learning. Avoid complacency. Your greatest asset as an applied academic research center is to be informed on the latest research and methods and then translate those into real-world applicable insights. Attend academic and practitioner conferences, dedicate

⁶⁵ More information available at <https://case-studies.advanced-hindsight.com/case-studies/RgynKcHQ/framing-encouraging-a-long-term-mindset>.

time every week to read newly published papers, and spend time just thinking deeply about the implications of new insights. This will help ensure that your work remains relevant and that you remain informed on the latest research.

In the next chapter we learn more about the unique challenges that come from working with governments on behavioral science projects. We will learn from the knowledge gained at the Centre for Social and Behaviour Change.

16.

Building Government Capacity for Behavioral Science

Gautam Patel, Centre for Social and Behaviour Change, Ashoka University

I have worked with over 200 government officials and functionaries at the national, state, and district level over the past decades, yielding over a thousand formal and informal interactions. With some officials, I interacted for a handful of meetings before they were transferred to a different department. In others, I identified early on that the collaboration was not viable. I worked with some officials in Gujarat for 15 years and in Bihar for over 7 years. I currently lead three Behavioral Insights Units (BIUs) in Delhi, Uttar Pradesh and Bihar, and regularly partner with governments.⁶⁶ Our interactions have covered the spaces

⁶⁶ The Centre for Social and Behaviour Change (CSBC) at Ashoka University, funded by the Bill and Melinda Gates Foundation (BMGF) set up three Behavioral Insights Units (BIUs) in India between 2019 and 2021. The BIUs were set up in Delhi with NITI Aayog, the Government of India's think tank; in Bihar with JEEViKA, the State Rural Livelihoods Mission; and with the Uttar Pradesh Technical Support Unit (UP-TSU) to the Government Health Department.

of their offices, their written communications, travels together in the field, and sometimes at their homes and family events.

This experience has allowed me to analyze:

- ▶ Effective ways of working with the government.
- ▶ How government officials make decisions.
- ▶ How context shapes the behavior of government officials.

Over the years, I both intervened (and sometimes interrupted) and investigated. Within an *intervention*, my aim was to understand problems in the government system and present evidence of impactful ideas.

Many organizations first introduce behavioral science via partnership or support from a third-party before ultimately creating an internal capability. Therefore, my *investigation* was to understand how to collaborate with governments and how governments can establish sustainable internal mechanisms to apply behavioral science. I'm grateful to every professional I've had the fortune to work with and learn from to build my understanding.

It is a privilege to get to share this knowledge with others who are ready to contribute to the strengthening of government systems for better public service.

I begin this article with a diagnosis of why working with governments in India is challenging. I've selected five features of government functioning that explain behaviors of officials towards collaboration. Next, I apply the insights from my diagnosis to three practical situations that you can expect to find yourself in if you plan to collaborate with the government. Finally, I offer long-term strategies governments can use to create and nurture their own internal capacity to apply behavioral science.

A Diagnosis

Identifying barriers to collaboration is important in finding ways to work with governments. Otherwise, the work of introducing

new thinking into large government systems is left to personal connections, privilege, luck, and sometimes, failure. The features of government functioning – which for the purposes of this chapter focus on collaboration for innovation – are listed below, along with the observed behaviors of senior government officials.

Table-1: Five features and behaviors of governments that restrict innovation.

#	Features of government institutions that restrict decisions for innovation	Associated behaviors of government officials
1.	Avoidance of risky decisions. Governments are set up for permanency and use a series of checks through the hierarchy on decision-making to reduce chances of mistakes, failures and embarrassment.	Having low interest in disruptive innovation/ Especially decisions that change, take away or add to the system. An interest in quick and easy reforms.
2.	Caution towards unfamiliar partners. As a default, government officials will avoid unfamiliar partnerships that are a potential hazard to their reputation. Past corruption scandals where people used government relations to make a private gain induce a powerful fear.	Demanding costly signals to trust. The assumption they might be duped by someone intending to gain privately from public resources places extra burdens on new partners.
3.	Innovation is under the purview of leaders. Ideas and decisions are the domain of only those with authority at the top of the hierarchy. These individuals can have power because of seniority and not necessarily expertise.	Valuing ideas only when voiced by authority. There is little space given to hear the voice of others.
4.	Commitment to public service not financial gain. Government salaries are much lower than for similar work in the private sector.	Disliking well-paid professionals. Government officials can prefer working with other government officials.
5.	Scarcity of time to plan deep reforms. Senior government officers often hold charge of multiple departments and must solve crises every day, giving little time to analyze and prepare for longer reforms. To show high performance, bureaucrats have a window of around two years, which is a short time for deep reforms.	Lacking time for deep thought decision-making. Preferring quick fixes for visible results, which often come down to efficiency upgrades.

The list of challenges in Table-1 is a starting point for designing how to set up new innovation units to collaborate with governments to provide behavioral solutions to persistent problems. I will use these insights to explain ways to overcome the barriers in three essential interactions that are part of collaborating with governments:

- ▶ The first meeting
- ▶ Building the partnership with review meetings
- ▶ Preparing for crisis management

Collaborating with Governments

Make the first meeting a start to a relationship.

The intuition of most expert teams — to use the first meeting to impress and maximize time to present their own high intellect — can lead them astray. For years, I'd slog hours into the night and on plane journeys to compress more smartness into my slide decks. The polite senior officers I met gave me a response that looked like hesitancy. That only made me work harder to present more fully worked out ideas and seek to amaze them with an innovation for their biggest problems.

In 2016, a senior officer in Patna informed me in the first two minutes of our conversation that he had no time to hear about a million-dollar rigorous randomized control trial and would only speak with me if I would work together to solve how the system can engage and uplift the poorest women in Bihar. The response forced me to close my laptop, discard the proposal I'd prepared, and actively listen. I told him that I had come a long way to see him, and I'd like to take the next 30 minutes to hear about the biggest problems he faced and what type of solutions the government needed.

That conversation started my seven-year relationship with the

Rural Development Department of the Government of Bihar. More than that, it switched me from *speaking to impress* in meetings to *listening and questioning to learn deeply*.

The government may have an aversion to considering disruptive innovation, reforms for outcomes in the long term, and listening to better-paid professionals. However, there is a deep commitment to social service and to solving persistent problems with multiple factors and changing conditions. This is the shared mindset that our Behavioral Insights Units (BIUs) and Government Departments have. We want to give our best time to solve the worst problems. Beginning conversations with questions to define a joint commitment is a strong start for an intellectual collaboration.

If we seat ourselves as inquirers, we can learn from senior government officials who have decades of experience operating large government systems and have access to information for decision making and functioning that is opaque to the public. I recommend using the first few minutes to present the ways your team can be of use and then ask about the toughest problems the decision-maker faces.

In India, government officials have a time scarcity. Many of these officials oversee multiple departments and are exhausted daily by the quantity of decisions. Their mindset is also on short-term fixes, amplified by the increased public vigilance and viral spread of grievances for governments on social media. Offering a government official the service of grounded data collection and analysis to inform how to solve a deep problem is a high value proposition to someone who has the intent but lacks the time.

The first meeting must center on an agreement to permit a deep inquiry into the problem. For this to be safe for the government, the findings must remain confidential and only for governmental use. Access to information regarding persistent

problems comes with the risk of public embarrassment. An assurance that your team will adhere to confidentiality can be demonstrated by providing a list of past projects with governments where you've maintained confidentiality on the insights. I often give the names of senior government officials and politicians that I've worked with as professional references. And ideally, I have my first meeting set up through a senior government official that I've worked with in the past to introduce me with their seal of trust.

The first meeting should close with:

- ▶ An agreement for our team to have access to government data
- ▶ Permission to interview managers and field functionaries
- ▶ A commitment on your part to make observations of government functioning without disturbing it, or without pre-announced visits

Once the senior government authority gives official communications to key informants in the government system, then our team can begin a systematic and transparent inquiry to understand the policy problem of interest. The next meeting may be convened after a *diagnostic report is ready to be shared and discussed*.

Build and rebuild alignment through regular review meetings.

The barriers to working with the government will not be broken down no matter how well your first meeting goes, nor will these barriers shrink after several good meetings. The barriers are a function of how the government system is set up to sustain and protect itself. This means there is a repeated need to check on the partnership and keep working to strengthen the alignment.

There must also be a commitment here to work in trust and confidentiality, along with a commitment to learn from both failures and successes without the unnecessary threat of public or political scrutiny.

The technical partner will remain under scrutiny as a suspect. The government's default will be to look out for signs of distrust, including signs that the partner is seeking prominence or proximity to higher officials for private gain. The huge size of public resources means there is a potential risk of pilfering from the government treasury, and large corruption cases from the past are highly memorable and fear-inducing to government officials who can be targeted as scapegoats.

Another systemic reason alignment won't remain in place are the differing objectives in the partnership. The government system is built to sustain and maintain extensive coverage along with uniform and long-term service delivery. However, the technical partner may seek to test disruptive innovations with high impact. This dichotomy needs careful management if the goal is to solve persistent problems of poverty. Both goals have importance, and a strong marriage of differences can create unique solutions to unsolved problems.

Disruptive innovation has low value if it can't be replicated at scale. Uniform standards implemented at large scale have low value if they don't lead to impact for the target group. A careful iterative process can enable both sides to strengthen common thinking. We need to establish comfort with experimentation and agreement to let data confirm what has value.

Expect and prepare for a crisis management meeting.

Regular review meetings will take care of the cracks and fissures in a partnership, but a different meeting is needed to address a crisis that threatens a breakdown and even a breakup of the

partnership. The aim isn't to eliminate every crisis, as sometimes a crisis is a good signal that the partnership isn't viable and it's time to wind up.

In 2019, I was part of a series of meetings when partners were considering disbanding a five-year partnership, after jointly signing an agreement for funding just six months earlier. Partners exchanged heated messages and shared lists of complaints and new demands. We all shared a certain amount of hope that we could rebuild the trust and continue working together.

After some months, we agreed on a new set of terms and indicators with a timeline to review after a quarter. In this case, the partnership strengthened and continued for five years. However, on reflection, the cost of the protracted discussion was high, the project was delayed in hiring and setting up the operations, and substantial leadership time went into crisis management and not into planning the program scale-up.

More efficient and timely handling of crisis management happens when we predict and prepare for the crisis. This requires inverting the optimistic and hopeful mindset we have to get through troubled times, and instead looking for where failure may set in, and when a crisis may surface. A way to overcome our confirmation bias is to agree early on a set of indicators to regularly gauge the health of the partnership. The health check should collect data across the period of the partnership to measure internal progress and not rely on biased opinions of key stakeholders.

Some basic measures include:

- ▶ Frequency and duration of partnership meetings
- ▶ Time taken before a decision is made or a request has a response
- ▶ The number of commitments that were followed through

The partners should agree that when a particular threshold is crossed, that this should trigger a crisis management effort with a series of regular senior leadership meetings to agree on repairs and reforms for the partnership to continue and succeed. By agreeing to the mechanism to address the crisis, it will ensure the action is time bound, and not avoided.

During crisis management meetings, partners can prioritize self-protection, as protecting their own organization and their own individual reputation can seem more important in the short-term. But each partner protecting themselves and looking to place blame on others is a clear recipe for disbanding. The best way to focus thinking on the task at hand is to draw the attention of all the partners immediately on to the functioning of the partnership.

I've found it is useful to present a process flowchart of the start-to-end process and use data to show where the breakdown is happening. Using data and facts to illustrate the malfunctioning of the partnership helps make decision-makers more responsible for addressing the blocks or defining new processes to bypass the challenges.

Collaborating with the government is associated with challenges, but using a diagnostic to understand five major behavioral barriers to working together provides insights to better handle the first meeting, review meetings and the crisis management meeting. A behavioral lens for working with governments is an advantage of Behavioral Insights Units (BIUs).

Table-2: Solutions to overcoming barriers to collaboration with government

#	Behavioral barriers	Solutions
1.	Having low interest in disruptive innovation	• Begin dialogue by proposing system improvements, not massive reforms. • After confidence in the partnership builds, substantial reforms can be proposed.
2.	Demanding costly signals to trust	• Signal sincerity, dedication and high effort for social service. • Build up a resume of successful projects with governments.
3.	Valuing ideas only when voiced by authority	• Engage officials at all levels in the government hierarchy and build rapport with the most senior officials so they may provide their authority when requested. • Document goal-alignment meetings, a series of interactions that cover all the officials in the hierarchy.
4.	Disliking well-paid professionals	• Hire staff who are local and not flying in from larger cities. • Build teams at academic or not-for-profit organizations (where salaries are much less than for-profit consultancies). • Collaboratively diagnose the problem, so there is a shared ownership of understanding the problem.
5.	Lacking time for deep thought decision-making	• Offer the service of an analytical report to inform their decisions. • Establish criteria to predict crisis in relations, so that interventions are timed and pre-committed.

Conclusion & Key Points

As governments see the benefits of behavior science to strengthen their programs and operating systems, a natural next step would be to internalize these approaches into their own systems, and not rely solely on the availability of technical partner organizations to provide essential services for problem solving. I have three recommendations for governments interested in building up their internal expertise in behavior science:

► ***Build technical capacity at local academic institutes.***

The work of applying behavioral insights to strengthen government systems and service provision is inherently scholarly. Academic institutes are established across locations, and building the technical expertise of local institutes can provide a long-term group of experts to support the government.

In Uttar Pradesh in April 2023, CSBC signed an MoU with the University of Lucknow, a century old institute, to advance behavior science research and local capacity. We will work with the university to conduct behavior science experiments, and to offer lectures and professional courses locally.

► ***Academies for administration should include behavior science in their curriculum.***

Government decision makers can use behavioral insights to improve their decision making for effective public service. Senior leaders in government have a scarcity of time and information, and there is a political interest in showing success rather than assessing where things may be going wrong. CSBC regularly runs behavior science courses at India's national academy for training officers of the Indian Administrative Service (IAS) at the Lal Bahadur Shastri National Academy of Administration (LBSNAA).

There are also academies for administration at the state-level. Training government officers to recognize cognitive biases and providing them with tools and behavior insights will strengthen their decision making.

► ***Set up data analytics units.***

Building behavior science expertise locally will take time. More immediately, governments can set up an

analytical unit to use government data from surveys and program data to prepare a briefing on the policy problems. Governments can use these briefings on behavioral problems to specifically invite organizations to prepare solutions for the problems, including through competitive selection. This setup will target the government's most important policy problems and will signal to behavior science organizations an intent to collaborate.

Behavior science provides remarkable insights that rethink how governments can provide services, design programs, and make decisions. For centuries economics has dominated policy recommendations, and we've made analyzes and predictions assuming humans will make decisions rationally to maximize utility. Understanding that human brains are also wired to make decisions with short-cuts, emotions, and often not in our best interest helps us to better predict and understand where large systems can fail.

These behavioral insights can provide governments with diagnostics to design solutions to persistent problems of poverty. The power to use behavior science for better decision-making should not remain in the domain of highly trained and scarce experts, rather it is a greater good if behavior science organizations can build capacity in government teams to use this knowledge to strengthen their systems through a scientific understanding and a well-informed process for problem-solving.

Join us in Part IV to learn from case studies that show the great impact and potential in the application of behavioral science strategies in this field.

Part IV:

AREAS OF IMPACT

This section is dedicated to showcasing practical use cases, where abstract concepts find concrete expression and, more importantly, tangible results. By navigating the nuanced challenges, inventive solutions, and the myriad outcomes from these real-world scenarios, readers are presented with a wealth of lessons that can inform their own endeavors.

Each use case has been selected to offer a blend of diverse challenges, contexts, and solutions, ensuring readers can draw parallels to their unique circumstances. Topics include financial wellbeing, the pivotal role of behavioral science in health and sustainability, and the complexities of governance, societal inclusivity, and domestic work. These chapters also discuss behavioral science applications at different altitudes -- both an individual and policy level.

For both seasoned practitioners and novices, these use cases offer a valuable opportunity. By dissecting each scenario, readers can extract key learnings, anticipate similar challenges, and formulate strategies that are both informed and innovative.

17.

Financial Wellbeing and Entrepreneurship: Working with SMEs in LAC

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While self-employment, especially gig work and business ownership, has provided a lifeline for many in Latin America, workers have struggled to build financial resiliency and thrive. The COVID-19 pandemic hit Latin America, and particularly independent workers and small business owners, hard.

Independent workers are the backbone of Latin American households and national economies. They account for about 40% of all workers in the region, a rate higher than Europe (14%) and the US (8%) and just below East Asia (43%) (Maurizio, 2019).⁶⁷ Micro, small, and medium-sized entrepreneurs account

⁶⁷ More information available at: https://www.ilo.org/wcmsp5/groups/public/---americas/---ro-lima/---sro-santiago/documents/publication/wcms_713802.pdf

for a staggering 99.5% of all firms in the region and employ about 60% of all formal workers (OECD/CAF, 2019).⁶⁸ While it varies from country to country, about 25% of gig workers in the Latin American region rely on gig work as their only or primary source of income.

These numbers are only poised to grow. According to recent estimates by Mastercard, the gig economy alone is estimated to reach \$455 billion in gross volume globally by 2023 (Mastercard & Kaiser Associates, 2019).⁶⁹ In this context, there is an equally valuable opportunity for digital work platforms and financial services providers to continue to foster financial resilience in the region. Organizations and platforms are uniquely positioned to help workers genuinely thrive. But this requires that they address behavioral barriers and design products that minimize the impact of inevitable financial shocks and maximize the chances of recovering afterward.

In this chapter, we share some insights of the Common Cents Lab's work to increase the financial resilience of gig workers and entrepreneurs in Latin America using applied behavioral science. We begin by outlining the concepts of financial wellbeing and resilience, which are key to understanding the mission of our work, and then walk readers through the process we've been following to identify opportunities for behavioral interventions, the design and testing process, and the implications of what we've learned.

⁶⁸ More information available at: <https://doi.org/10.1787/d9e1e5f0-en>.

⁶⁹ More information available at: https://fs.hubspotusercontent00.net/hubfs/7834608/Flagship_Advisory_Partners_December_2021/docs/Gig-Economy-White-Paper-May-2019.pdf

Financial Wellbeing and Resilience

Financial Resilience

The first component underlying our understanding of the financial challenges of gig workers and entrepreneurs is financial resilience. The ability to “absorb disturbance and reorganize while undergoing change so as to still retain essential functions,” is what defines “resilience.” The word “resilience” is derived from the Latin “resilio”, which means “to jump back,” which implies that the system experiencing a shock be able to get back to its core functions. A key lesson the Covid-19 pandemic taught us is that our context and circumstances can change dramatically in a very short span of time.

In this context, we’ve observed how communities recover and create new opportunities. They did this by transforming their usual spaces, organizing to demand better working conditions where this was possible, and carving out new economic spaces for themselves.

Unfortunately, as we also saw during such a difficult period, not everybody is adequately prepared to face unexpected challenges. It is often those in the lowest income brackets who are the least well prepared but are hit the hardest. They also find it more difficult to get back on their feet, let alone thrive.

Resilience has been connected to financial wellbeing and management, at both the household and individual level. Financial resilience has become one of the indicators to measure financial health. It implies the ability to:

- ▶ Build savings and achieve short-term goals
- ▶ Hold insurance to cover for property losses, life, and health emergencies and losses
- ▶ Have a diverse safety net, including an emergency fund and access to affordable credit or social network

- Recover from financial problems and having a self-perception of financial control (Gutman & Hogarth, 2014)

The concept has even been systematized to prescribe having enough savings to cover the equivalent of 3-6 months of expenses or a minimum of \$400 for emergency expenses (Gravier, 2023; Board of Governors of the Federal Reserve System, 2017).

The Covid-19 pandemic brought to the surface the urgency to expand the application of resilience measurements in policies and services to improve the financial wellbeing of individuals and households. However, a comprehensive resilience analysis must also address the question of the kinds of disturbances the system of analysis must be resilient to, or “resilient to what?” These are the elements that represent potential disturbances or uncertainty in the system’s equilibrium.

In the realm of financial resilience, potential disruptions include economic factors like recessions or job losses, or a policy context that may not provide sufficient opportunity to overcome adversity. Therefore, if the goal is to improve the resilience of gig workers and entrepreneurs, the analysis must also look at the contextual and systemic factors that cause the disruptions. Rather than pushing gig workers and entrepreneurs to “have to” be resilient on their own, we must improve the design of systems and contexts so that resilience is embedded in them.

Financial Wellbeing

One path to analyzing and improving financial resilience is understanding financial wellbeing. Achieving financial wellbeing is arguably the goal of most (if not all) of our financial decision-making. Yet, despite the widespread and growing popularity of the term “financial wellbeing” in academia, industry

and popular press, there is still no universal consensus regarding what it actually means.

Only recently has there been efforts to conceptualize, define and measure “financial wellbeing” in a comprehensive and meaningful way. Specifically, the US Consumer Financial Protection Bureau’s stands out as the first, most methodologically robust and perhaps most comprehensive definition: *“Financial wellbeing is a state of being wherein a person can fully meet current and ongoing financial obligations, can feel secure in their financial future, and is able to make choices that allow them to enjoy life”*.⁷⁰ Similar subsequent definitions provided by other researchers often build upon this initial work by the CFPB.⁷¹

Financial well-being refers to how people perceive and feel about their own financial lives, both in the present and future. Importantly, the key characteristic of this conceptualization of financial wellbeing is its subjective nature. Often referred to as “subjective financial wellbeing”, this concept is based upon the notion that only the individual can assess their own wellbeing. Focusing on the subjective aspect of financial wellbeing allows us to tap into a unique dimension of people’s financial lives that has been completely overlooked in the past.

By extension, financial wellbeing is inherently different from making external judgements based on observing people’s financial behaviors and outcomes. As shown by research measuring financial wellbeing both subjectively and with observable data on financial decision making, these are tightly related yet entirely different constructs which are not necessarily always aligned.

⁷⁰ For detailed overview of how the definition was developed see CFPB *Financial Well-Being Scale Scale development technical report*

⁷¹ Similar subsequent definitions were provided by other researchers, often building upon the initial work by the CFPB. For example, Bruggen and colleagues (2017) define subjective well-being as “the perception of being able to sustain current and anticipated desired living standards and financial freedom”.

It is perfectly possible and not uncommon for some individuals to display high levels of observable financial wellbeing but low levels of subjectively perceived financial wellbeing. Similarly, an individual reporting high levels of subjective financial wellbeing can display behaviors indicative of low objective financial wellbeing. Recent evidence from the Australian context shows that only about 40% of people have matching levels of observed and subjective financial wellbeing (Botha et al., 2021).

The way people experience and perceive their financial lives is not always aligned with external objective observation of financial outcomes and behaviors. Financial wellbeing is a complex and multifaceted concept, with a network of interrelated factors and determinants, including:

- ▶ Financial behaviors
- ▶ Household characteristics
- ▶ Social contexts
- ▶ External conditions

Despite its conceptual complexity, the measurement of subjective financial well-being is easy, fast, and cheap to implement. The current best way of measuring it is via physical or online surveys, where people directly provide self-reports of their own financial well-being based on a standardized scale. The CFPB has developed and validated one of the most robust measurement scales which measures financial wellbeing in two dimensions: control over present finances and future shocks and the financial freedom to make choices today and in the future (CFPB, 2015).⁷²

Measuring financial well-being contributes to overcoming major methodological challenges faced by behavioral accounts of financial well-being, given that observable behavioral data

⁷² More information available at: https://files.consumerfinance.gov/f/201512_cfpb_financial-well-being-user-guide-scale.pdf

from financial products usually captures only one part of people's financial lives. Self-reported financial behaviors have also been repeatedly shown to be biased and inaccurate accounts of actual behavior.

Focusing on people's subjective financial well-being comes with its own unique challenges. First, the measurement of financial wellbeing is hard to interpret by itself, primarily because it is a relative scale rather than an absolute one. Financial Well-being Indexes place individuals along a continuum to determine how one individual compares to others, but numbers do not necessarily provide a straightforward interpretation of financial well-being.

The second and most important challenge is that we still know very little about its determinants. There have only been a handful of academic studies that focused on trying to understand what drives financial well-being, and the most promising finding is the fact that financial behaviors seem to be the most important predictor of financial wellbeing, even more so than level of income and financial knowledge (Comerton-Forde et al., 2022).

This is consistent with previous findings that the level of income is only weakly correlated with people's overall happiness and well-being (Dolan et al., 2008), as well as robust meta-analytical evidence showing that information and education-based interventions have a tiny and often negligible effect on actual financial decision making (Kaiser et al., 2022).

A Behavioral Science Approach to Financial Resilience

Combining the insights from financial resilience and well-being, behavioral science is uniquely positioned to serve gig workers and entrepreneurs, by focusing on key behaviors that

contribute to financial well-being as well as on the contexts that influence those behaviors and their resilience to financial shocks. Traditional approaches have focused their efforts on providing financial education or financial coaching. While financial coaching programs seem to be effective at improving people's financial behaviors, their cost per user is usually higher and programs struggle to scale (Theodos et al., 2018).

Conversely, traditional informative financial education programs have shown to have a powerful effect on improving financial literacy, but only a minor effect on improving downstream financial behaviors (Kaiser et al., 2022). Research suggests that the effects of these types of programs are even smaller or non-existent for low-income and vulnerable groups.

Behavioral science shifts the intervention focus from the individuals (people receiving financial education or coaching programs) to the systems, products, and services that they interact with. A robust body of research has proven that making small but meaningful changes to the environment in which people make decisions can have a larger impact than providing more information alone (Beshears & Kosowsky, 2020). If we consider the cost of implementing behavioral science interventions, researchers have found a higher impact-to-cost ratio than traditional information-based policies and programs (Benartzi et al., 2017).

In other words, a behavioral science approach can lead to cheaper and more impactful solutions to increase the financial resilience of gig workers and entrepreneurs.

At the Common Cents Lab, we follow a five-phase model to implement studies that test behavioral science interventions in the field. In the following sections, we will briefly describe each of those research phases and share key insights we have identified in our work with gig workers and entrepreneurs in Latin America. We believe these insights can help other

organizations use a similar approach for social impact in their domains of work.

Partnering with Financial Organizations and Digital Platforms

Most of the success of behavioral science relies on redesigning the environment in which people make decisions. Rather than working directly with gig workers or entrepreneurs, we've partnered with two types of organizations that are key to improving their financial resilience: digital platforms that connect them with demand and work opportunities, and financial organizations that help them manage their resources. These organizations are uniquely positioned to help workers genuinely thrive, but this requires that they address behavioral barriers and design products that minimize the impact of inevitable financial shocks and maximize the chances of recovering afterward.

Diagnosis: Understanding People and Their Environments

Gig workers and entrepreneurs are in a complex and unique situation. They are attracted to the flexibility of working hours that allow them to focus on multiple priorities. However, it means that they may experience higher uncertainty (of demand and income) and have to make more complex decisions that affect their outcomes. All of this occurs in a regional landscape, Latin America, which has a weak social security net, changing regulations and is highly affected by informality.

In Latin America, we've identified three core behavioral challenges that must be addressed with priority to support their financial resilience.

Too much choice and fewer benefits.

Workers have to make multiple decisions under uncertainty, like:

- ▶ “How many orders will the business receive this month?”
- ▶ “How many hours should I drive or work today?”
- ▶ “How likely is it that I will have an accident?”
- ▶ “Should I buy insurance?”
- ▶ “How much should I save?”

What the seminal work in behavioral sciences shows is that under uncertainty, we rely on heuristics or rules of thumb to make choices (Tversky & Kahneman, 1974). These heuristics, however, do not always guide us toward the best decision. Having to choose between many options can lead to decision paralysis and revert to the status quo.

For instance, research from the retirement savings domain shows that increasing the number of retirement funds from 2 to 59 reduces participation in retirement savings from 75% to 60%; a 20% drop caused by adding choices to the mix (Sethi-Iyengar et al., 2004). Likewise, when we think about gig workers and entrepreneurs choosing how much to save for emergencies, how often and through which savings products, there’s a high likelihood that the number of choices they need to make could lead to decision paralysis.

Planning failures and reduced income.

The second challenge has to do with the inevitable volatility and uncertainty around workers’ income. While surveys of gig workers in Latin America have found that one of the attractiveness of platforms is the flexibility of hours and control over their time and income, recent studies have found that the volatility of income is associated with poorer health outcomes and induced

scarcity mindset (Sayre, 2022; Azuara et al., 2019).⁷³ Potentially because of focusing on the temporary scarce resource (income), gig workers engage in a practice called “income targeting”, focusing on income goals regardless of how efficient or inefficient their efforts are.

In a classical study, researchers found that because of daily income goals, taxi drivers in New York worked longer hours on days that had a lower demand (i.e., sunny days) and worked fewer hours on days with higher demand (i.e., rainy days) (Camerer et al., 1997). This phenomenon has shown to decrease the labor supply of gig work platforms by about 17% (Sinchaisri et al., 2019). Research on small businesses has found that business owners may overestimate how well they perform compared to other similar businesses. These insights show that while flexibility is attractive, it comes at a cost of not maximizing our potential income and our wellbeing (Seither, 2021).⁷⁴

Forecasting, shocks, and preparing for the future.

Gig workers and entrepreneurs need to manage their volatile income, and they also need to estimate future expenses and plan for unexpected ones. Unfortunately, research shows (Howard et al., 2022) that all of us fail to accurately predict our future expenses by about 15%. While people who experience financial constraints tend to be more accurate (Berman et al., 2016), failing to estimate our expenses accurately could have harmful impacts on those with a smaller safety net. Complementary research (Sussman & Alter, 2012) shows that rather than

⁷³ More information available at: https://publications.iadb.org/publications/english/viewer/Who_Uses_Ride-Hailing_Platforms_to_Drive_in_Latin_America_A_Profile_of_Uber_Drivers_in_Brazil_Chile_Colombia_and_Mexico_en.pdf

⁷⁴ More information available at: <https://novafrica.org/wp-content/uploads/2021/04/2108.pdf>

failing to predict how much an unexpected event would cost, we cannot predict how often we will have to cover unexpected expenses. Anecdotal evidence from our work with gig drivers shows they are accurate in predicting how much a small car crash would cost them (i.e., repairs), but fail to predict how many small crashes they may have over a period of time.

Being overly optimistic about our future behavior (optimism bias) can lead us to being underprepared and even saving less for our retirement (Balasuriya et al., 2014). This challenge is paramount to understanding how to help gig workers and entrepreneurs build financial resilience. We fail to predict our future expenses. We underestimate how many shocks we will have, and because of an already weakened safety net and volatility around income, gig workers and entrepreneurs are more susceptible to financial shocks.

Designing Behaviorally-Informed Interventions

Understanding the challenges from a behavioral science perspective is key to designing the right solutions for gig workers and entrepreneurs.

Simplify decisions and choose the right defaults.

In a study with a fintech in the U.S., we found that prompting users to create a recommended savings goal during their onboarding (effectively giving them a savings amount by default), more than doubled the number of people saving for an emergency (Common Cents Lab, n.d.).⁷⁵

In our discussions with financial organizations and digital

⁷⁵ Full Case Study available at: <https://case-studies.advanced-hindsight.com/case-studies/DIsAqBqR/prompting-in-onboarding-to-increase-emergency-savi>

platforms, we are seeing more efforts being made to simplify the decisions for gig workers and entrepreneurs, by integrating financial services into platforms (linking financial tools to their income source) and setting-up default choices that will benefit them in the long-run. Defaults have proven to be successful even in less researched societies (Blumenstock et al., 2018).

Establish new reference points.

One key challenge described above reflects how gig workers anchor themselves to reference points that guide their behaviors and decisions, even when those reference points are not the most useful ones. Interventions that help workers establish new reference points at key moments can drive behavior change.

A study in Mozambique with small business owners found that comparing their performance to that of other business owners similar to them increased the likelihood of making strategic business decisions that increased their performance in the coming months (Seither, 2021). Learning about what others do, part of what we understand as social norms, can have a powerful influence in our behavior, though more research is necessary in the financial domain.

Likewise, in a study with a fintech in the U.S. that supports gig workers, we found that adding an income goal to their income tracker increased the earnings of gig workers by \$7-20 per week (a 6% lift) (Common Cents Lab, n.d.).⁷⁶ Establishing a new reference point to compare their income helped gig workers make work decisions throughout the week. In Latin America, our experience shows that rideshare drivers are typically anchored either to a daily or weekly income goal (to cover a set of weekly expenses) or to a daily time goal.

⁷⁶ Full Case Study available at: <https://case-studies.advanced-hindsight.com/case-studies/nezX02WT/using-goals-to-enourage-account-linking-and-increa>

However, when weekly expenses are reduced, testimonials reflect that drivers may choose to reduce their income target to keep a weekly take-home pay constant (the anchor). While surge pricing seems to be effective, less is known about the boundaries of its effects and the interaction with expense changes (Allon et al., 2023).

Pre-commit Today for the Right Choice Tomorrow

One of the most challenging barriers identified by behavioral science is temporal discounting. We value the present much more than the future. To overcome this, successful interventions have nudged people to make decisions today that would be reflected in the future (Thaler & Benartzi, 2004). In a study with a fintech, we found that asking people to commit to saving a percentage of their tax refund before receiving it increased the amount saved compared to those asked after receiving their refund (Common Cents Lab, n.d.).⁷⁷

An additional challenge with gig workers and entrepreneurs, however, is the volatility of income in the present and future. This, combined with the still nascent implementation of automatic tools in Latin America, makes automatization challenging to implement. In a set of studies with financial organizations in Chile and Costa Rica, we are testing whether people are more likely to create a savings prepaid card when prompted to do so before their income or after receiving it, as well as whether small business owners can be motivated to choose to automatically save a percentage of every sale they make. We hope these studies will shed light on how to help gig workers and entrepreneurs overcome present bias to increase their financial resilience.

⁷⁷ Full Case Study available at: <https://case-studies.advanced-hindsight.com/case-studies/WlaalUOh/increasing-savings-at-tax-time>

Testing in the Field: The Power of RCTs

Designing behaviorally-informed interventions for gig workers and entrepreneurs is only one part of the solution. What a large body of evidence in behavioral science shows is that, while the field has grown exponentially, we are still learning about the robustness of behavioral interventions, their long-term effects, the mechanisms underlying interventions and, not less importantly, the effect on understudied populations (Beshears & Kosowsky, 2020).

This last point motivates our work with gig workers and entrepreneurs. While more and more researchers partner with traditional organizations, very few efforts have been done to explore academically what works and what doesn't work to increase the financial resilience of gig workers and entrepreneurs.

At the Common Cents Lab, we use Randomized Controlled Trials (RCTs) to test interventions and answer research questions. By randomly assigning participants (users, clients, workers, entrepreneurs) to a control group (business as usual) and to one or more treatment conditions (interventions to be tested), we can confidently learn about the effect of these interventions on a set of predetermined outcomes. While most behavioral principles have been tested in laboratory settings, more and more field studies are aiming to test similar principles or explore these mechanisms in the field.

Designing and running experiments with digital platforms and financial organizations, however, has its challenges. Throughout our partnerships we've identified challenges at three key levels of implementing RCTs:

- **First**, some organizations are still skeptical about the benefits of running an RCT compared to rolling out an intervention and measuring before and after outcomes. The proliferation of behavioral sciences and

experimental designs is changing this perception at a fast pace.

- ▶ **Second**, the tools and structures that organizations have in place for design testing, data collection and development are not always readily available or have the required features for implementing an RCT. In the most common scenario, backend processes pose a challenge for showing users two different versions of a product, a screen or a set of tools. While motivated organizations allow for creative workarounds, this is a challenge that increases the risk of RCTs not being implemented properly, or the perception of how costly they can be.
- ▶ **Third**, regulations and legal constraints also influence the implementation of RCTs and behaviorally-informed interventions. While evidence shows that defaults and automation are among the most powerful nudges that can be implemented, financial regulations in some countries restrict institutions or act against offering differentiated services to users at random. While the intentions behind these regulations are to protect customers, we hope that the growing body of behavioral science research in Latin America improves how regulations protect customers and allow for innovation and research to serve them better.

Conclusion & Key Points

A final point of reflection that comes from our work with gig workers and entrepreneurs in Latin America has to do with the implications of doing behavioral science and where it is rightfully located. Most of the financial resilience challenges described in the introductory section and throughout this

chapter can be addressed with behavioral interventions implemented by financial institutions and digital platforms. They can also be addressed with evidence-based policy change and regulations that support increased financial resilience for workers. Regulating digital platforms has been a heavily discussed topic in the policy domain (Baldwin & Cave, 2020).

In line with these concerns, a recent publication has questioned whether behavioral science is placing the burden of behavior change, especially for vulnerable groups, on individuals rather than on systems (Chater & Loewenstein, 2022). In their own terms, behavioral science has focused extensively on “i-frame solutions” (focused on individual behavior change) at the expense of “s-frame solutions” that focus the attention on redesigning systems and environments to support better outcomes. We share this concern, but also have a few reflections of our work with digital platforms that can contribute to this discussion.

On the one hand, considering the challenges that low- to medium-income households face when dealing with financial shocks and the devastating effects of the pandemic, gig work platforms served as a lifeline to generate additional income, including in underrepresented sectors such as home care and freelance work (Arroyo et al., 2021; Financial Access Initiative, 2020).⁷⁸ In this context, behavioral science is uniquely positioned to help digital platforms and financial services that have already contributed to this lifeline implement small but meaningful changes that will better serve users.

On the other hand, by learning through applied research what works and what doesn’t in supporting behaviors and decisions that increase workers’ financial resilience, behavioral

⁷⁸ More information available at: <https://www.financialaccess.org/blog/2020/5/21/how-resilient-are-we-a-dive-into-the-global-data-on-how-people-deal-with-unexpected-shocks>

science also can, and should, inform policy decisions that expand the impact of interventions to larger groups. For instance, what a meta-analysis (Beshears & Kosowsky, 2020) of behavioral interventions shows (which aligns very well with an analysis of our own field interventions) is that structural interventions can have a larger impact than communication-based interventions. This means interventions that change the set of options that people choose from and automate decisions that have a positive impact or that set-up default options can increase financial resilience. These interventions, by definition, can also serve as evidence for policy changes that directly improve the context in which people make decisions.

Gig workers and entrepreneurs in Latin America are an understudied group from a behavioral science perspective. We believe that this work will be the starting point for stronger collaborations between researchers, platforms, practitioners, and policymakers to continue understanding what works and what doesn't in increasing their financial resilience.

In the next chapter we learn more about behavioral science in the health field.

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18.

Behavioral Science for Health Impact

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Over the last 30 years, USAID's health programming has invested heavily in interventions aimed at changing behaviors.⁷⁹ In the early 1970s and 1980s USAID invested in social marketing through projects such as Social Marketing for Change (SOMARC) I, II, and III. Those projects used the principles of marketing to promote the use of products such as contraceptive pills, condoms, oral rehydration salts, and other health products. The products were sold at subsidized prices in commercial outlets to maximize access.

At the same time, Information, Education, and Communication (IEC) was used to raise the awareness of health information, including the importance of handwashing or the use of family planning through communication methods and principles.

⁷⁹ The content of this chapter is not based on a comprehensive portfolio review. The contents of this article are those of the authors and do not necessarily represent the official views of the U.S. President's Emergency Plan for AIDS Relief, the U.S. Agency for International Development, or the U.S. government.

Often projects developed posters, flyers, leaflets, brochures, booklets, messages for health education sessions, radio broadcast, TV spots, and more.

In the 1980s and 1990s, USAID's programming evolved with the introduction of behavior change communication (BCC). The field of BCC recognized that raising awareness was important but not sufficient to meet priority public health objectives. While BCC projects such as HealthCom or AIDSCom also focused on communication methods to promote positive health behaviors, the work was grounded in proven theories and models of behavior change, such as the Health Belief Model, the Stages of Change or the Socio-ecological model.

In the 2000s, the field embraced the importance of social change as essential to efforts in behavior change and began using the term social and behavior change communication (SBCC). SBCC, through projects such as C-CHANGE or the Health Communication Capacity Collaborative (HC3), emphasized the use of communication-based approaches to promote behavior change and facilitate social change to improve health outcomes. SBCC projects worked to change social norms to change behaviors, recognizing that often behavior is governed by the groups to which people belong.

As evidence has grown around the importance of structural and non-communication-based approaches to behavior change, USAID began using the term social and behavior change (SBC). SBC better reflects the breadth of approaches (communication and non-communication) and is inclusive of many social science fields of practice. USAID considers SBC a systematic, evidence-based approach to improve and sustain changes in behaviors, norms, and the enabling environment. SBC interventions aim to affect key behaviors and social norms by addressing their individual, social, and structural determinants and are grounded in several disciplines, including systems thinking,

strategic communication, marketing, psychology, anthropology, and behavioral economics (BE).

USAID's health-related SBC programs have predominantly targeted System 2 drivers of beneficiary populations for the desired behaviors. In 2016, USAID developed an agreement with the U.S. government's Social and Behavioral Science Team (now known as the Office of Evaluation Sciences) to explore the use of BE within USAID health programs. Initially, this effort focused on different health areas in two countries, including a feasibility study on increasing family planning uptake in Nigeria, increasing the uptake of intermittent preventive treatment of malaria during pregnancy (IPTp) in Nigeria, and increasing retention in care and antiretroviral therapy adherence in Ethiopia in female sex workers.⁸⁰

Many lessons learned from these initial efforts, including:

- ▶ The complexity of creating good data sources
- ▶ The importance of sample sizes for detection of significant change
- ▶ The need for testing and co-creating with the target audience

These and other lessons have fed into the continued integration of BE and behavioral insights (henceforth referred to as behavioral science) into the work USAID does in SBC.

Use of Behavioral Science at USAID

With the launch of USAID's project, Breakthrough ACTION, in 2017, behavioral science applications gained further momentum. An early example comes from addressing the mistreatment of women during childbirth in health facilities, a documented problem worldwide and prevalent across all country income levels (Bohren, M. et al., Lancet 2019). Through Breakthrough

⁸⁰ More information available at: <https://oes.gsa.gov/usaaid/>

ACTION, the behavioral consulting firm Ideas42 collected insights in several countries, including Liberia and Zambia.

The work uncovered a range of behavioral biases impeding respectful maternal care among health care providers, including unfavorable norms and mental models (such as pain being considered a natural part of the birthing experience), a lack of cues for standard operating procedures, and an absence of feedback on performance.

To overcome these barriers, the project developed a “provider-client promise” introduced during intake, a pain management toolkit, a client feedback box, and a reflection workshop for health providers. In a quasi-experimental trial, results showed a 15-percentage point decrease in reported instances of disrespect and abuse among women in the intervention sites. There was also an increase in pain management support provided to women and providers placed a greater importance on pain management as part of the routine care they provided.

Behavioral science has also been applied to address malaria testing and treatment in Nigeria.⁸¹ While most health care providers know the protocols for testing and treating patients for malaria, they often prescribe malaria medication even when test results come back negative, wasting stretched resources and potentially contributing to resistance to current treatment regimens. Root causes for this behavior include persistent mental models, anchoring (the first symptom shared by patients is often fever), and decision fatigue.

Solutions underway include:

- ▶ A simple redesign of client flow at the facility level.
- ▶ A revised consultation packet including pediatric evaluation form, fever care card, data, and accountability devices.

⁸¹ More information available at: <https://breakthroughactionandresearch.org/wp-content/uploads/2020/05/Behavioral-Design-PBC-Malaria-Nigeria.pdf>

Together, these interventions led to better adherence to testing and treatment protocols, shorter waiting times for patients, and greater efficiency for providers (Haqqi et al., 2022).

Another example of applied behavioral science, again through Breakthrough ACTION, is the work in Zambia to improve provider-client confidentiality, a key barrier for adolescents to access reproductive health services and for men to get HIV testing.⁸² The project introduced a confidentiality pledge and a mnemonic job-aid called “SECRET” spelling out key provider behaviors (like C for “Close your door”) that confer a reassurance of confidentiality to clients.

To further explore the opportunities to leverage behavioral science in its HIV programs, USAID’s Office of HIV/AIDS contracted a consultant to carry out an internal literature review of applications (Long, 2022) in Lower- and Middle-Income Countries (LMICs). 36 studies were reviewed. Some findings worth highlighting include:

- ▶ **Gain-framing using “Undetectable = Untransmittable”:** Embedding a positive outcome through the “U=U” (having an undetectable viral load when you are taking HIV treatment also stops HIV transmission to a sexual partner) message doubled testing rates in one study with men in South Africa (Smith et al., 2021). Men test for HIV at lower rates than women. Globally, 88% of women aged 15 years and older living with HIV are aware of their status. This percentage stands at 82% among men in the same age group (UNAIDS, 2020).
- ▶ **Social proof:** The use of social proof has proven to deliver promising effects along the HIV continuum. It suggested a way to increase testing among men in

⁸² More information available at: https://zambia.thecompassforsbc.org/sites/default/files/project_examples/142-1235_BABrief_Zambia_NiZii_final.pdf

Uganda (Perkins et al., 2018). Along with priming, it increased attendance and adherence in Tanzania (McCoy et al., 2017) while it pointed (but not conclusively) to increased adherence in youth and adults in Uganda (Perkins et al., 2022; MacCarthy et al., 2020).

- ▶ **Simplification:** Reducing the hassles and hurdles that clients face to access services may nudge them into prevention and treatment options.
 - ▷ For example, taking services closer to adolescent girls and young women, who are more at-risk of HIV than adolescent boys and young men, increased sexual and reproductive health visits in Eswatini (Brault et al., 2021).
 - ▷ Providing male partners of pregnant women with self-test kits that can be used at home increased testing rates by 39 percentage points in Kenya (Masters et al., 2016). Distributing self-test kits also increased periodic testing in hard to reach male truckers (Kelvin et al., 2019) and female sex workers in Kenya (Kelvin, 2019).
 - ▷ Policy changes that enable differentiated service delivery models, including community-based distribution of medication, have also shown that simplification is important for continuity of treatment.

Several studies are underway, including one examining the effect of “Fresh start” text messaging to support reengagement in treatment in South Africa (see indela.com).

We should note that, though motivated by clinical and medical reasons, the Universal Test and Treat introduced by WHO represented a powerful change in default where individuals

testing positive for HIV were automatically eligible for and linked to treatment, regardless of CD4 levels and clinical stage. Other structural interventions such as multi-month dispensing of antiretroviral (ARV) medication are simplification nudges to support adherence and continuity in treatment among clients living with HIV.

Challenges and Limitations

As interest grows in the application of behavioral science, USAID Global Health programs face several challenges. One of the more difficult ones is the importance placed on randomized control trials (RCTs) and A/B testing in behavioral science. Often, limited data systems and cost prohibit the use of RCTs and/or A/B testing for evaluating an intervention. Good data, however, is crucial for iterative programming.

The behavioral science field will have to confront this challenge with creative, affordable and realistic methods (including leveraging program data) for measuring the effectiveness of an intervention. The behavioral science field will also need to address how to handle poor results. While often these provide learning opportunities, there is less support by stakeholders in the health and development field who may equate testing results with success or failure.

Equally challenging is the limited capacity of applied behavioral science in LMICs. This challenge is two-fold. There are both few behavioral economists or local BE organizations in LMICs and a limited number of external researchers with experience applying BE in these regions. With a premium placed on BE activities being led by researchers in the countries in which implementation is occurring, capacity building will require an intentional effort.

There has been limited experience integrating BE partners

within SBC projects. For USAID programs, BE is best applied in a layered, multi-pronged SBC program. Therefore, finding common language and platforms for partners to collaborate and learn together will be essential for future USAID programming.

As more implementing partners integrate behavioral science approaches into their work, questions around the ethics of BE, in particular, have arisen. Concerns expressed have included:

- ▶ When does changing an individual's environment to create more predictable behaviors become manipulation?
- ▶ Does BE limit autonomy and freedom of choice?
- ▶ Can we improve the wellbeing of some people while unintentionally harming others?

Conclusion & Key Points

As the evidence base for behavioral science applications in global health grows, there is a potential risk that nudges are viewed as a silver bullet that can be used in isolation of other behavioral change approaches or as a substitute. Given this risk and the challenges and limitations cited earlier, when is the application of behavioral science most appropriate to advance health outcomes in lower-resource settings? We suggest the following:

1. **Embed behavioral science as part of a holistic approach.** While behavioral science approaches target System 1 drivers, critical System 2 drivers may be just as if not more powerful. For example, systemic stigma and discrimination faced by people living with HIV cannot be reversed through nudges alone. A comprehensive structural and behavioral program that includes policy changes, health facility-based improvements, and community-level interventions are required, among others.

2. **Address last mile challenges.** Given their relatively limited effect sizes (outside of defaults), behavioral science applications may be more relevant in reaching population segments whose behavioral determinants have not been adequately addressed through other behavioral and structural interventions. Nudges such as social proof may be useful to move the needle on health outcome targets as desired behaviors become prevalent.
3. **Ensure that core principles and values are applied and established as minimum standards.** Behavioral science interventions should be based on a diagnosis of relevant barriers and drivers for the desired behaviors. They should also be designed with ethical and equity concerns in mind. “First do no harm” remains fundamental. The interventions should be co-created with the target populations or at the very least, pre-tested.
4. **Commit to testing.** When you can’t test, accept alternatives such as quasi-experimental designs, case control studies or other observational methods of measurement when a RCT or A/B testing is not feasible due to sample size requirements, timing, cost, and other considerations.

Finally, agree with stakeholders including government counterparts, donors, and implementing partners on what success looks like and embed learning into monitoring and evaluation efforts.

In the next chapter we explore behavioral science initiatives addressing challenges around climate change, the environment, and sustainability.

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19.

Environment and Sustainability

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We need to invest in and build a world that makes sustainable and equitable changes easier for everyone. Applied behavioral science looks at how daily choices — such as eating less meat, using less water, or using less energy at home — can be effectively encouraged at scale.

Behavioral policy design includes, but also looks beyond, traditional “nudges” to explore ways to encourage behaviors at a population level. We must think bigger. We will not nudge our way out of the climate crisis.

Background

The Climate Emergency and Behavior Change

During the COVID-19 pandemic, we learned with piercing clarity the cost of undervaluing the role of human behavior in tackling global emergencies. Indeed, when the outgoing director of the United States' National Institutes of Health (NIH) was asked what the NIH could have done differently in their fight against COVID, he said: "Maybe we underinvested in behavioral research." (PBS, 2021) We should not make the same mistake in the climate crisis.⁸³

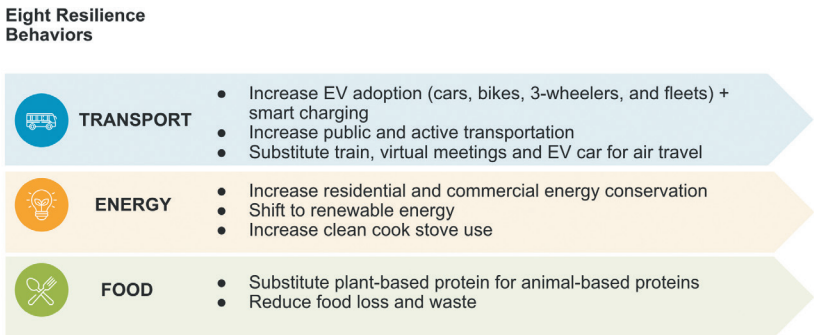
The latest report from the Intergovernmental Panel on Climate Change (IPCC) estimates that comprehensive behavior changes have the potential to reduce greenhouse gas emissions by 40–70% by 2050—the same amount as eliminating emissions from China, the United States, India, the European Union and Russia *combined* (Creutzig et al., 2022).

What behaviors matter most? At the World Resources Institute, we use the term (from the Finnish innovation fund, Sitra) *hotspot behaviors* to target the most impactful behaviors for the climate. These include how we eat, how we travel, and how we light and heat our commercial and residential buildings. Each hotspot behavior has a corresponding *resilience behavior*. These are actions that, if adopted at scale, would produce drastic emissions reductions. See Figure 1.

Resilience behaviors have benefits that reach beyond climate. These shifts can produce wellbeing impacts like food and nutrition security, increased household savings, and increased economic opportunities and improved health, especially for those most marginalized.

⁸³ Full quote available at: <https://twitter.com/NewsHour/status/1473082652181807106>

Figure 1: Eight Resilience Behaviors



Source: Extracted from World Resources Institute (2023)

How to Apply Behavioral Science to the Climate Crisis

A Global and Local Approach

The climate crisis is global, but its causes and impacts vary drastically by region and income. Effective climate action requires acknowledging that high-income countries bear significant responsibility. For example, around 80% of the planet's population has never set foot on a plane. Addressing aviation emissions is the responsibility of the 1% (led by the U.S.) that make up 50% of air travel emissions.⁸⁴

Increasing energy efficiency in growing, emerging economies like India is critical, but we also must recognize that significant portions of the population in the global south still lack — and need — access to reliable electricity. And while much of the global north can shift away from meat, it's important to recognize that many people around the world need access to more protein, *including* meat.

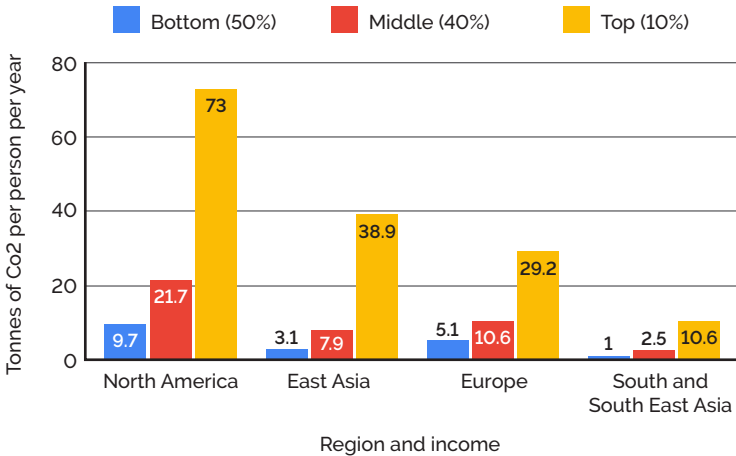
⁸⁴ See <https://ourworldindata.org/co2-emissions-from-aviation>

Therefore, applied behavioral science needs to take a tailored approach to behavior change. For example, decreasing meat eating makes sense in the U.S. and European Union (two of the highest consumers of meat), but not in India.

Besides a regional approach, we must consider socioeconomic differences. Intra-country emissions vary drastically by income. The emissions of the top 10% of the population in South-East Asia have approximately the same emissions as that of the median European. In East Asia, those with incomes in the bottom 50% produce only 3.1 tons of CO₂e per person per year. But those with incomes in the top 10% produce 28.9 tons per person per year; more than the median American. See Figure 2.

Figure 2

Tonnes of CO₂ Per Person Per Year, 2019



Source: Extracted from World Inequality Report (2022)

Consider both Systems and Individual Change

Applied behavioral science has been narrowly focused on changing the behavior of individuals, rather than changing the larger policy context that often determines how individuals behave. In the fight against climate change, this focus on the individual is not accidental.

Powerful political and business forces have pushed the narrative that reducing emissions is the role of individuals and not industries or governments. As applied behavioral scientists working in this space, we have to be very careful not to perpetuate this narrative.

As Chater and Lowenstein (2022) describe in their recent paper *The i-Frame and the s-Frame: How Focusing on Individual-Level Solutions Has Led Behavioral Public Policy Astray*, the field has “mistaken deep systemic problems of political economy and conflicts of interest, for problems of individual human folly and responsibility.” The article exposes the initial roots of the carbon footprint calculator.

In the midst of the climate wars, British Petroleum, or BP, developed a clever strategy to promote the idea of a personal “carbon footprint.” They put out a carbon footprint calculator to help people determine their personal impact on the planet. For BP, the campaign was a resounding success, and individuals, campaigners, media organizations, and government agencies all provided their constituents with access to carbon calculators to help *them* reduce *their* impact on the planet.

The implicit message was obvious. If people decreased their personal carbon footprints, then we could collectively reduce the problem of climate change. By framing carbon reduction as an issue of individual responsibility, BP encouraged people to focus on their personal actions and overlook the role of broader systems and policy interventions in driving change. This framing ultimately detracted from the need for systemic change.

Individual and systems change do not have to be mutually exclusive. Instead, they should be mutually reinforcing. We need to shift downstream individual behaviors *and* the upstream system-level behaviors (e.g., of industry leaders, policymakers, employers) that determine what choices are available.

For example, behavior change can mean “nudging” households to modify diets, reduce flights and use energy more efficiently. But it should not stop at individual actions. The food industry can invest in plant-based meat alternatives, employers can institute flight reduction policies, and policymakers can invest in solar and wind solutions to free energy grids’ dependency on fossil fuels.

Some work should focus on the individual, but our aim should always be to push for neighborhoods, workplaces, industry practices and policies that make it easier for people to live healthier, greener lives.

Focus on Social Justice

We must consider how historical and structural barriers and inequities impact behavior. Our failure to tackle bias, racism and stigma will sabotage our fight to protect the planet. When behavioral interventions are simply laid over the roots of unjust policies and practices, “nudges” can make those marginalized even worse off. The scholars Crossman, Jurcevic, Holmes, Hall and Allison lay out an illustrative scenario in their recent article, *An Equity Lens on Behavioral Science for Conservation* (Crossman et al., 2022).

Imagine that a conservation organization uses a behavioral approach to enforce a ban on fishing for grouper during spawning season by providing information and then asking each family to make a commitment to comply with the ban. The wealthier families have the resources to move to other sources of income and comply with the ban, while the poorest families violate the ban because they have no other options.

The intervention has created a norm that some families cannot

afford to meet. An evaluation of the intervention does not consider socioeconomic status and misses the effects on poor households. Although the intervention exacerbated existing disparities and the poorest families faced ongoing social and psychological costs because of violating the ban, the evaluation looks like a success.

A few years later, a major golf resort financed by an international property development group was constructed in the area, altering reef-land interactions and rendering the area an unsuitable habitat for grouper. By focusing on the least powerful actors and neglecting the larger power dynamics in this natural resource system, the intervention misunderstood the larger threats to the grouper population and left the grouper and the fishing community, especially the poorest, worse off.

The authors explain that a more successful strategy would involve supporting existing village institutions and assisting the community to engage successfully with more powerful actors. Applying behavior change tools to powerful tourism-sector actors could have been a more successful strategy.

Overlooking existing power dynamics can have negative impacts, but the opposite is also true. By leaning into our ambition and optimism to design for the world we want, we advance the field and fight for justice. We must change the game, not just help individuals be better players.

For example, besides assessing the climate impacts of behavior change, we should explore the social impacts of resilience behaviors on health, income, and equity. Behavioral shifts can produce wellbeing impacts in terms of food and nutrition security, increased household savings, increased economic opportunities and improved health, especially for those most marginalized.

By demonstrating that climate actions aren't only investments in long-term planetary health but are also near-term investments in the people they serve, we will equip policymakers and industry leaders with the evidence they need to invest in the highest impact changes.

Our goal should be to use behavioral sciences as a tool to support the transition toward a more sustainable and equitable society.

Case Studies of Insights to Action

Energy Use in India

From an equity and economic development perspective, it's critical for more households — especially in emerging economies like India — to become electrified. However, electricity use in India is a major contributor to rising greenhouse gas emissions and other externalities (like air pollution) because of the Indian grid's dependence on fossil fuels.

VidyutRakshaka (VR), a joint project between Technology Informatics and Design Endeavor (TIDE) and WRI India, produced “behaviorally designed” household energy reports that reward more sustainable energy use.

Even before “nudge” became a well-known concept, TIDE and its consortium partners had shown small but significant specific energy reduction through simple housekeeping and behavior change measures in tea processing factories in South India.⁸⁵ TIDE extended the idea to the urban residential sector, with a hypothesis that informed behavior change measures should lead to a reduction in electricity use.

Sumathy Krishnan, who was Executive Director of TIDE at the time, conceptualized the program VidyutRakshaka (VR) along with her team with dual goals of helping households reach an optimal and sustainable electricity spend and to keep in check the rising per capita electricity consumption in India.

The household electricity reports established a baseline for the home and then, because insights from behavioral science tell us that our behavior is informed by what others do, the reports

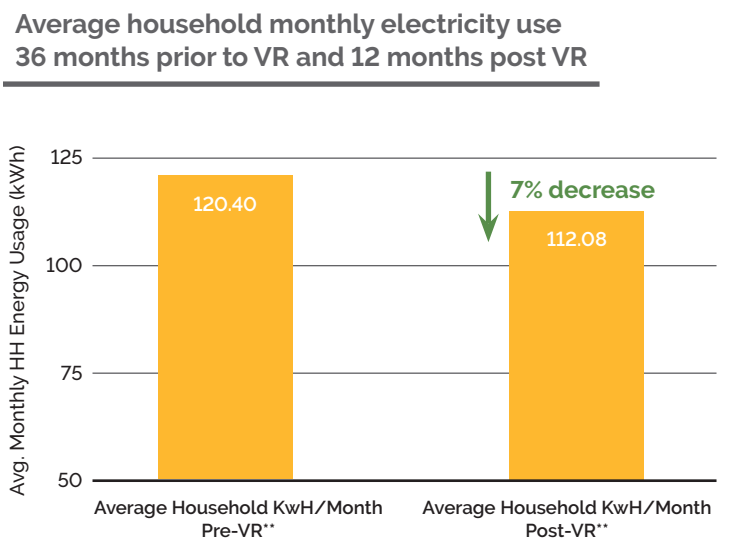
⁸⁵ More information available at: <https://tide-india.org/energy-conservation/>.

benchmarked a household’s energy use against neighboring households. The report also provided customized, actionable recommendations to help people bring down energy use and costs. These were categorized into Nil cost, low cost and RoI (Return on Investment) based recommendations, like increasing the temperature setting on air conditioner, replacing incandescent lights with LED bulbs, and investing in solar water heaters.

Social scientists have found that because people procrastinate on tasks that seem complicated or unclear, simplifying the steps can increase positive behaviors.

The project’s findings suggest that receiving these behaviorally informed energy reports decreased energy use by an average of 7%. See Figure 3.

Figure 3.



Note: *p<.05, **p<.01, ***p<.001
Source: WRI Authors.
Source: Extracted from Hernandez et al. (2022)

This may not seem like a lot, but individual actions can add up quickly when expanded to entire populations. If the entire city of Bangalore received and acted on these reports, 604 million kilowatts of energy a year would be avoided and households would save almost \$60 million per year collectively.

Similar results have been found with energy audits elsewhere, including in Delhi, India, U.S. and the United Kingdom. But implementing these at scale is still difficult in the global south. Practical difficulty in continuously accessing reliable data in a timely manner is an issue. There is also an opportunity for the system to reinforce the individual's behavior choices through the right incentives.

Imagine if presenting energy bills this way was the default for all major utility companies. That would be a structural change built off an understanding of human psychology with the potential for massive and lasting impact.

Water Use in Mexico

The situation in northeastern Mexico is a warning call for drought-risk communities around the world, particularly those in northern Mexico and the southern United States. A growing population, a strong water dependent industry and a large agricultural area has helped to over-exploit water resources. A particularly dry summer during 2022, pushed the already low reservoirs to the brink. Large swaths of the population did not have access to water for days. Given excessive water consumption by many residents and the increasing heat as a result of climate change, this situation could get even worse.

With local partners, WRI is designing interventions that could sustainably reduce the consumption of water at the residential level. Interventions are still in the design stage, but our formative research provides some insights into the practice of

behavioral design in tackling climate issues built on partnership and participation.

First, we are doing several interviews with local and international experts on water and behavioral change to identify key elements necessary for a potential intervention, policies, forces and other kind of limitations that could create factors to promote specific behaviors. Next, we are spending time understanding how different communities perceive the situation by holding focus groups with diverse households. These focus groups will take into account socio-economic and geographical factors that could have an impact on water consumption.

Finally, we are considering both system level changes (policies, infrastructure improvements) as well as behavior change techniques focused on individual level change. Depending on the results of our formative research, interventions focused on changing individuals' behavior might include elements such as consumption reports, simplifying and explaining the water bill and the cost of water, and/or promoting the use of native plants that require less water in gardens. But to have a wider impact it is important to include infrastructure changes as well. This might include controlling the water pressure in some areas of the city or providing subsidies on water saving technology like water saving shower heads and water efficient washing machines.

Conclusion & Key Points

Through The Living Lab for Equitable Climate Action (TLL), WRI is committed to working with partners to apply behavioral research to help the people and planet thrive, with a focus on those too often left behind. Future work is inspired by WRI's "count it, change it, scale it" framework.

► ***Count It: Identify and Quantify***

Lack of sufficient research has prevented policymakers and the public from understanding the true potential of behavior changes. Upcoming research will model the climate and socioeconomic benefits of adopting resilience behaviors.

By demonstrating that climate actions aren't only investments in long-term planetary health but are also near-term investments in the people they serve, we will equip policymakers and industry leaders with the evidence they need to invest in the highest impact changes.

► ***Change it: Design and Test***

Research is necessary but insufficient. TLL is working with key partners (industry leaders, policymakers, and civil society) to co-design and test behavioral change solutions in the field, with a focus on improving the lives of those traditionally marginalized.

► ***Scale it: Drive Pathways to Scale***

The urgency of the climate crisis demands that we shift behaviors at a population-level, which requires identifying pathways to rapid scale. TLL will partner with industry leaders, civil society, and policymakers to scale successful interventions into actionable policy packages that provide behavioral policy solutions across the spectrum of high-impact behaviors from transport to food to energy.

In the next chapter we take a look at how behavioral science can address issues of democracy and governance.

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20.

Democracy and Governance

Laura van Berkel, USAID ⁸⁶

Democracy and Governance (DG) are core goals of international development. It is a developmental goal for people to live freely, have a say in how they are governed, and live under accountable, transparent governments. Democracy is also positively associated with a host of other development goals including economic growth, improved health, increased peace, and action to mitigate climate change (Acemoglu et al., 2019; Bollyky et al., 2019; Reiter, 2017).⁸⁷ Given the importance of democratic governance to improving the lives of people around the world, it is important to understand and shape patterns of human behavior that allow democracy to thrive.

The exact characteristics and definitions of democracy have

⁸⁶ The author's writing and opinions are their own and do not reflect official guidance or stances from their organization.

⁸⁷ Review https://www.v-dem.net/media/publications/pb_31.pdf. While democracies are better at taking action to address climate change they don't necessarily have better climate outcomes (see Selseng, Linnerud & Holden, 2022).

long been debated by scholars. According to USAID Primers, democracy is “in the most minimal sense, [...] a system of government in which the people can choose and replace their leaders in regular, free, fair and competitive elections.”.⁸⁸ Generally, a high-quality liberal democracy also maximizes political competition, participation and responsiveness, protection of individual rights and political equality, civil society and independent media, and governance in the public interest (minimizes corruption). There is also no universal definition of ‘good governance.’

One definition of good governance offered by the United Nations is “the process whereby public institutions conduct public affairs, manage public resources and guarantee the realization of human rights.”.⁸⁹ It includes elements such as rule of law, effective participation, transparent and accountable processes and institutions, and institutions that can effectively and efficiently provide public goods and services.

Why Do Democracy and Governance Need Behavioral Science?

It is a common misconception that democracy and governance are simply too complex and too abstract to clearly define and measure at the level of individual human behavior. Democratic governance, however, fundamentally depends on the behaviors of people. There is no democracy without inclusive voter turnout, citizen participation in public fora, civil servants that respond to citizen needs, and service providers that deliver quality services among many other crucial behaviors.

In this way an understanding of social and behavioral science

⁸⁸ USAID Democracy and Governance Primer Series (2018). https://pdf.usaid.gov/pdf_docs/PA00TMSQ.pdf

⁸⁹ More information available at: <https://www.ohchr.org/en/good-governance/about-good-governance>

is critical to assess the drivers of human behavior that prevent or enable DG goals. Even in the best-case scenario when people have intentions to engage in democratic behaviors, there are many potential, often non-intuitive, barriers that can stand in the way to create an “intention-action gap” (Sheeran & Webb, 2016).

Let’s take the hypothetical case of Fatima as an example; she cares deeply about her municipality but fails to participate in municipal forums. The absence of this desired behavior could be because she simply forgets about the forums, lacks the confidence to speak in front of an audience, fears reprisals from government officials, or believes that it is inappropriate for women to be involved in formal political affairs, among many potential barriers. Psychological and social obstacles to intended behavior appear throughout the DG sector, for example:

- ▶ Civil servants may not respond to citizen requests or complaints because no specific department or person is assigned to respond (diffusion of responsibility).⁹⁰
- ▶ Civil society organizations (CSOs) may struggle to ignite collective action movements because people do not have a shared sense of identity with each other or with the CSO (Zomerren, Postmes, & Spears, 2008).
- ▶ Members of legislative bodies may fail to cooperate to effectively solve national problems because they do not trust colleagues’ motives (Tyler and De Cremer, 2006).
- ▶ Judges may not make fair sentencing decisions upholding the rule of law because they find it difficult to ignore or forget inadmissible information (Teichman & Zamir, 2014).

⁹⁰ More information available at: https://www.ideas42.org/wp-content/uploads/2020/01/142-1198_GovernmentResponsivenessPaper_final.pdf

Interventions based on evidence from behavioral science can help to overcome psychological, social, and structural barriers to desired behavior in DG objectives. Returning to the example of Fatima, if she intends to participate in municipal forums but forgets about them, then an intervention like reminder texts or postcards may be effective at making meetings more salient and help compensate for the natural limits of human memory (Milkman et al., 2021).

If Fatima believes that it is inappropriate for women to be involved in governance and politics, then it may help to highlight positive examples of women involved in governance, especially within or close to her community, social norms that support women's participation, and to work with officials to restructure meetings to invite participation from women.⁹¹

A particularly thorny problem in the DG sector is corruption. Superficially, corruption seems like a problem of individual greed and insufficient deterrence. However, corruption often becomes much more embedded in a culture, including public institutions, through social norms, organizational structures, and institutional failures that make corruption both collective and functional (Marquette & Peiffer, 2020).

For example, a police officer may personally find it immoral to solicit bribes at traffic stops, but the social norm, or prevalent behavior or expectation for behavior, among police officers is to solicit such bribes. Non-compliance with this norm can result in informal sanctions from colleagues, such as expressed disapproval and threats or acts of violence, or more formal sanctions such as demotion, reduced opportunity for promotion,

⁹¹ More information available at: <https://www.povertyactionlab.org/evaluation/changing-teenage-girls-aspirations-and-educational-attainment-through-increased-female>; <https://www.sciencedirect.com/science/article/pii/S0002929719302290>

and transfers to undesirable locations (Scharbatke-Church & Barnard-Webster, 2017).

There is no single driver of corrupt behavior and as with any other behavior change design, the solution depends on the specific behavior (e.g., bribes for service, nepotistic hiring, misuse of public funds, etc.) and the particular drivers of that behavior (e.g., social norms, ineffective government services, perceived inefficacy of whistleblowing, etc.).

FHI 360 recently identified recommendations from social and behavioral science to combat corruption including leverage positive social norms and positive deviance, avoid messaging that corruption is pervasive (i.e., the norm), promote perceived efficacy of anti-corruption actions, and build social cohesion (FHI 360, 2022). But as FHI 360 notes, these recommendations are built on a relatively limited evidence-base and much more rigorous, formative research is needed to better understand and interrupt drivers of corrupt behavior.

How Do You Integrate Behavioral Science?

At first glance, concepts like transparent, accountable governance or free and fair elections are legal and policy affairs that have nothing to do with behavior. However, on closer inspection, many challenges in the DG sector can be broken down into behavioral components by asking questions such as “who needs to do what?”

For example, free and fair elections depend on voters to turn out to vote, partisans to refrain from violence, civil society organizations and citizens to advocate for electoral rights, election workers to accurately follow administration procedures, and citizens to refuse attempts at vote buying among other behaviors.

The following examples illustrate how three particular DG

challenges, voter participation, women's political leadership, and voluntary tax compliance, have been approached from a behavioral science lens.

Example 1: Voter Turnout in Tunisia (Democracy International, 2020)

Since Tunisia's transition to democracy in 2011, voter turnout has declined—plummeting about 25 percentage points between 2011 and 2014 according to International IDEA's Voter Turnout Database.⁹² Democracy International sought to boost voter turnout through a canvassing voter mobilization campaign. They developed four possible messages to be delivered via leaflets or in-person canvassing based on experimental behavioral science research on psychological and social motivations to vote and focus group discussions:

1. **Planning:** “Do you have a plan for Sunday, May 6?” with prompts to articulate when they will vote, how they will get to the polling place, with whom they will go, and what they will do after they vote.
2. **Social Pressure:** “Did you know that some neighborhoods vote more than others? Don't let your neighborhood vote less than others.”
3. **Gratitude:** “Thank you for having an interest in the destiny of your region by participating in the municipal elections.”
4. **Sarcasm** (delivered only via leaflet, not in-person canvassing): “If you are satisfied with the situation of the country, don't vote.”

DI randomly allocated these interventions to 1,750 polling center “catchment areas” and assigned 400 polling centers to a

⁹² Review full information at: www.idea.int/data-tools/country-view/284/40

“pure control” condition (i.e. no message at all). They found that 1) canvassing was more effective than leafleting alone—it generated 10.5 percentage points greater turnout than leafleting and 2) messages prompting people to make a plan for election day delivered via in-person canvassing were the most effective—they increased voter turnout by 15.5 percentage points compared to people who received no message.

Example 2: Women's Political Leadership in Cambodia (Cruz & Wendt, 2019)

Women in Cambodia face many barriers to running for political office, both extrinsic (women are not encouraged to run) and intrinsic (women lack political efficacy or interest). NORC at the University of Chicago in partnership with the National Democratic Institute (NDI) developed and tested a related intervention to lower the social barriers to enter politics: they asked individuals to nominate women in their communities to run for office by SMS text. Researchers randomly assigned 50 communes to only the text intervention, 50 communes to the text intervention and CSO partner workshops and programming (“My Own Commune”) and 50 communes to no intervention (the control).

For the text intervention, the researchers notified the women that they were nominated and encouraged them to put themselves forward for the party lists for commune council. Communes in which women were nominated to run had an increase in women candidates of 19 percentage points (average of 3 more women candidates per commune) over communes in which women were not nominated. The workshops did not significantly increase the number of women candidates, but also faced additional implementation difficulties.

Example 3: Increasing property tax compliance in Kosovo (Ideas42)

Residents in Kosovo often fail to pay municipal taxes or pay taxes on time. This creates a problem for both municipalities as they are unable to plan or make investments and citizens who face increasing debt and fines. Ideas42 partnered with municipal governments to encourage taxpayers to pay their taxes on time by sending reminder letters that leveraged several behavioral insights. Behavioral insights included drawing attention to the deadline and consequences of inaction (making them salient) and prompting residents to make a concrete plan for paying their bill.

Out of 52,652 taxpayers who still owed property taxes, 4,000 were randomly assigned to receive the letter. Residents who received the letter were 15.8 percentage points more likely to make an on-time property tax payment. Those who received the letter were more likely than those who did not to have made payments towards their property taxes both 10 weeks (15%-points) and 5 months later (7%-points).

As these three examples illustrate, DG-related behaviors can be identified, measured, and changed. In each case behavioral science successfully encouraged people to engage in a given behavior using relatively “light-touch” interventions and notably without forcing or coercing changes.

What Are the Limitations?

Behavioral science is a powerful tool to advance democracy and good governance. However, behavioral science is not a silver bullet that will easily solve intractable, complex, systemic issues. It is one tool in the toolbox. There are two important misconceptions about its capabilities for democracy and governance worth noting.

The first misconception is that behavioral science can and

should be used to change the behavior of individual people, especially political elites. Rather, behavioral science focuses on the patterns of behavior in a society or group of people. It does not mean that every person exposed to an intervention changed their behavior. Behavioral science cannot predict and precisely change the behavior of a single individual, nor should it, if we uphold the ethical principle of autonomy (Sunstein, 2015).

The second misconception is that behavioral science focuses on individual change as a substitute for systemic change. Both individual and systemic change are necessary and often need to work in coordination for societal change and development to succeed. People cannot engage in behaviors that are blocked by laws, policies, or other structural impediments. No “get out the vote” campaign will change voting behavior in an autocracy where there are no elections for example. Rather, behavioral science can help to fill in gaps between policy intentions and implementation and is integral to designing effective systemic policy changes that are informed, supported and understood by the public. Without the backing of the public, systemic changes may be ignored or rejected and could even backfire (Jenny & Betsch, 2022). Behavioral science can help to design the right systems that fit the way humans live and work as individuals and as social groups.

Challenges and Opportunities

Challenges in Security and Safety.

There are unique challenges to implementing behavioral science in the DG sector due to the often-sensitive nature of the work. People who benefit from corruption or hold formal or informal power through undemocratic processes will likely find DG activities threatening and may resort to violence or other forms of repression to preserve the status quo. In this context, it is extremely important to protect the safety of local partners and

beneficiaries. It may be risky for beneficiaries to answer sensitive questions such as those related to gender-based violence, corruption, or violent extremism.

In such cases, practitioners and researchers can establish additional safeguards to protect participant confidentiality, ask indirect questions about behavior (e.g., list experiments), or adjust the focus of interventions if it is not possible to measure change in the initial behavior of interest. It is also important to work with local partners (as lead implementers when possible) to help ensure that interventions are culturally sensitive and would not inadvertently indicate partisanship or impose other undue risks.

Challenges in Promoting Behavioral Science with Partners

DG activities often include partnerships with host governments and civil society. One of the challenges in promoting the value of behavioral science to these partners is in translating behavioral science language into understandable terminology for non-experts. In my work with government and civil society partners across a diverse set of countries, native languages, and roles, I have found that terms like “behavioral intervention” can be interpreted in a patronizing manner to mean correcting behavior as a teacher might discipline students.

It is best practice not to assume that a term like “behavior” is readily understood in the way that behavioral scientists use it, to adjust language to fit the local culture, and to provide interpreters and translators a glossary of behavioral science terms in advance so they can prepare for the way terms are meant to be used.

Another counterintuitive challenge is that partners may become too enthusiastic and want to set up their own behavioral science units or activities before building up the expertise needed to design effective, ethical interventions. This can risk harm to participants or waste resources. Rather than discouraging partners from engaging in behavioral science, it would be helpful for

practitioners, international donors, or experienced government behavioral science units to develop guidance on how to best form and develop behavioral science practices within government and civil society and to facilitate relationships among governments, civil society, and local behavioral science researchers and practitioners.

Opportunities

DG programs are more important than ever to counter the trend of resurgent authoritarianism. As Administrator Power stated in her remarks at the United Nations, “it is not enough to win elections based on promises to fight corruption and strengthen democracy, we have to deliver real results for our people. Democracy must deliver.” Behavioral science offers a path to improve the efficacy of existing programs and innovate rigorous new approaches.

However, this also requires willingness and commitment from researchers, practitioners, and donors to transparently share their successes and failures to build on knowledge and avoid duplicating (unsuccessful) efforts. Exciting opportunities in behavioral science fall on a spectrum of external validity from research that generates or is based in generalizable knowledge to research that generates or is based in highly specific knowledge about particular people and contexts. On one end, where the goal is to produce generalizable knowledge, a promising approach to build evidence is “megastudies,” in which researchers can test one intervention across contexts or multiple interventions simultaneously to better understand what works, where, and why (Duckworth & Milkman, 2022).⁹³

Practitioners can use learning from megastudies to develop

⁹³ The Evidence in Governance and Politics (EGAP) network coordinates tests of common treatments and outcomes in the DG space through their “Metaketa” initiative, though not necessarily using behavioral science.

common best practices for behavioral science to promote democracy and good governance, including information about how context characteristics affect intervention efficacy. At the other end of the spectrum, there is a substantial opportunity to amplify and empower the work of local behavioral scientists to design bespoke interventions that fit the culture, history, and environment. This means both building relationships with existing local experts in behavioral science and building local expertise in behavioral science—working with local partners as true collaborators, not just as research informants.

Conclusion & Key Points

Democratic governance depends heavily on having the right structural framework, laws and policies for the context. It also depends on humans to act in ways that support democracy and governance informed by public interests. Behavioral science is a tool that can support democratic governance by using research and evidence to identify effective target behaviors, diagnose psychological, social, and structural factors that impede behavior change and design and evaluate appropriate interventions.

While DG challenges are often complex and multifaceted, behavioral science can move the needle toward liberal democracy through target impact on key behaviors.

In the next chapter we learn how behavioral science strategies can help overcome problems related to the discrimination and exclusion of minority groups to achieve more inclusive, fair and prosperous societies.

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21.

Preventing the Exclusion of Minority Groups

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The exclusion of minority groups comes at the price of economic, social, and humanitarian prosperity. Because of their ethnicity, class, religion, sexual orientation, or health condition, minorities may be marginalized by barriers preventing access to opportunities, rights and resources (Akerlof and Kranton, 2000; Tajfel and Billic, 1974). The design of policies breaking the vicious circle of discrimination, exclusion, and marginalization of minorities has become a hot topic in academia and in policymaking.

As an example, the United Nations identifies as Goal #10 in the *Sustainable Development Goals 2030* a commitment to reduce inequality by designing, implementing and evaluating strategies to ease social inequalities *across* and *within* countries (UNDP, 2023). Despite the strong efforts in reshaping legislative and political decisions, traditional approaches have not provided optimal, cost-effective and sustainable solutions as they struggle

to integrate underlying conscious and unconscious biases from local communities in the design of effective policies.⁹⁴

Fortunately, recent work in behavioral science has accumulated solid evidence that we can reach this goal and promote the inclusion of minority groups. By conceptualizing a wider and richer framework on human decision making, we are in a much better position to understand the origins and causes of discrimination, and to design policies to tackle discrimination *across* and *within* communities.

For example, behavioral science extends standard approaches to further examine the role of cognitive mechanisms such as: prejudice, stereotypes, and outgroup dynamics.

Set against this background, in this chapter, we present recent theoretical and empirical insights in behavioral science that offer new solutions to improve the living standards of both minority groups and local communities around the world. Our starting question is straightforward: how can we provide innovative, cost-effective and sustainable ways to promote the inclusion of marginalized groups in 2023? The policy relevance of this research question comes from multiple sides and angles.

First, its magnitude. Numerous studies have found substantial and recurrent evidence of unequal treatment against minority groups as a direct consequence of their identities. Academic and policy literature has carefully quantified discriminatory outcomes towards:

- ▶ First-generation migrants and their offspring
- ▶ Women, single mothers and their children
- ▶ Young and low-skilled workers
- ▶ Rural communities

⁹⁴ Traditional approaches assume that people lack the information to change their behavior, or assume that better regulation can automatically induce individuals to comply with the right codes of conduct.

- Ethnic and religious groups
- People living with disabilities, chronic or mental diseases (*Artavia-Mora, 2020; Layland et al., 2020; Neumark, 2018; Becker, 2010*).

Because of their mistreatment, these groups consistently report lower income, and face barriers to accessing a higher standard of wellbeing and a better quality of life. Besides the intrinsic barriers they face, their insufficient knowledge of the institutional laws, language, and local norms also prevents them from escaping discriminatory traps, protecting themselves from biased behavior or asking for help. This occurs in both democratic and non-democratic countries.

Second, the dynamics of global inequality. The surge in income inequality in recent decades offers evidence to quantify the challenges faced by historically disadvantaged groups and the importance of better protecting their livelihood. According to the United National Development Program 2022, the richest decile has about 40% of global income. The poorest decile holds less than 7%. Population growth inequality has also increased by 11% in developing countries over the past decades.

This acceleration in income disparities highlights the need to design interventions that empower groups with fewer opportunities to access basic resources, and in this effort, to promote the inclusion of everyone, regardless of their profile.

On top of these factors, the past decades conceal a host of events and trends that have intensified the prejudice and exclusion towards minority groups, creating growing social and economic tensions between and within societies. These diverse events include the polarizing rhetoric in political campaigns, the Russian invasion of Ukraine in 2022, the global Covid-19 pandemic, recurrent economic crises across countries, a stronger flux of migrants and refugees across regions (*Vidal and Laczko, 2022*).

The impact of Climate Change is also intensifying world-wide, raising the competition to access resources in local and global markets that consequently create additional pressures on the opportunities of already vulnerable groups. The combination of all these events is adding negative tensions that will hit disadvantaged groups harder as we progress in this century.

In this chapter, we present recent work on how behavioral science may promote the inclusion of minority groups in our societies. The next section contextualizes the theoretical literature around in-group and out-group dynamics, and the third section reviews empirical case studies putting forward tools to mitigate discrimination and promote the inclusion of minority groups. The last section concludes and highlights avenues for future research.

Theoretical Insights

A growing body of literature in social sciences focuses on the analysis of cooperation between individuals from different social groups. When individuals prefer to associate and/or cooperate with individuals of their own group (ingroup), rather than with members of other (out)groups, ingroup favoritism in fact implies outgroup exclusion. Ingroup favoritism may be linked to the formation of cultural groups (Efferson et al., 2008), the prevalence of ingroup conformism and outgroup differentiation (Fatas et al., 2018), and is analyzed by social identity theory (Akerlof and Kranton, 2000; Tajfel and Turner, 1979; Tajfel, 1974).

Public-goods dilemmas exemplify this situation. Creation of a public good depends on individual contributions, but once it exists, the public good is available to all members of a collective regardless of their level of contribution. Thus, each individual maximizes self-interest by benefiting from the resource while contributing nothing. But if everyone in the collective fails

to contribute, the public good is lost, and benefits no one. Collectively, everyone is better off if all contribute, even though cooperation involves self-sacrifice at the individual level.

Public-goods dilemmas have been studied extensively by behavioral economists, political scientists, and social psychologists, and laboratory and field research is instructive about the conditions under which individuals will behave cooperatively. Levels of cooperation are significantly higher when shared in-group identity is made salient or group members strongly identify with the collective than when no shared identity is available or group identification is weak (e.g., Brewer & Kramer, 1986; De Cremer & Van Vugt, 1999; Wit & Kerr, 2002; Buchan et al., 2011).

Although researchers have proposed several mechanisms to explain why social identification with the relevant collective enhances self-sacrificial cooperation on behalf of that group (Brewer, 2008), we focus here on two mechanisms that have received the most attention: expectancies and values.

One theory is based on the idea that individuals expect that other people will be cooperative within the in-group (Yamagishi & Kiyonari, 2000; Brewer, 1986) and that people cooperate with each other because they are members of the same group (Foddy, Platow, & Yamagishi, 2009). Such norms of generalized reciprocity are strong in intragroup interactions and weaker or absent across group boundaries (Buchan et al., 2009; Tanis & Postmes, 2005; Yamagishi, Jin, & Kiyonari, 1999).

Expecting that other people will behave cooperatively reduces the fear that one's own cooperation will only benefit others, making cooperation compatible with individual incentives. Cooperation in these settings relies heavily on how members of one group expect others to contribute to the collective good (Croson et al., 2015, 2005). Thus, it is tempting to believe that if shared group membership increases the expectations that other

in-group members will cooperate, that would account for the effect of ingroup identity on intragroup cooperative behavior.

However, expectations that other people intend to cooperate does not generate cooperative behavior, especially in large, dispersed groups (e.g., De Cremer, Dewitte, & Snyder, 2001). Although trust reduces fear, it does not eliminate the self-interested benefit of noncooperation (i.e., greed). If everyone else is expected to cooperate, noncooperation takes advantage of other people's contributions to group welfare and maximizes personal outcomes. In large collectives, in which monitoring and reproach for non-reciprocity are minimal, this temptation to take advantage of other people's cooperation is large.

Interestingly, between-group competition for resources may play a significant role in promoting ingroup prosociality (Eckel et al., 2016). As Darwin (1874) famously noted, "groups with numerous members who were always willing to support and sacrifice for one another would achieve victory over most other tribes, leading to natural selection". Intergroup competition may amplify the value of altruistic behavior within a group and reduce the availability of resources to share with the outgroup, facilitating outgroup exclusion.

Recent work suggests that this reinforcement of ingroup favoritism may be stronger when group identities are linked to cultural traits like language (Espinosa et al., 2019), intergroup interaction associated to an unequal distribution of political rights (Eckel et al., 2022), common even between vulnerable groups with salient identities (Restrepo-Plaza & Fatas, 2022), and reduced by ingroup and outgroup exposure to violence (Restrepo-Plaza & Fatas, 2023).

Case Studies and Empirical Insights

Recent contributions in social and behavioral sciences offer a variety of insights to improve the inclusion of minority groups in our societies. In this section, we present a series of empirical case studies around two central areas of paramount importance:

1. Improving access to labor and housing markets.
2. Promoting cultural and religious integration.

The following cases present instruments to mitigate historical inequalities and enhance the access to essential services and public goods.

Improving Access to Labor and Housing Markets

The discrimination in labor and housing markets has historically affected the living standards of minority groups by impeding their access to essential services such as education, security, healthcare, and financial services (Brevoort, 2011; Urban, 2009; Aalbers, 2007; Buck, 2001). The literature typically identifies a large and persistent discrimination towards migrants and their descendants (produced by their foreign, non-native names, see Oreopoulos, 2011), while there is also substantial evidence of exclusion towards women, young and/or unskilled workers, and disabled people only as a consequence of their identity group association (Bertrand & Mullainathan, 2004).

For instance, the evidence confirms that applicants with Moroccan, Turkish, Black or Hispanic backgrounds require significantly more effort in the application processes to access their labor and housing targets (in comparison to their native counterparts) in countries such as Germany, The Netherlands, Sweden, France, Spain, Italy, Greece, the United Kingdom and the United States of America (for a review and indicative literature see Artavia-Mora, L., 2020).

Two studies in South Africa give hints about how to alleviate this situation. In a study of 1,100 young job seekers, Abel et al. (2019) found that subjects who create and follow a detailed job search plan submit 15% more applications while not spending more time on the task, helping to close this intention-behavior gap.⁹⁵ Participants in the plan-building treatment group were found to use more formal search channels and to diversify their search strategy than the control group. Treated subjects showed higher efficiency and effectiveness by getting 30% more job offers and reaching 26% more employment.⁹⁶ The research found that sending weekly reminders or having peer support groups did not stimulate the creation of such plans among these job seekers.

In a related study, Abel, Burger and Piriano (2020) find that including a reference letter can increase the chances of low-skilled youth and women to receive an invitation for a job interview. In a sample of 2,050 candidates, the group that attached the reference letter showed an increase of 60% in call-backs, higher employment rates after three months, and demonstrated more gender equality (boosting employment chances of women).

The reference letter may help mitigate asymmetries in candidates' information which improves screening processes and the consequent employability of low-skilled and minority groups. This latter insight is valuable as frictions in the labor market of developing countries may produce larger mistreatments, misallocations and inequalities towards vulnerable groups.

Furthermore, Fang, Guess and Humphreys (2019) found

⁹⁵ On top of the standard 90-minute career-counseling workshop conducted by the Department of Labour.

⁹⁶ Hence, a job search planning intervention can help overcome a typical behavioral barrier of job seekers that are already in a disadvantaged position, as they must submit more job applications compared to their more experienced counterparts.

that punitive messages can mitigate housing discrimination towards ethnic minorities by investigating strategies to improve the access of Blacks and Hispanics (compared to White) in the United States.

In their research design, landlords or brokers were assigned to receive one of three types of messages after a candidate visited nearly 700 apartments in New York: a *punitive* condition stressing the potential monetary and legal sanctions of engaging in discriminatory practices, a *monitoring* condition reminding the party about the city's mission to prevent discrimination (without stating legal penalties), and a *control* group with no message.

The punitive condition reduced discrimination towards Hispanics but it was fully ineffective to help Blacks.⁹⁷ The monitoring treatment produced similar qualitative results.

Another subsequent study indicates that using an online housing platform created by and designed for minority groups may help to promote their access to services. Artavia-Mora (2020) finds that a specially-tailored website for immigrants can prevent discrimination towards Turkish and Moroccan residents in Amsterdam. While conventional platforms can perpetuate discriminatory outcomes, a secondary website used by non-native groups generates a safer space to promote collaboration and integration among minority groups in the society. Conversely, attaching a reference letter in the application does not have an additional positive effect. The fact that this platform is mainly composed of non-native minority groups may explain part of its success (Abel, Burger and Piriano, 2020).

⁹⁷ Hispanics were 6.8 percentage points (p.p.) more likely to receive callbacks and 2.5 p.p. more likely to have an offer.

Promoting Cultural and Religious Integration

In a world where people of different backgrounds, beliefs and traditions coexist, promoting understanding, respect, and acceptance is of utmost importance, especially if cultural and religious differences are large and apparent. Particular difficulties arise in contexts where there are historical confrontations between groups because of political, class or religious opposition. Achieving ways to promote positive relations is essential to foster cultural norms of inclusivity and tolerance.

In a post-war scenario in the Middle East, Mousa (2020) finds that intergroup contact when playing sports builds social cohesion. Her research employs a soccer intervention randomly assigning Iraqi Christians displaced by the Islamic State of Iraq and Syria (ISIS) to an all-Christian team or a mixed team (with Muslims players). The results show that players that participated in mixed teams improved the integration of Muslims as they were more likely to:

- ▶ Vote for a Muslim (outside of their own team) to win a tournament prize.
- ▶ Register in another mixed team the following season(s).
- ▶ Keep training with Muslims six months after study completion.
- ▶ Believe that coexistence is possible.

Despite these positive outcomes, there wasn't much improvement in behaviors in other social settings such as participating in mixed social events, visiting Mosul, or donating to an agency that assists both groups. Thus, this evidence indicates that such intervention is effective in changing attitudes in a particular context, but has a limited effect on behavior change outside of it.

Kirgios et al. (2022) study how marginalized groups rarely get the support and advice they seek, severely obstructing their professional development. By collecting over 5,000 observations

in a series of audit and online experiments among city councilors and students, the research finds that stating their own (vulnerable) demographic identity helps women and racial/ethnic minorities to receive the support they ask for when submitting help-seeking emails inquiries (for instance: “As a Black woman...”). City councilors and students in treated groups gave more responses to emails sent from these vulnerable groups by 7.42 percentage points (p.p.) and 2.73 p.p., respectively.

The authors recommend that mentioning one’s vulnerable identity improves the outcomes of marginalized individuals when seeking support, as this information may activate helpers’ motivations to avoid prejudiced choices and increase their willingness to answer requests.

In the discriminatory Indian caste system, Lowe (2021) finds that the type of contact between castes shapes the benefits of their integration. His study examined the impacts of men’s intergroup contact (from different castes) by assigning them to mixed cricket leagues (or a control group). Then, players were randomly assigned to scenarios in which collaborative (within their cricket teams) or adversarial (between cricket teams) interaction was possible, studying variations of games with teams composed of members of either the same or different castes.⁹⁸

Collaborative contact was found to significantly improve positive cross-caste behavior and to significantly reduce favoritism towards each player’s own caste; while adversarial games actually harmed cross-caste relations.⁹⁹

⁹⁸ Collaborative games (contact) refer to interactions with individuals from different castes in a cooperative manner, while adversarial contact indicates interactions with individuals from different groups in a competitive manner.

⁹⁹ Lowe (2021) indicates that collaborative contact increased an omnibus index of pro cross-caste behavior by 0.23 standard deviations (p-value < 0.01), but adversarial contact reduced it 0.38 standard deviations (p-value = 0.07).

Overall, participating in the tournament reduced intergroup differences, distilling that collaborative contact reduces barriers to integration of different communities, while adversarial interactions may instead enhance exclusion.

Conclusions & Key Points

This chapter has discussed the challenges that marginalized groups face in achieving optimal access to essential socioeconomic activities. By examining theoretical factors that contribute to their exclusion and empirical case studies, we have also identified a range of applied behavioral strategies that have shown to improve their outcomes in two areas: improving their access to labor and housing markets, promoting their cultural and religious integration.

The strategies we presented include:

- ▶ signing a reference letter to reduce information asymmetries about the ability of minority applicants;
- ▶ establishing a job-search plan to increase employability and system efficiency;
- ▶ creating platforms (by and for minority groups) or communicating potential penalties to improve housing outcome,
- ▶ including one's marginalized identity to obtain responses in help-seeking message,
- ▶ creating collaborative contact to promote intergroup relations.

Importantly, these strategies belong to a wider set of studies delivering more insights to protect minority groups from other vulnerabilities and inequalities. Related research has also explored insights to improve their: physical and mental health, use of financial and digital services, participation in political and

leadership roles, and even in teaching evaluations (see a review in Artavia-Mora, 2020; Boring & Philippe, 2021; Beaman et al., 2009).

As an example, using a system of message reminders (SMS) may improve life-long compliance to medical treatment, adherence to therapy, and prevent stigma for patients living with chronic and life-threatening diseases such as HIV and/or tuberculosis (WHO, 2011).¹⁰⁰

Despite the significant progress in finding new cost-efficient solutions to improve the wellbeing of minority groups it is crucial to continue the research across contexts and subgroups. Doing so becomes important to prevent unintended negative consequences such as boomerang effects.

For instance, Dur et al. (2022) shows that, while sending messages that had the intention to better integrate ethnic minorities in Sweden, the intervention actually increased the rates of discrimination towards these groups in some circumstances. Restrepo-Plaza and Fatas (2022) show that an intervention targeting the inclusion of ex-combatants in Colombia may trigger the discrimination of victims of conflict by members of other vulnerable groups. Likewise, Mousa (2020) mentions that interventions can help build intergroup relations within the context of the intervention, but they have limitations to influence settings outside of it.

To contribute to this goal, the authors of this chapter are working in a series of consulting and academic projects to find new insights. In partnership with leading organizations like the Centre for Social and Behaviour Change in India, we are currently investigating ways to improve the access of migrants to

¹⁰⁰ A recent systematic review finds that 60% of SMS programs have positive results to improve retention and adherence in low- and middle- income countries (Demena et al., 2020).

financial services and, to promote consumer redressal behavior of vulnerable populations when using digital technologies.

By examining underlying social and behavioral factors (e.g. biases in beliefs, financial knowledge, normative and empirical expectations, and general perceptions) we aim to put forward insights that may help to shape the decisions of bankers and financial advisors, policymakers, and consumers towards more positive and sustainable behaviors.

In the future, we expect to lead extensions in other areas such as social integration, health, civic collaboration, and female empowerment. In short, by addressing the behavioral factors driving the exclusion of minority groups, we can work towards creating a more equitable, cohesive, and just society, where all individuals and communities have an opportunity to thrive.

In the next chapter we learn how behavioral science can be used to impact the challenges facing domestic workers from practitioners and researchers working at leading policy and academic organizations.

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22.

Domestic Work and Complex Challenges

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Compliance with labor and social security legislation is a key human rights issue. In too many places, having a job does not necessarily provide an escape from poverty. Achieving this goal involves creating “decent” work, meaning work that:

- ▶ Is productive
- ▶ Is fairly remunerated
- ▶ Occurs in a safe and secure environment
- ▶ Provides freedom from discrimination with access to social protection

¹⁰¹ This chapter was developed from our interventions in the “Breaking ground: generating behavioural insights on decent work” panel at the 2022 United Nations Behavioural Science Week. We are grateful to the other panel members for their valuable insights and comments both during the event and through their comments to earlier versions of this text (including Mario Herberz and Elva Lopez Mourelo).

- ▶ Offers opportunities for development
- ▶ Recognizes freedom of association¹⁰²

Adequate and effective labor and social security laws can help to realize the human right to be free from slavery (Universal Declaration of Human Rights, 1948 – art. 4); have access to social security (art 22); work under just and favorable conditions and have protection from unemployment; equal pay for equal work without discrimination and just and favorable remuneration (art 23); rest and leisure, including reasonable limitation of working hours and periodic holidays with pay (Art 24).

These rights form the basis of SDG 8 on Decent Work and Economic Growth. In the world of work, behavioral science has frequently been used to improve management and workplace performance, prevent discrimination and violence and harassment, and promote employment. Beyond that, however, it has rarely been explored as a tool for promoting formalization, including by improving compliance with labor and social

¹⁰² The concept of “decent work” sums up the aspirations of people in their working lives, including productive employment for a fair income, security in the workplace, social protection, prospects for personal development and integration, freedom to express concerns, organize, and participate in the decisions that affect their lives, and equality of opportunity and treatment. These principles were institutionalized as the ILO’s “Decent Work Agenda” when the International Labour Organization unanimously adopted the ILO Declaration on Social Justice for a Fair Globalization on 10 June 2008. The Decent Work Agenda calls on Member States to promote policies that advance “opportunities for women and men to obtain decent and productive work in conditions of freedom, equity, security and human dignity” (International Labour Office, 2007, p. viii). It is composed of four strategic objectives or “pillars,” namely: 1) full and productive employment, 2) rights at work, 3) social protection and 4) the promotion of social dialogue. These became integral elements of the new 2030 Agenda for Sustainable Development, codified in SDG 8 on the promotion of sustained, inclusive and sustainable economic growth, full and productive employment and decent work.

security legislation (with some notable exceptions such as Horodnic & Williams, 2018, 2019).

The systematic, evidence-based understanding and intervention techniques that behavioral science can produce have the potential to accelerate progress towards realizing these rights and considerably increasing the real impact of other regulatory and policy initiatives. The appeal of these techniques is that, when used smartly in articulation to other regulation and policy interventions, behavioral initiatives have the potential to boost real-world impact in a cost effective, measurable, and sustainable manner. They can be prepared in relatively short timelines in either the absence of regulation, or in complement of regulation or other strategies.

In this chapter, we discuss how the International Labour Organization has used behavioral perspectives and tools to inform its work to make decent work a reality for a group of workers particularly exposed to poor working conditions.

After describing the policy challenge, we discuss some of the major obstacles we've faced when applying behavioral approaches in this area and ways in which we have tackled them. We argue that the effective application of behavioral science to complex developmental objectives such as achieving decent work for domestic workers often requires an approach that goes beyond light-touch interventions such as nudging, including:

- ▶ Multilayered diagnostics
- ▶ Capacity-building of local actors
- ▶ Building multidisciplinary teams
- ▶ Engaging with decision-makers and other stakeholders throughout the process

While pertaining specifically to our experience promoting compliance with labor and social security legislation, we believe many of the challenges and lessons are like those faced by

development practitioners in many other policy areas (especially those in which regulatory compliance is key).

Making Decent Work a Reality for Domestic Workers Worldwide

For several decades, organizations have been attempting to make decent work a reality for domestic workers. Domestic workers are those who perform work in or for a household or households (C189 - Domestic Workers Convention, 2011 (No. 189), 2011), typically providing both direct and indirect care services, including cleaners, nannies, cooks, gardeners, guards, chauffeurs, and many others.

Globally, around 75.6 million women and men over the age of 15 are employed as domestic workers, representing 4.5% of employees. A large majority (76.2 per cent) are women, and around 8.8% of female employees are employed as domestic workers (International Labour Organization, 2021).

Despite the key services they provide to households, their working conditions are among the poorest, working either extremely long hours or very short and unpredictable hours, for very low wages (just 56% of the average monthly wage of other employees). Domestic work is also a sector with a high rate of informality (81 percent), meaning that employers of domestic workers are not contributing to social security, either because domestic workers are not covered by labor or social security laws, or because of a lack of compliance with these (International Labour Organization, 2021).

In recognition of these persistent deficits in human and labor rights, the International Labour Organization adopted in 2011 its Convention No. 189 on decent work for domestic workers (International Labour Organization, 2021). The Convention seeks to ensure that all domestic workers are free from human

rights abuses, and that they enjoy rights equal to those of other workers regarding key labor rights like working time, wages, and social security.

These rights should be enjoyed not just in law, but also in practice. Thus, we aim to make decent work a *reality* for domestic workers. Achieving this goal would contribute to several of the Sustainable Development Goals, but most notably target 8.3, 8.7, 8.8, 8.10 and several targets of SDG 5 on gender equality.

To implement these rights, governments, often with ILO help and guided by international labor standards, have adopted, reformed or extended laws and policies, strengthened the capacity of labor market institutions to monitor and enforce compliance, and adopted fiscal incentives. Trade unions have played an essential role in promoting and enforcing these rights as well.

While these approaches are necessary, their capacity to influence the behavior of individuals at the household level has been limited. Even where domestic workers are covered by labor and social security laws, compliance among household employers remains low, indicating a long-standing, entrenched behavioral norm and perception that domestic work is not a legitimate form of employment.

While few studies have been conducted to identify specifically what causes these behavioral norms among employers of domestic workers despite the adoption of the aforementioned measures, scholars, policymakers and activists alike have observed deep-rooted discriminatory notions that domestic work is but a natural extension of women's unpaid work in the home. Domestic workers also face other forms of discrimination as they frequently are from groups that are marginalized based on their race, status in migration, belonging to indigenous groups, or because they suffer from multigenerational poverty.

Another factor often referred to as a driver of informality in domestic work is the low awareness among households that

they are employers with legal responsibilities. The low level of self-regulation among household employers puts pressure on the limited capacity of public institutions to promote, monitor and enforce compliance with legislation. Indeed, that domestic work takes place in private households makes it difficult for labor inspectors to identify and pursue possible violations in hundreds of thousands of households or more.

To reinforce the policy measures taken by governments, the ILO turned to behavioral science as an additional tool to increase legal compliance and improve working conditions in the domestic work sector. Here, behavioral science is a potentially effective tool to reach households that employ domestic workers. Once those households are reached, behavioral scientists can:

- ▶ Convey information on rights and obligations more effectively.
- ▶ Design behaviorally informed incentives.
- ▶ Encourage self-regulation to complement the more traditional tools of regulation, enforcement and incentives.

Because in this domain we often deal with contexts in which compliance is relatively low even when regulatory and financial incentives are already in place, behavioral tools seem to be an especially promising way to increase compliance rates. To date, the ILO has developed interventions to diagnose and address the behavioral drivers of non-registration to social security (Argentina, Ohaco & Vello, 2019) (Guatemala, Zambia), of a household employer's willingness to help a domestic worker (Hobden, 2017) and their selection of employment agencies (Hong Kong - (International Labour Organization, 2022)), and of violence and harassment in domestic work (Argentina).¹⁰³

¹⁰³ Unless otherwise mentioned, the reports of these studies are yet unpublished, but are available with the authors of this chapter.

Complex Challenges, Complex Interventions: Experiences in Domestic Work

The application of behavioral science to an international development goal such as pursuing decent work for domestic workers presents several challenges. There is very little application of behavioral science to this policy challenge, providing almost no base of knowledge on which to build. The target behavior is underpinned by complex and widespread norms and perceptions. The behavior takes place in various specific contexts that can differ significantly from one another, making it impossible to apply a one-size fits all approach. And finally, interventions typically take place in contexts with limited human resource and financial capacity.

Hence, the approaches developed by the ILO respond to those challenges while accounting for limited resources, local capacity and access to data, sociopolitical context, and regulatory developments. These issues are rarely discussed in the specialized literature.

Behaviors linked to the employment of domestic workers are complex, underpinned by multiple, intersecting forms of discrimination, deep power and gender relations, and long-standing undesirable norms and beliefs. These manifest in the design and implementation of laws, policies, and programs, in the willingness of the system actors to support significant change, in the willingness of household employers to comply with legal obligations, and even in the awareness of domestic workers themselves as employees with rights.

That issues are often also traversed with deep-rooted traditional roles, and with important financial implications for employers and workers, makes it even more complicated.

The complexity of these conditions make “nudging” or

light-touch behavioral intervention techniques that have become dominant in behavioral science research and practice unlikely to succeed even for apparently restricted decisions like social security registration. Rather, behavioral approaches that combine a broader understanding of human behavior and deeper intervention strategies are needed.

By this, we mean diagnostics and strategies that go beyond incentives and biases to understand and transform a broader set of physical, psychological, and social determinants of behavior.

Domestic workers many times have reported wanting the recognition that comes with formal employment, but also fear that registering for social security would crowd out other benefits they may receive through social welfare programs or prefer to use their limited salary for immediate expenses rather than for future security (present bias).

Instead, employers may hold a self-concept that they are good employers but remain unwilling to register their domestic workers to social security using thinly veiled excuses to justify an underlying perception that domestic workers are not real workers deserving of social protection or claiming that the domestic worker is not interested in such benefits.

Neither party is normally aware of their respective rights and obligations. Social security systems are also complex in that they incur actual costs and require often lengthy registration procedures that are ill-adapted to the capacities of households and domestic workers. More comprehensive intervention designs, drawn from more complex systematic research into local behavioral determinants, are required.

ILO's agenda is international, requiring work to take place in diverse geographic and economic contexts. This implies that work is executed across countries at various levels of economic development, urban and rural environments, mono or multicultural, etc. This reality escapes the inter-national and

inter-cultural generalization efforts common in nudge work (Komatsu et al., 2022; Mertens et al., 2022).

Often, particularly in low- and middle- income economies, governments have limited human, financial and institutional resources to implement and rigorously evaluate ordinary laws and policies, let alone behavioral interventions.

Rather than having unlimited funds and patience to test viable solutions, very often development practitioners must carefully balance limited timelines and financial restrictions to address complex challenges in complex local and political contexts. There is also a deficit of behavioral science expertise in most countries, challenging quick and smooth design and implementation of interventions.

A third category of challenges is a lack of prior research on behavioral science in the domain. While domestic work appears to be a prime area of application for behavioral science, to date, there is very little research on behavioral drivers and interventions to change behavior in the sector. The lack of prior research on which to build presented important risks, such as making false assumptions, not knowing which behavioral levers or “nudges” to test, and, of course, not knowing whether a behavioral intervention of any kind would work. The ILO has had to develop such a knowledge base from the ground up.

Finally, in most countries, there remains little technical capacity to carry-out behavioral science. Indeed, in many cases, public institutions have limited resources and staff to implement policies, and do not necessarily have the capacity to support the application of complex behavioral science interventions.

Based on this context, the ILO has developed approaches that consider these realities, many of which are common to international development. While still a work in progress, these methods can inform behavioral initiatives beyond domestic work issues, especially those that share the challenges described

above and those where citizen compliance with formal regulation is important.

The most important include:

Conducting comprehensive research on the application of behavioral science to similar domains.

To narrow down what behavioral insights might apply to domestic work, and to develop hypotheses, the ILO worked with behavioral science experts to identify the key characteristics of domestic work, and to conduct literature reviews on domains of application that shared those characteristics. For example, prior to designing social norms research and interventions in domestic work, we identified specific characteristics of the target behaviors that resembled behaviors and interventions in other domains (such as the fact that behaviors happen inside the household instead of public spaces, that they imply financial costs, that neighbors and family are important referents, and so on).

Similarly, work in Argentina to address violence and harassment began with a broad, global review of determinants and interventions that had worked in similar contexts and shared basic characteristics with those we wanted to address in Argentina. These reviews of literature and interventions provided frameworks within which to design research and interventions to change social norms in domestic work.

Supporting multi-layered and multi-method diagnostics that build a sufficiently broad and solid knowledge base around the problem and the context of intervention.

Understanding local beliefs, customs, and power relations through qualitative data is as important as measuring their prevalence through quantitative methods. In Hong Kong,

Zambia, and Guatemala for example, we used focus groups, open-ended interview questions and key informant interviews (complemented by a user experience map in Guatemala) to better understand the context and the problem before designing and applying quantitative methods (Davis et al., 2019).

In Hong Kong in particular, very little knowledge had been developed on the decision-making process by which employers of domestic workers selected employment agencies through which to recruit domestic workers. Our team developed an open-ended questionnaire to identify a rough scope of steps and factors involved in the decision from the user perspective. Next, we designed an experiment to test the effectiveness of various message frames on the employer's willingness to conduct due diligence on the employment agencies they were considering.

The qualitative research that preceded the experiment allowed us to develop treatments that had higher chances of having statistically significant effects, as they were firmly rooted in evidence derived from the qualitative research results. It was also strengthened because we created collaborations between international and local researchers to leverage both situated and comparative areas of expertise (International Labour Organization, 2022).

Another useful tool is to employ a systematic exploration of the physical, psychological and social determinants that regulate behavior in local contexts (Cislaghi & Heise, 2019; Lahlou, 2015, 2017). Unfortunately, most current intervention efforts focus on only one type of determinant, and thus miss important interdependent factors that might affect both the current behaviors and their transformation. This is essential to capture the interaction of factors at physical, social, psychological, and other levels.

For example, such a broad diagnosis of local determinants around registration to health coverage for domestic workers in

Guatemala allowed us to uncover a wide range of factors. This includes barriers in registration procedures, platforms and access to service points, financial barriers, low knowledge about the system, inexact beliefs and perceptions, laws and regulations that didn't make health coverage mandatory, social perceptions around how atypical and even undesirable registration was, barriers linked to power relations.

The following diagram provides an overview of the factors and levels identified:

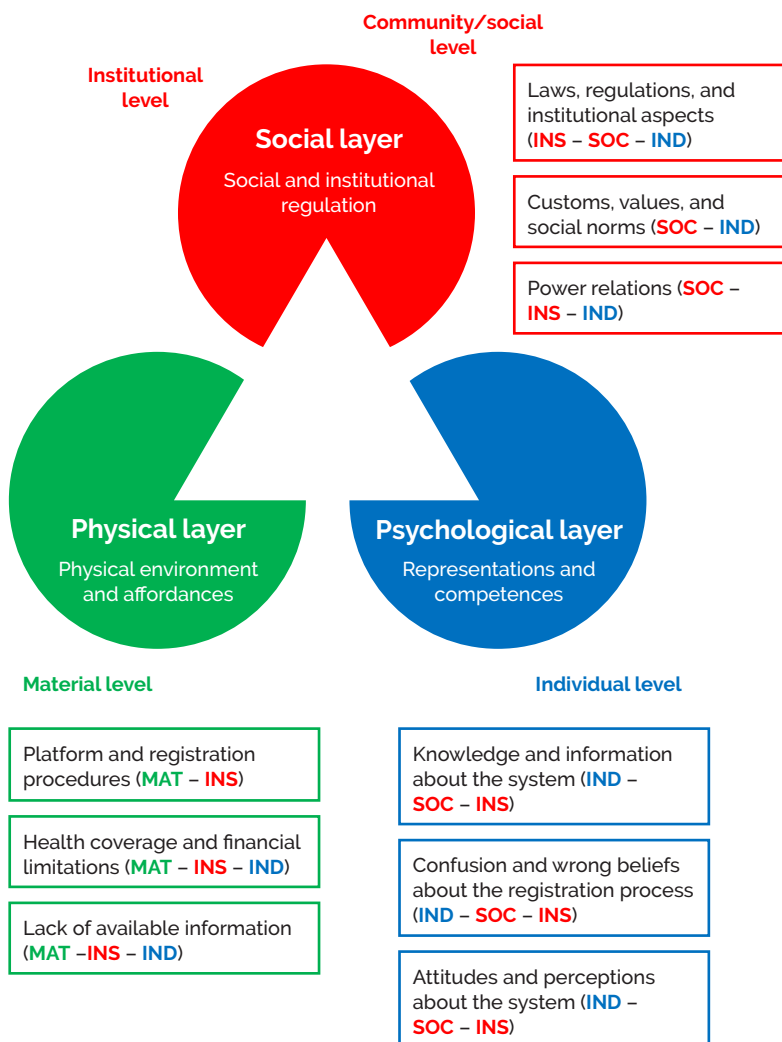
While academic experiments often favor one or two very restricted factors (such as knowledge only, or social perceptions only) to tease out their specific effects on behavior, this is not always the right approach in field applications. Policy interventions that target such restricted factors are often ineffective or don't live up to investment vs. impact appraisals. The "weight" of the unaddressed factors usually minimizes any change.

In the example above, what kind of short- or long-term impact can a behavioral intervention to increase knowledge or change social norms achieve when registration procedures, economic incentives, access to registration points and power relations are not even analyzed, and are in practice all pushing in the wrong direction? This point might seem obvious, of course, but it's unfortunately common in many policy interventions.

A broad and systematic approach like this has two additional advantages in policy interventions. First, it makes it easier to identify whether a behavioral intervention could address the policy challenge in question, or if alternative or complementary strategies would be necessary.

In Guatemala, a thorough investigation into the main behavioral determinants linked to social security registration revealed important access and knowledge barriers that would make behavioral interventions applied by themselves largely ineffective. This was confirmed in a large survey experiment, in

Diagram 1. Main determinants and influence levels – registration to health insurance for domestic workers in Guatemala



Source: "Norms and behaviors around adoption of health insurance among domestic workers in Guatemala" report for ILO - Paulius Yamin and Sania Ashraf

which simple information about the existence and benefits of health insurance was as effective to motivate both employers and domestic workers to access registration forms as an intervention leveraging descriptive and injunctive perceptions (social norms).

Because we had conducted a complete systematic diagnostic of the multi-layered determinants in the physical, social, and psychological levels we could provide local actors with a “menu” of different policy, regulatory, and behavioral interventions that could address the main barriers and opportunities that both employers and domestic workers had identified.

Second, it helps to tease out the particularities of the target challenge and the target behaviors, which might reveal links to successful experiences in other behavioral domains and potential intervention avenues. For the specific problem of promoting social security registration among employers of domestic workers, beyond local contexts we seem to face a behavior that happens mostly in private spaces (the household), has a strong economic and a strong traditional dimension, and where what we believe others are doing seems to matter a lot.

In this way, while the specific challenge of social security registration has not been extensively explored in the behavioral literature, exploring such characteristics can help to better understand the context of action and to draw factors of differentiation or similarity with other behavioral policy challenges and experiences.

Engaging with stakeholders informs the policy objective and intervention design, and ensures ownership and sustainability of the results.

In Argentina and Zambia, the policy objectives were first agreed upon between the ILO’s tripartite constituents, including representatives of government, trade unions, and employer’s organizations. These constituents and other stakeholders were

also asked to share their views as experts on and users of the policies and programs and processes of concern. This fed into the multilayered diagnostic stage. They were then consulted after the diagnostic had been conducted to discuss the results, and, based on these, to take part in ideation workshops to identify methods of intervention that were adapted to the realities of the country and sector.

Once the pilots had been tested through experiments, the results were again brought to the constituents to discuss how the findings might apply to policies beyond what might have been envisioned at the outset. Involving these stakeholders and local staff throughout the process can also significantly build their capacity to participate in or even initiate and implement behavioral interventions in the future.

Building behavioral science capacity at the country level.

If behavioral science is to be applied regularly towards a specific policy objective, it is important to have a pool of local experts in the field. In the absence of these, the ILO has built ad-hoc teams across country and international contexts to design and carry out diagnostics or interventions.

For example, in Argentina, Guatemala and Hong Kong, international behavioral science experts worked on the development of the research design and tools, while local researchers or research institutes were tasked with data collection. Analysis was shared between the two teams, bringing together the local and technical knowledge necessary to get a comprehensive picture. In doing so, the local research teams were exposed to behavioral science principles and methodologies, further developing their interest, skills and capacities.

The presence of behavioral science teams embedded in or attached to governmental departments are helpful collaborators

in promoting and supporting the application of behavioral science to public policy. For example, the ILO has begun collaborating with the Behavioral Science and Public Policy Unit of the government of Argentina, increasing the capacity of both institutions to design effective research and interventions.¹⁰⁴

Conclusion & Key Points

In sum, rather than limiting behavioral interventions to nudges or other light-touch applications of behavioral science, and rather than transferring interventions that have worked elsewhere (sometimes for different challenges). Our approach should integrate behavioral policy work with local conditions as well as the local regulatory and policy context.

Instead of assuming behavioral interventions must be applied in every case, it takes a broader behavioral perspective to understand local activity and identify a range of potential interventions or redesigns of local policies, processes or conditions that might or might not involve nudges (but that can hopefully impact behavior at scale). Conducting a multi-layered diagnostic that looks at the physical, psychological and social layers that regulate behavior can help identify if a behavioral intervention could address the policy challenge in question, or if alternative or complementary strategies would be necessary.

In Guatemala, a multi-layered diagnostic revealed that the social security system would likely benefit from significant changes to its design, and that merely nudging employers to

¹⁰⁴ The Behavioral Science and Public Policy Unit was inaugurated in 2021 by the President of Argentina. Its mission is to promote projects based on human behavior studies to improve people's quality of life. To do so, the Unit coordinates with various areas of the national government, provincial and municipal governments, organizations, and experts in the field. <https://www.argentina.gob.ar/consejo/unidad-ciencias-del-comportamiento>

register their domestic workers would be insufficient as a stand-alone intervention. However, because of its comprehensiveness, the diagnostic results served as a menu of potential measures for policymakers to test and deploy by identifying the areas of the social security system that could be revised (for example, identifying new and revised channels of service for different actors, or identifying barriers in procedures).

If behavioral science is a tool to face the important societal challenges of our time, it cannot become a list of restricted biases and nudges that are applied uncritically in widely divergent social, cultural, and political contexts.

Rather, it must capture and adapt itself to the particularities, the power relations, and the legal and policy landscapes of the local contexts in which it aims to have an impact. It must act as a new lens to understand, interpret, and innovate in those challenges - not as another small awareness campaign or design tweak, but as a more impact-oriented systematic approach to understand and design for enduring transformations.

In the next section we present the final Part V where we discuss the future of behavioral science in development and conclude by reviewing the main ideas shared in this book.

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Part V:

LOOKING TO THE FUTURE

23.

What the Future May Hold

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Our purpose in creating this book was to meet a need we repeatedly heard from behavioral science practitioners: to learn from other experts how best to incorporate behavioral science into international development work. This gave us an opportunity to gather experts and groups with the most experience and impact around the world to share their lessons learned, challenges, and best practices. It also presented an interesting set of challenges, as behavioral science combines a broad toolkit and international development spans a myriad of applications and interested parties around the world.

Our attempt to navigate this was to work with leading practitioners with decades of broad experience and crystalize that knowledge into chapters with relatively niche expertise. This allowed us to both go deep in targeted areas and wide across tools and topics. As with the book *Building Behavioral Science in an Organization*, we expect that while some readers may find the entire book useful for them, others may find more interest in some select chapters that are most relevant for their work.

In constructing the book, we presented it in a way that felt intuitive to us. We started with foundational tools and methodologies, presented an overview of the key types of organizations in the field and their best practices, and concluded with examples of the impact behavioral science is having in important areas of development.

One of the key themes we see across the chapters on tools and organizations is the increasingly vital role of behavioral science in international development – and how far the field has come. The once nascent discipline, often seen as simply an adjunct to economics, has now blossomed into a thriving field in its own right with significant implications for the future of global development.

One clear sign of the maturation of behavioral science in international development is the rise of formal roles for behavioral scientists within governmental and non-governmental organizations alike.

A key example is the United Nations (UN), which has taken significant steps to integrate behavioral science into its mandate delivery, program implementation, and administration. In 2021, the UN Secretary-General issued a Guidance Note on behavioral science, which aimed to introduce the practice of behavioral science for enhancing UN operations, promoting evidence-based approaches, context-specific solutions, and ethical considerations.

To support the integration of behavioral science into its work, the UN launched a behavioral science fellowship program, providing UN entities with direct access to behavioral science expertise. The UN Innovation Network Behavioral Science Group serves as a platform for sharing knowledge, resources, and best practices, facilitating collaboration and learning among UN entities and behavioral science practitioners. To help operationalize behavioral science, the UN also developed a comprehensive

Practitioner's Guide for UN Entities, providing practical advice, tools, and resources for applying behavioral insights across various contexts.

We expect this momentum will continue to grow and that the field will see a proliferation of formal and informal organizations promoting behavioral science. In 2023, the World Health Organization adopted a global resolution calling for the establishment of behavioral science teams, stating that it “highlights the importance of behavioural sciences in understanding the factors that affect healthy behaviours and in tackling key public health challenges.”¹⁰⁵

This shift and the prominence of these initiatives show that the insights and techniques offered by behavioral science are being valued and integrated across the development sector. As the landscape of organizations engaging with behavioral science expands and diversifies, the exchange of ideas and the ability to learn from one another's successes and challenges are increasingly important.

By studying these organizations and their work, we can glean valuable lessons and insights that help to guide future efforts in the field. This book is an attempt to do that, but we are not alone in developing guidance and innovative resources for behavioral scientists in international development.

Another interesting resource is currently in development in India. A team from Dalberg and the Ashoka Center For Social and Behaviour Change, in collaboration with the Bill & Melinda Gates Foundation, are developing a technology platform that would help organizations implement high quality behavioral interventions.

The Atlas of Behavior Change in Development (ABCD) is an

¹⁰⁵ More information available at: <https://www.who.int/europe/news/item/20-06-2023-new-global-resolution-calls-for-establishment-of-behavioural-science-units-or-teams>

innovative platform designed to assist NGOs, governments, and funders in learning, designing, and scaling behavioral programs. The platform is organized around behaviors relevant to social impact and provides a host of resources across different stages of a program's journey.

The Learn section of the platform allows users to understand behaviors, barriers, and solutions to overcome them. This includes a Behavior 101 page which serves as a crash course on changing a behavior, a Barrier & Solution Identification repository, a Knowledge Library of case studies and research, and a Behavior Journey highlighting the process of behavior change.

The Design section offers resources for creating high-quality behavioral interventions, such as a Collateral Library of campaign materials, an RFP Library for creating behaviorally smart and successful RFPs, and a Workspace for users to curate resources and collaborate on projects. In addition, the Connect section provides directories of funders, experts, and partners to foster collaboration and partnerships. Looking ahead, ABCD will introduce new features, including a Behavior Strategy Tool, a chatbot, and a Whatsapp-based messaging system.

These formal tools, repositories and resources are a sign of the maturation of the field beyond anecdotal and diffuse lessons to a more systematic approach to integration. The tools and frameworks used to apply behavioral science in international development are becoming more structured and varied, as practitioners move beyond mere nudging to employ a broader toolkit of behavioral sciences. This shift is a testament to the recognition that complex global challenges often require multifaceted solutions, and that a narrow focus on nudging or individual decision-making may not always be the most effective or equitable approach.

In the final section on applications, we explored a variety of areas where behavioral science is making a tangible difference in

the lives of people around the world. From financial wellbeing and entrepreneurship to health decisions, democracy and governance, and creating opportunities for decent work and protecting vulnerable minority groups, behavioral insights are being leveraged to create innovative solutions to seemingly intractable problems. As the field continues to grow and mature, we can expect to see behavioral science applied to an even wider range of issues and sectors.

It is worth noting, though, that the growth of behavioral science in international development is not without its challenges. The most immediate of these challenges are the areas in which behavioral science and international development are both weakest. We attempted to frame this in the introductory chapter “We’re WEIRD,” which gets to the core of some of the biggest challenges we face.

Both behavioral science and international development independently receive criticism for being overly paternalistic. This critique is particularly salient as we think about who sets the agenda and designs interventions to change behavior. That theme is woven throughout this book, as practitioners attempt to strike a balance in designing behavioral interventions without undermining the autonomy of individuals and communities.

As the field continues to evolve, it will be crucial to change not just how initiatives are run but who is involved and where they occur to ensure that behavioral science is applied responsibly, ethically and in a way that affirms individual dignity.

Another critique is that the field may be susceptible to overreach, applying behavioral solutions where they may not be appropriate or effective. The chapters highlighting lessons from the field demonstrate a sensitivity to this. Behavioral scientists are looking for the right altitude to integrate behavioral insights, which may be at the level of individual behavior or organizational behavior.

This is not just a question of logistics and impact, it is also an ethical question of when is it appropriate to place responsibility with an individual versus an institution that is meant to represent or serve individuals. These are complicated questions and will have different answers in different contexts. A behavioral lens can help to better understand and inform the work.

Behavioral scientists are most successful when approaching projects and partners with humility, curiosity, and an openness to learning. As is highlighted in the chapter on health decisions, behavior change in development has a long history and a behavioral science lens is a complement rather than a replacement for other skills and ways of knowing.

The potential of behavioral science to effect positive change in the world is undeniable. As the field moves forward, it will be important to continue refining the tools at our disposal and learning from the successes and failures of organizations already engaged in this work. By doing so, we can better understand how to harness the power of behavioral science to create more effective, sustainable, and equitable interventions that improve the lives of individuals and communities globally.

The conclusion of this book, therefore, is less of an ending and more of a jumping-off point for the next stage of growth and exploration in behavioral science and international development. The journey is just beginning and the potential for progress is immense.

Together, we can continue to push the boundaries of what is possible, advancing our understanding of human behavior and using that knowledge to make a meaningful difference in the world.

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15 United Nations entities on applying behavioral science to improve policy objectives. His primary domains of interest include public administration reform, conflict reduction, peace process support, early childhood education, economic mobility and transitional justice. Josh has a Masters degree in Public Policy from the Kennedy School of Government at Harvard University and Bachelors Degree in International Development Economics and Middle Eastern Studies from New York University.

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Deborah is a Senior Behavioral Economics Fellow at the Behavioral Economics Group at the IDB. She works with academics and governments in the Latin American and Caribbean regions to develop field experiments based on behavioral economics. She holds a Ph.D. in Economics from George Mason University. Her publications are published in academic journals, including the Journal of Economic Behavior & Organization. Deborah has held various roles as Senior Consultant for the American Cancer Society, Senior Associate at ideas42, and Research Assistant at the Interdisciplinary Center for Economic Science at GMU and the Federal Reserve of Dallas.

Ana Maria Munoz Boudet, The World Bank, Mind, Behavior and Development Unit

Ana Maria Munoz Boudet is a senior social scientist and co-lead of the Mind, Behavior, and Development Unit (eMBeD) at the World Bank. Her work focuses on gender, poverty and inequality issues. Her work spans over 38 countries across all world regions. As part of eMBeD Ana Maria's work has focused on socio-emotional skills development to close gaps in education and labor markets, social norms and gender inequality, and

on interventions aimed at changing behaviors associated with negative human capital outcomes. She has led analytical research at the country and regional level, impact evaluations of policies and programs, as well as technical assistance to client countries. Sociologist by training, she holds a master's degree from the London School of Economics and doctorate studies from the University College of London.

Stanley Ngugi, Director People and Operations Busara

Stanley Ngugi (GSGS) is Director of People and Operations at Busara in Nairobi, Kenya, where he is responsible for Busara's policies to support the organization's commitments to diversity, equity and inclusion. Stanley holds a bachelor's degree in law and a master's in diplomacy, which supported his interest and earlier work in negotiation and mediation and training of both skills. In more recent years, his focus has been on developing and implementing innovative talent management programs that help organizations attract, retain, and develop top talent and do so in a manner that is appropriate for an international organization operating in the Global South.

Tahirat Omolara Eniola, Research and Innovation Manager, International Rescue Committee

With a passion for qualitative research, I am driven to explore the intricacies of human behavior and social dynamics. Armed with a toolbox of behavioral techniques and qualitative methodologies, including thematic analysis and grounded theory, I uncover patterns, meanings, and underlying motivations in data. Through my research, I aim to give voice to marginalized populations, challenge existing assumptions, and generate knowledge that has practical applications in policy, healthcare, and social justice initiatives.

Cristina Parilli, Development Effectiveness Officer at IDB Invest

Cristina Parilli is a Development Effectiveness Officer at IDB Invest, the private sector arm of the Inter-American Development Bank (IDB), focusing on measuring the impact of the bank's private sector investments. As a Fulbright Scholar, she earned a Master's degree in Behavioral and Decision Sciences from the University of Pennsylvania. Additionally, she is a Chevening Scholar and holds a Master's degree in Public Policy from University College London (UCL). Cristina has also worked as Behavioral Economics Consultant at the IDB and has held positions at the Wharton Behavioral Lab, CAF - Development Bank of Latin America, and the International Initiative for Impact Evaluation (3ie).

Gautam Patel, Deputy Director (Lead - State and NITI Behavioural Insights Units), Centre for Social and Behaviour Change (CSBC), Ashoka University

Gautam Patel has worked with governments for two decades, currently with the Government of Bihar, the Government of Uttar Pradesh, and NITI Aayog in Delhi. Prior to joining CSBC, Gautam worked for 8 years with J-PAL South Asia, where he led partnerships with several state governments in India. Earlier, Gautam worked in the Chief Minister's Office in Gujarat, and before that, in the Chief Minister's Office in Rajasthan. Before settling in India, he worked for the government in the UK for education reform, and in the evenings as a trained youth worker. Gautam graduated from the London School of Economics and Political Science with a Master's in Development Management (2007), and holds a Master's in Molecular and Cellular Biochemistry from Oxford University (2003).

***Neela Saldanha, Executive Director at the Yale Research Initiative on Innovation and Scale (Y-RISE) at Yale University
Co-Founder and Principal, Venn Advisors***

Neela Saldanha is the Executive Director at the Yale Research Initiative on Innovation and Scale (Y-RISE) at Yale University and Co-Founder and Principal at Venn Advisors. Previously, Neela was the Founding Director of the Centre for Social and Behaviour Change (CSBC) at Ashoka University India. She has also consulted with several organizations working in the area of behavior change and development such as Innovations for Poverty Action, Johns Hopkins Center for Communication Programs, the Busara, Surgo Ventures and Noora Health. Neela combines her social sector expertise with extensive experience in the private sector. Over 15 years, she led global teams in strategy, consumer insight, sales and marketing at multinationals like Nestle, Unilever and PepsiCo and Accenture. Neela was mentioned in Forbes magazine as “Ten Behavioral Scientists You Should Know”. Her writing has appeared in Harvard Business Review, Behavioral Scientist, Apolitical, Nature Human Behavior, Lancet Regional Health among others. She is a board member of The Life You Can Save, Grameen Foundation USA, Bescy and Her Move Next. She is also on the advisory board at the Master of Behavioral and Decisions Sciences Program at the University of Pennsylvania. Neela has a PhD in Marketing from the Wharton School, University of Pennsylvania and an MBA from the Indian Institute of Management, Calcutta, India.

Carlos Scartascini, Head of the Development Research Group at the Research Department and Leader of the Behavioral Economics Group of the Inter-American Development Bank

Carlos Scartascini is Head of the Development Research Group at the Research Department and Leader of the Behavioral Economics Group of the Inter-American Development Bank. He has published eight books and more than 60 articles in academic journals and edited volumes. He is a member of the Executive Committee of IDB's Gender and Diversity Lab, member of the Scientific Committee of Elcano Royal Institute, member of the Board of Advisors of the Master of Behavioral and Decision Sciences at the University of Pennsylvania, Associate Editor of the academic journal *Economía*, and Founding Member of LACEA's BRAIN (Behavioral Insights Network).

Laurenz Karim Franziskus Scheunemann, The World Bank. Mind, Behavior and Development Unit and ifok

Laurenz Scheunemann works in the nexus of (behavioural) science, citizen engagement and policymaking. As a consultant at the World Bank's 'Mind, Behavior and Development (eMBeD)' unit as well as 'ifok', a leading public sector consultancy for citizen and stakeholder engagement, his work is focused both on the research and implementation of novel, participatory approaches to more effective and human-centred policymaking. Laurenz worked at the EU Commission's Policy Lab/Behavioural Insights unit, the Behavioural Insights Team (BIT) and the human rights division at the German UN Mission in Geneva. He completed the prestigious Mercator Fellowship for International Affairs, run by the German Federal Foreign Office and the Mercator Foundation. Laurenz holds degrees in Social Intervention & Policy Evaluation from the University

of Oxford (M.Sc., with distinction), as well as a Politics & Public Administration (B.A.) and Psychology (B.Sc.) from the University of Konstanz. During his studies, he was a fellow of the Friedrich Ebert Foundation (FES).

Mareike Schomerus, Vice President, Busara

Mareike Schomerus (GNGN), (PhD, London School of Economics and Political Science) is Vice President at Busara in Nairobi, Kenya. She is a widely published researcher, with a body of work on violent conflict, political contestation and peace processes in South Sudan and Uganda and across borders, as well as behavioral mechanisms in post-conflict recovery, for which she has developed a body of work on the ‘mental landscape’ of lives in or after situations of violence conflict. She has conducted applied research for a wide range of international organizations and is a visiting lecturer at the Harris School of Public Policy at the University of Chicago. Formerly Director of Programme Politics and Governance at ODI in London, she oversaw the Secure Livelihoods Research Consortium (SLRC), also at ODI, as Research Director. She is the co-editor of two volumes (on African secessionism and South Sudan’s borderlands (Palgrave Macmillan) and author of the monographs *The Lord’s Resistance Army: Violence and Peacemaking in Africa* (Cambridge University Press 2021) and, most recently, *Lives Amid Violence: Transforming Development in the Wake of Conflict* (Bloomsbury, 2023).

Mallika Sobti, Chief of Staff to the CEO, IDinsight

Mallika Sobti is the Chief of Staff to the CEO, based in Nairobi, Kenya. As a Senior Associate and subsequently Senior Manager prior to taking on this role, Mallika led projects in India and Africa focused on generating demand-responsive, rigorous evidence for governments, funders, and non-profits, particularly in

the domains of health and nutrition, community empowerment, and data systems. Mallika holds a Masters in Development Studies from the Graduate Institute of International and Development Studies (IHEID), Geneva and a bachelor's degree in Economics (Honours) from Shri Ram College of Commerce, Delhi University. She is passionate about helping decision-makers in the social sector amplify their impact and dismantling power asymmetries in the aid world. She was closely involved in IDinsight's Dignity Initiative in its early months and remains a keen supporter of efforts to better uphold the dignity of those impacted by the work of global development actors.

Carter Timon, Associate Director, Master of Behavioral and Decision Sciences at the University of Pennsylvania

Carter E. Timon is Associate Director of the Master of Behavioral and Decision Sciences at the University of Pennsylvania, and former Social Science Director at CHILD USA, a national think tank for child maltreatment prevention. His work has been published in *Anthropological Theory*, *Signs and Society*, *BMJ Open*, and others.

Britt Titus, Behavioral Insights Lead, The International Rescue Committee

Britt Titus' expertise lies at the intersection of behavioral insights and humanitarian action. She leads the behavioral science team at the International Rescue Committee's Airbel Impact Lab, where her team applies behavioral insights to practical issues affecting humanitarian outcomes ranging from women's health to education in emergencies, nutrition, and mental health. She has over a decade of experience in the humanitarian sector working in dozens of countries and previously worked at the United Nations World Food Program, the Behavioral Insights Team,

and Nudge Lebanon. She has a Master of Public Policy (MPP) from the University of Oxford.

Laura van Berkel, Behavioral Economist, USAID

Laura Van Berkel is a Behavioral Economist at the US Agency for International Development's (USAID) Center for Economics and Market Development. She advises and supports USAID Office, Bureau, and Mission staff on the use of behavioral science and experimental methods in USAID programming; serves as a liaison to the USAID behavioral science and experimental economics community; and develops and manages staff training programs in behavioral science and experimental economics. Dr. Van Berkel's prior experience includes USAID's Center for Democracy, Human Rights, and Governance; Democracy International, Inc.; the National Science Foundation; and academic research in political psychology. Laura is a proud alumna of the AAAS Science and Technology Policy Fellowship and the American Psychological Association Executive Branch Science Fellowship. She earned her PhD in Social Psychology from the University of Kansas with a minor in Quantitative Psychology and completed postdoctoral research at the University of Cologne in Germany. She holds a B.A. in Psychology from Saint Louis University with minors in History and Spanish. She lives in Silver Spring, MD with her husband, son, and dog.

Tom Wein, Director of the Dignity Initiative at IDinsight

Tom is Director of the Dignity Initiative at IDinsight, based in Nairobi, Kenya. He joined IDinsight in January 2022. Tom has 13 years experience leading research in the Global South. Prior to joining IDinsight, he founded The Dignity Project, a campaign for more respectful international development. With The Dignity Project, he gave evidence to the UK Parliament, and

wrote several academic papers; a book on dignity and organizations is forthcoming. He previously led Raising Voices' work to prevent violence against children in schools in Uganda. Before that, he worked with the Busara, where he built the CREME research agenda on culture, research ethics and methods, and led work on activism, civil society and edutainment.

Tom holds a Master's degree in Communication for Development from Malmö University in Sweden, and an undergraduate degree in War Studies from King's College, London. He works to advance justice and create better governance through useful research, and believes research is most useful when it grapples with questions of power. He is coauthor of the forthcoming book from Wharton Business Press, 'The Dignity Imperative'.

***Steve Wendel, Vice President, Busara
Co-Founder, Bescy***

Dr. Wendel is an applied behavioral scientist who helps organizations effectively use behavioral science to the benefit of their users. He currently serves as a Vice President at the Busara, where he supervises diverse areas of the non-profit organization's work. He previously founded and led the behavioral science teams at Morningstar and HelloWallet, developing and field testing hundreds of interventions to improve financial behavior. Stephen has authored three books on applied behavioral science, *Designing for Behavior Change*, *Improving Employee Benefits*, and *Spiritual Design*. He also founded the non-profit Bescy, which has educated the public across fifteen cities on how to apply behavioral research. He has two wonderful kids and a wife, who don't care about behavioral science at all.

***Allison White, Distributed Work Intervention Lead,
Team Anywhere, Atlassian***

Allison White started her career as a User Experience Designer at Deloitte Consulting, and her background in human-centered design continues to shape her work as a behavioral scientist. She entered the world of behavioral science by joining the Common Cents Lab at Duke University's Center for Advanced Hindsight, where she partnered with financial institutions to design and test solutions to help people make healthier financial decisions. Most recently, Allison was a Senior Behavioral Scientist on the Product Team at Humu: an HR product that aims to make work better for everyone, everywhere by empowering people to improve themselves, and the teams around them through behavioral science. She is the author of the Habit Weekly Pro case study on behavioral mapping, and has been featured on Action Design Radio, in Fintech Futures, and in UX Collective.

***Paulius Yamin, Scientific Director, Paris Institute for
Advanced Study
Visiting Fellow, London School of Economics and
Political Science***

Paulius is an anthropologist and behavioral scientist with broad experience working on behavioral and social norm interventions in the public, community, and international development sectors. He is currently the Scientific Director at the Paris Institute for Advanced Study and a Visiting Fellow at the Department for Psychological and Behavioral Science at the London School of Economics.

In the past, Paulius has worked as Managing Director of the Center for Social Norms and Behavioral Dynamics at the University of Pennsylvania (US), the Co-founder and Managing Partner of the Behavioural Lab Lithuania, and as consultant on social norms, behavioral change and governance for USAID,

the World Bank, ILO, and the IOM. In 2015-2016, he also created and headed one of the first behavioral teams of the Colombian Government, coordinating among other interventions the creation of the new integrity code for public servants with the OECD.

Allison Zelkowitz, Founder and Director CUBIC

Allison created and launched the Center for Utilizing Behavioral Insights for Children (CUBIC <https://www.cubic-sci.org>) in April 2020, supported by the Save the Children Innovation Accelerator. Currently based in Kampala, Uganda, Allison previously served as Save the Children's Lebanon and Thailand Country Directors, and held senior leadership positions in Pakistan and Aceh, Indonesia. She has been working in international aid and development for over 17 years. Allison holds a Master's of Pacific and International Affairs from the University of California, San Diego, where she specialized in International Development, Nonprofit Management, and Southeast Asia. She graduated from Bowdoin College with a bachelor's degree in English and Theater. She is a certified member of the Global Association for Applied Behavioral Scientists (GAABS).

